Analytic Number Theory Note

Introduction to the Theory of Numbers

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Chapter 1 Introduction

1.1 Basic Definitions

Definition 1.1 (Natural Numbers)

The set of natural numbers is denoted by $\mathbb{N} = \{1, 2, 3, \ldots\}$.

♣

Definition 1.2 (Prime Numbers)

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself. And the set of prime numbers is denoted by $\mathbb{P} = \{2, 3, 5, 7, 11, \ldots\}$.

♣

Definition 1.3 (Prime Counting Function)

We using the notation $\pi(N) = \#\{p \le N : p \in \mathbb{P}\}$ *to denote the number of primes less than or equal to* N. *Sometimes, we also use* $\pi(N) = \sum_{p \le N} 1$ *to denote the same function.*

♣

Definition 1.4 (Euler Totient Function)

For any natural number n*, the Euler totient function* $\phi(n)$ *is defined as the number of integers from 1 to* n *that are coprime to* n*. In other words,*

$$\phi(n) = \#\{k \in \mathbb{N} : 1 \le k \le n, \gcd(k, n) = 1\}.$$

♣

1.2 Fundamental Theorem of Arithmetic and Infinitely Many Primes

Theorem 1.1 (Fundamental Theorem of Arithmetic (FTA))

Every natural number greater than 1 can be uniquely factored into prime numbers, up to the order of the factors. In other words, for any $n \in \mathbb{N}$ *with* $n > 1$*, there exist unique primes* $p_1 < p_2 < \cdots < p_k$ *and positive integers* $\alpha_1, \alpha_2, \ldots, \alpha_k$ *such that*

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}.$$

♡

Proof First, we show existence. We proceed by induction on n. The base case $n = 2$ is prime itself. Assume true for all integers less than n. If n is prime, we are done. If not, let $n = ab$ with $1 < a, b < n$. By the induction hypothesis, both a and b can be factored into primes. Combining these factorizations gives a prime factorization of n.

Next, we show uniqueness. Suppose n has two distinct prime factorizations:

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k} = q_1^{\beta_1} q_2^{\beta_2} \cdots q_m^{\beta_m},$$

where p_i and q_j are primes. Without loss of generality, let p_1 be the smallest prime in the first factorization. Since p_1 divides n, it must divide the right-hand side. By the property of primes, p_1 must equal one of the q_j. Repeating this argument for all primes in both factorizations shows that the multisets of primes must be identical, establishing uniqueness.

Theorem 1.2 (Infinitely Many Primes)

Over 2,300 years agao, Euclid showed that there are infinitely many primes. (i.e. $\lim_{N \to \infty} \pi(N) = \infty$*)*

♡

Proof Assume, for the sake of contradiction, that there are only finitely many primes: $p_1, p_2, \ldots, p_k$. Consider the number

$$N = p_1 p_2 \cdots p_k + 1.$$

By construction, N is not divisible by any of the primes $p_1, p_2, \ldots, p_k$, since it leaves a remainder of 1 when divided by any of them. Therefore, either N is prime itself or it has prime factors not in our original list. In either case, this contradicts our assumption that we had listed all primes. Hence, there must be infinitely many primes.

Proposition 1.1 (Lower Bound for $\pi(n)$)

For any natural number $n \geq 1$, we have the following inequality for $\pi(n)$:

$$\log\log n \leq \pi(n)$$

Exercise 1.1 Prove the Proposition 1.1. (workshop 1)

(Hint: Let p_n be the n-th prime number. Show that $p_n \leq 2^{2^{n-1}}$ for all $n \geq 1$.)

Solution *First, we prove by induction that $p_n \leq 2^{2^{n-1}}$ for all $n \geq 1$.*
The base case $n = 1$ holds since $p_1 = 2 \leq 2^{2^0} = 2$. Assume the statement is true for some $k \geq 1$, i.e., $p_k \leq 2^{2^{k-1}}$. We need to show it holds for $k + 1$. Consider the number

$$N = p_1 p_2 \cdots p_k + 1.$$

By construction, N is not divisible by any of the primes $p_1, p_2, \ldots, p_k$. Therefore, either N is prime itself or it has a prime factor greater than p_k. In either case, we have

$$p_{k+1} \leq N = p_1 p_2 \cdots p_k + 1 \leq (2^{2^0})(2^{2^1}) \cdots (2^{2^{k-1}}) + 1 = 2^{\sum_{i=0}^{k-1} 2^i} + 1 = 2^{2^k - 1} + 1 \leq 2^{2^k}.$$

Thus, by induction, the inequality $p_n \leq 2^{2^{n-1}}$ holds for all $n \geq 1$.

Now, to prove the proposition, let $n \geq 1$ and let $k = \pi(n)$, which is the number of primes less than or equal to n. By the definition of $\pi(n)$, we have $p_k \leq n < p_{k+1}$. Using the inequality we just proved, we get

$$n < p_{k+1} \leq 2^{2^k} \implies \log n \leq 2^k \log 2 \implies \log\log n \leq k \log 2 + \log\log 2 \leq k = \pi(n).$$

Therefore, we conclude that $\log\log n \leq \pi(n)$ for all $n \geq 1$.

Theorem 1.3 (Prime Number Theorem (PNT))

As N approaches infinity, the prime counting function $\pi(N)$ is asymptotically equivalent to $\frac{N}{\log N}$: $\pi(N) \sim \frac{N}{\log N}$ as $N \to \infty$. In other words,

$$\lim_{N\to\infty} \frac{\pi(N)}{\frac{N}{\log N}} = 1.$$

The proof of Theorem 1.3 is one of the main goal of this course and will be *given in later chapters* after introducing more tools from complex analysis.

Lemma 1.1

To prove Theorem 1.3, it equivalent to show that for any positive constants c_1, c_2 with $c_1 < 1 < c_2$, we have

$$c_1 \frac{N}{\log N} \leq \pi(N) \leq c_2 \frac{N}{\log N}$$

Proof Suppose Theorem 1.3 holds. Then for any $\varepsilon > 0$, there exists N_0 such that for all $N > N_0$,

$$\left| \frac{\pi(N)}{\frac{N}{\log N}} - 1 \right| < \varepsilon.$$

Choosing $\varepsilon = \min(1 - c_1, c_2 - 1)$, we have

$$1 - \varepsilon < \frac{\pi(N)}{\frac{N}{\log N}} < 1 + \varepsilon,$$

which implies

$$c_1 \frac{N}{\log N} < \pi(N) < c_2 \frac{N}{\log N}.$$

Conversely, suppose the inequalities hold for all sufficiently large N. Then for any $\varepsilon > 0$, we can choose $c_1 = 1 - \varepsilon$

and $c_2 = 1 + \varepsilon$. Thus, for all sufficiently large N,

$$(1-\varepsilon)\frac{N}{\log N} < \pi(N) < (1+\varepsilon)\frac{N}{\log N},$$

which leads to

$$-\varepsilon < \frac{\pi(N)}{\frac{N}{\log N}} - 1 < \varepsilon.$$

Therefore,

$$\left|\frac{\pi(N)}{\frac{N}{\log N}} - 1\right| < \varepsilon,$$

proving Theorem 1.3.

Remark A more precise version of Theorem 1.3 states that

$$\pi(x) = \mathrm{Li}(x) + E(x) \tag{1.1}$$

where $\mathrm{Li}(x) = \int_2^x \frac{dt}{\log t} \sim \frac{x}{\log x}$ and the error term $E(x)$ satisfies $E(x) = \mathcal{O}(\sqrt{x}\log x)$.

✍ **Exercise 1.2** Prove that equation (1.1) implies Theorem 1.3.

Solution *From equation (1.1), we have*

$$\pi(x) = \mathrm{Li}(x) + E(x).$$

Dividing both sides by $\frac{x}{\log x}$, we get

$$\frac{\pi(x)}{\frac{x}{\log x}} = \frac{\mathrm{Li}(x)}{\frac{x}{\log x}} + \frac{E(x)}{\frac{x}{\log x}}.$$

As $x \to \infty$, we know that $\mathrm{Li}(x) \sim \frac{x}{\log x}$, so

$$\lim_{x\to\infty} \frac{\mathrm{Li}(x)}{\frac{x}{\log x}} = 1.$$

Next, we analyze the error term:

$$\left|\frac{E(x)}{\frac{x}{\log x}}\right| = \left|E(x)\cdot\frac{\log x}{x}\right| = \mathcal{O}\left(\sqrt{x}\log x \cdot \frac{\log x}{x}\right) = \mathcal{O}\left(\frac{(\log x)^2}{\sqrt{x}}\right).$$

Since $\frac{(\log x)^2}{\sqrt{x}} \to 0$ as $x \to \infty$, we have

$$\lim_{x\to\infty} \frac{E(x)}{\frac{x}{\log x}} = 0.$$

Combining these results, we find

$$\lim_{x\to\infty} \frac{\pi(x)}{\frac{x}{\log x}} = 1 + 0 = 1.$$

Thus, equation (1.1) implies Theorem 1.3.

Definition 1.5 (Riemann Zeta Function (Outline))

The Riemann zeta function $\zeta(z)$ is defined for complex numbers z and for $\Re(z) > 1$ is given by the infinite series

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}.$$

for the other values of z, $\zeta(z)$ is defined by analytic continuation as it has an analytic continuation to the whole complex plane except $z = 1$.

In the following chapters, we will show that

$$\zeta(1-z) = 2^{1-z}\pi^{-z}\cos\left(\frac{\pi z}{2}\right)\Gamma(z)\zeta(z) \quad \textit{for all } z \in \mathbb{C}.$$

and thus $\zeta(-2n) = 0$ which are called the trivial zeros of $\zeta(z)$. ♣

Moreover, if we let $\{\rho\}$ be the set of non-trivial zeros of $\zeta(z)$, then we have the following explicit formula for $\pi(x)$:

$$\pi(x) = \mathrm{Li}(x) - \sum_{\rho} \mathrm{Li}(x^\rho) + \text{smaller order terms}.$$

Proposition 1.2 (Analytic Continuation of Zeta Function)

The Riemann zeta function $\zeta(z)$ is holomorphic for all z with $\Re(z) > 1$.

Exercise 1.3 Show the proposition 1.2 by Weierstrass M-test.

Solution *To show that the Riemann zeta function $\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}$ is holomorphic for all z with $\Re(z) > 1$, we will use the Weierstrass M-test.*

Let $z = x + iy$ where $x = \Re(z)$ and $y = \Im(z)$. For $\Re(z) > 1$, we have $x > 1$. We can write

$$\left|\frac{1}{n^z}\right| = \frac{1}{|n^z|} = \frac{1}{n^{\Re(z)}} = \frac{1}{n^x}.$$

Since $x > 1$, the series $\sum_{n=1}^{\infty} \frac{1}{n^x}$ converges.

Now, we can choose $M_n = \frac{1}{n^x}$ as our bounding sequence. Since $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{1}{n^x}$ converges for $x > 1$, by the Weierstrass M-test, the series $\sum_{n=1}^{\infty} \frac{1}{n^z}$ converges uniformly on any compact subset of the half-plane $\Re(z) > 1$.

Since each term $\frac{1}{n^z}$ is holomorphic in z for $\Re(z) > 1$, and the series converges uniformly, it follows that the sum $\zeta(z)$ is also holomorphic in this region.

Therefore, we conclude that the Riemann zeta function $\zeta(z)$ is holomorphic for all z with $\Re(z) > 1$.

Postulate 1.1 (Riemann Hypothesis (RH))

All non-trivial zeros ρ of the Riemann zeta function $\zeta(z)$ have real part equal to $\frac{1}{2}$, i.e., $\Re(\rho) = \frac{1}{2}$.

Remark One can find that the non-trivial zeros of $\zeta(z)$ are only appear in the critical strip: $0 < \Re(z) < 1$.

Definition 1.6

Usually, we using the notation $(RH)_\sigma$ to denote the following statement:

$$\zeta(\rho) \neq 0 \quad \text{for all } \rho \text{ with } \Re(\rho) > \sigma.$$

where ρ is the set of non-trivial zeros of $\zeta(z)$.

By extend Euler Product Formula 1.4 to complex variable, one can show that $\zeta(z) \neq 0$ for all z with $\Re(z) > 1$. Hence, $(RH)_1$ is true. And the Riemann Hypothesis 1.1 is equivalent to $(RH)_{\frac{1}{2}}$.

A fact is that for $\sigma < 1$, $(RH)_\sigma$ is equivalent to the following statement about the error term $E(x)$ in equation (1.1):

$$|\pi(x) - \mathrm{Li}(x)| = \mathcal{O}(x^\sigma \log x).$$

and by using this fact, it can be shown the Prime Number Theorem 1.3. But unfortunately, we have no ideal if $(RH)_\sigma$ is true for any $\sigma < 1$.

However, Hadamard and de la Vallée Poussin (1896) independently proved that by using the fact that $(RH)_1$ is true (i.e. $\zeta(z) \neq 0$ for $\Re(z) > 1$), one can prove the Prime Number Theorem 1.3.

Note *Why the zeros of $\zeta(z)$ is important for the distribution of primes?*

From the Complex Analysis course, we know that a holomorphic function is completely determined by its zeros and poles (i.e. Singularities). In order to using complex analysis to study $\zeta(z)$, one introduces a fixed function which has the same zeros and poles as $\zeta(z)$, called the Riemann Xi function $\xi(z)$:

$$\xi(z) = \frac{1}{2} z(z-1)\pi^{-\frac{z}{2}}\Gamma\left(\frac{z}{2}\right)\zeta(z).$$

where $\Gamma(z)$ is the Gamma function.

A famous result about $\xi(z)$ is the Hadamard Factorization Theorem:

$$\xi(z) = e^{A+Bz}\prod_{\rho}\left(1 - \frac{z}{\rho}\right)e^{\frac{z}{\rho}}$$

where the product is over all non-trivial zeros ρ of $\zeta(z)$, and A and B are constants.

So once we know the distribution of zeros of $\zeta(z)$, we can understand the behavior of $\xi(z)$ and hence $\zeta(z)$, which in turn gives us information about the distribution of prime numbers via the explicit formula for $\pi(x)$.

1.3 Euler Product Formula and Dirichlet Theorem

Theorem 1.4 (Euler Product Formula)

Thinking the zeta function as a function of real variable $x > 1$, we have the Euler Product Formula:

$$\zeta(x) = \sum_{n=1}^{\infty} \frac{1}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-x}}. \quad \textit{for } x > 1.$$

♡

Proof Recall the geometric series formula: for $|r| < 1$, we have

$$\frac{1}{1-r} = \sum_{k=0}^{\infty} r^k = 1 + r + r^2 + r^3 + \cdots.$$

Setting $r = p^{-x} < 1$ for each prime p, we get

$$\frac{1}{1-p^{-x}} = \sum_{k=0}^{\infty} p^{-kx} = 1 + p^{-x} + p^{-2x} + p^{-3x} + \cdots.$$

Consider two distinct primes p and q. Multiplying their corresponding series, we have

$$(1 + p^{-x} + p^{-2x} + \cdots)(1 + q^{-x} + q^{-2x} + \cdots) = \sum_{j,k \geq 0} \frac{1}{[p^j q^k]^x}.$$

and so by the Fundamental Theorem of Arithmetic 1.1, the number of the form $p^j q^k$ are precisely the number n whose prime factors are only p and q. Hence

$$\frac{1}{1-p^{-x}} \cdot \frac{1}{1-q^{-x}} = \sum_{n \in N_{p,q}} \frac{1}{n^x} \quad \text{where } N_{p,q} = \{n \in \mathbb{N} : n = p^j q^k, j, k \in \mathbb{N}_0\}.$$

Extending this argument to all primes less than or equal to integer N, let $E_N = \{n \in \mathbb{N} : \text{all prime factors of } n \text{ are } \leq N\}$, we have

$$\prod_{p \leq N} \frac{1}{1-p^{-x}} = \sum_{n \in E_N} \frac{1}{n^x}.$$

Now, we show that as $N \to \infty$, we can get Euler Product Formula. Let

$$S_N = \sum_{n \in E_N} \frac{1}{n^x} \quad T_N = \zeta(x) - S_N = \sum_{n \notin E_N} \frac{1}{n^x}.$$

For any $n \notin E_N$, n must have a prime factor greater than N. Thus, $n = pm$ for some prime $p > N$ and some $m \in \mathbb{N}$. Therefore,

$$T_N \leq \sum_{\substack{p>N \\ p \in \mathbb{P}}} \sum_{m=1}^{\infty} \frac{1}{(pm)^x} = \sum_{\substack{p>N \\ p \in \mathbb{P}}} \frac{1}{p^x} \sum_{m=1}^{\infty} \frac{1}{m^x} = \zeta(x) \sum_{\substack{p>N \\ p \in \mathbb{P}}} \frac{1}{p^x}.$$

Since the series $\sum_{p \in \mathbb{P}} \frac{1}{p^x}$ converges for $x > 1$ by p-series test and comparison test, the tail $\sum_{p>N} \frac{1}{p^x} \to 0$ as $N \to \infty$. Hence, $T_N \to 0$ as $N \to \infty$, which implies that $S_N \to \zeta(x)$. Thus we have

$$\zeta(x) = \lim_{N \to \infty} S_N = \lim_{N \to \infty} \prod_{\substack{p \leq N \\ p \in \mathbb{P}}} \frac{1}{1-p^{-x}} = \prod_{p \in \mathbb{P}} \frac{1}{1-p^{-x}}.$$

Corollary 1.1 (Divergence of the Sum of Reciprocals of Primes)

By taking the logarithmic derivative of the Euler Product Formula, we can get the following fact

$$\sum_{p \in \mathbb{P}} \frac{1}{p} = \infty.$$

♡

Proof Taking the logarithm of both sides of the Euler Product Formula, we have

$$\log \zeta(x) = -\sum_{p \in \mathbb{P}} \log(1 - p^{-x}) = \sum_{p \in \mathbb{P}} \sum_{k=1}^{\infty} \frac{p^{-kx}}{k}$$

where the last equality follows from the Taylor series expansion of $\log(1-y) = -\sum_{k=1}^{\infty} \frac{y^k}{k}$ for $0 \le y < 1$.

For the inner sum, we have

$$\sum_{k=2}^{\infty} \frac{p^{-kx}}{k} < \sum_{k=2}^{\infty} p^{-k} = \frac{p^{-2}}{1-p^{-1}} = \frac{1}{p(p-1)}.$$

Thus,

$$\log \zeta(x) = \sum_{p\in\mathbb{P}} \left(\frac{p^{-x}}{1} + \sum_{k=2}^{\infty} \frac{p^{-kx}}{k} \right) = \sum_{p\in\mathbb{P}} p^{-x} + \sum_{p\in\mathbb{P}} \sum_{k=2}^{\infty} \frac{p^{-kx}}{k} \le \sum_{p\in\mathbb{P}} p^{-x} + E(x)$$

where $E(x) = \sum_{p\in\mathbb{P}} \frac{1}{p(p-1)}$ is a convergent series, and

$$E(x) \le \sum_{n=2}^{\infty} \frac{1}{n(n-1)} = \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1.$$

Therefore,

$$\log \zeta(x) \le \sum_{p\in\mathbb{P}} p^{-x} + 1.$$

As $x \to 1^+$, $\zeta(x) \to \infty$, so $\log \zeta(x) \to \infty$ by continuity of logarithm. Hence,

$$\sum_{p\in\mathbb{P}} p^{-x} \to \infty \quad \text{as } x \to 1^+.$$

and thus,

$$\sum_{p\in\mathbb{P}} \frac{1}{p} = \infty.$$

Definition 1.7 (Arithmetic Progression (AP))

An arithmetic progression is a sequence of numbers in which the difference between consecutive terms is constant. It can be expressed in the form:

$$\mathrm{AP}_{a,q} = \{a+q, a+2q, a+3q, \dots\} = \{nq + a : n \in \mathbb{N}\}$$

where a is the first term and q is the common difference.

Proposition 1.3 (Finiteness of Primes in Certain APs)

If $d = \gcd(a,q) > 1$, then the arithmetic progression $\mathrm{AP}_{a,q} = \{nq + a : n \in \mathbb{N}\}$ contains only finitely many primes.

Exercise 1.4 Prove the proposition above.

Solution *If $d = \gcd(a,q) > 1$, then every term in the arithmetic progression can be expressed as*

$$nq + a = n(dk) + (dm) = d(nk + m)$$

for some integers k and m. This means that every term in the progression is divisible by d. Since $d > 1$, none of these terms can be prime (except possibly the first term if it equals d itself). Therefore, the arithmetic progression contains only finitely many primes.

Theorem 1.5 (Dirichlet's Theorem on Arithmetic Progressions)

If a and q are coprime (i.e., $\gcd(a,q) = 1$), then the arithmetic progression $\mathrm{AP}_{a,q} = \{nq + a : n \in \mathbb{N}\}$ contains infinitely many primes.

The proof of Dirichlet's Theorem requires more advanced tools from analytic number theory, including Dirichlet characters and L-functions, which will be introduced in later chapters. However, various special cases and consequences of this theorem can be explored with more elementary methods. The following exercise provides an opportunity to investigate one such special case.

Exercise 1.5 Give an elementary proof of Dirichlet's Theorem for the AP $\{4n-1 : n \in \mathbb{N}\}$ by completing the following outline:

- Argue by contradiction as Euclid did for the infinitude of primes and suppose there are only finitely many primes of the form $4n-1$, say $p_1, p_2, \ldots, p_k$.
- Consider the number $N := 2^2 \cdot 3 \cdot 5 \cdots p_k - 1$. (all odd primes up to p_k).
- Note that $N > p$ and $N = 4m - 1$ for some $m \in \mathbb{N}$.
- Hence N is not prime. Deduce that N has a prime factor of the form $4n-1$ to obtain a contradiction.

Solution *We proceed by contradiction. Suppose there are only finitely many primes of the form $4n-1$, say $p_1, p_2, \ldots, p_k$. Consider the number*

$$N := 2^2 \cdot 3 \cdot 5 \cdots p_k - 1.$$

Note that $N > p_k$ since $p_k \geq 3$. Also, we can express N in the form $4m-1$ for some integer m.

Since N is greater than any of the primes $p_1, p_2, \ldots, p_k$, it cannot be prime itself (as it would contradict our assumption). Therefore, N must be composite and have prime factors.

Now, we analyze the prime factors of N. Any prime factor of N must be either of the form $4n-1$ or $4n+1$. (This is because all primes greater than 2 are odd, and odd primes can be expressed in one of these two forms.) However, if all prime factors were of the form $4n+1$, then their product would also be of the form $4n+1$. This is because the product of two numbers of the form $4n+1$ is also of that form:

$$(4a+1)(4b+1) = 16ab + 4a + 4b + 1 = 4(4ab + a + b) + 1.$$

Since we have established that N is of the form $4m-1$, it cannot be expressed as a product of only primes of the form $4n+1$. Therefore, at least one prime factor of N must be of the form $4n-1$. But this contradicts our initial assumption that $p_1, p_2, \ldots, p_k$ were the only primes of that form and all of them not dividing N.

Hence, we conclude that there are infinitely many primes of the form $4n-1$.

Lemma 1.2

Let $\mathbb{1}_{a,q}(n)$ be the indicator function for the arithmetic progression $\mathrm{AP}_{a,q}$, defined as

$$\mathbb{1}_{a,q}(n) = \begin{cases} 1 & \text{if } n \equiv a \mod q, \\ 0 & \text{otherwise}. \end{cases}$$

Then once we have

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \mathbb{1}_{a,q}(p) p^{-x}} \quad \text{for } x > 1.$$

we can show that the series

$$\sum_{\substack{p \in \mathbb{P} \\ p \equiv a \mod q}} \frac{1}{p} = \infty$$

and hence Dirichlet's Theorem 4.5 holds.

Note *Although our assumption in Lemma 4.1 may not true, but it gives us the main idea of Dirichlet's proof of Theorem 4.5.*

Proof Dirichlet was inspired by Euler's proof that the series of reciprocals of primes diverges (Corollary 1.1) to prove his theorem. Let $\mathbb{1}_{a,q}(n)$ be the indicator function for the arithmetic progression $\mathrm{AP}_{a,q}$, defined as

$$\mathbb{1}_{a,q}(n) = \begin{cases} 1 & \text{if } n \equiv a \mod q, \\ 0 & \text{otherwise}. \end{cases}$$

Then once can show that the series

$$\sum_{\substack{p \in \mathbb{P} \\ p \equiv a \mod q}} \frac{1}{p} = \infty$$

for any a, q with $\gcd(a, q) = 1$, which implies that there are infinitely many primes in the arithmetic progression $\mathrm{AP}_{a,q}$

and hence Dirichlet's Theorem 4.5 holds. Then one might hope extend Euler's Product Formula 1.4 to

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x} = \prod_{p\in\mathbb{P}} \frac{1}{1-\mathbb{1}_{a,q}(p)p^{-x}} \quad \text{for } x > 1.$$

On the other hand, since

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n} = \sum_{m=1}^{\infty} \frac{1}{mq+a} \geq \frac{1}{a+q}\sum_{m=1}^{\infty} \frac{1}{m} = \infty.$$

wee see that $L(x;\mathbb{1}_{a,q}) := \sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x}$ tends to infinity as $x \to 1^+$. Thus, do the similar things as in Corollary 1.1, we have

$$\log L(x;\mathbb{1}_{a,q}) = -\sum_{p\in\mathbb{P}} \log(1-\mathbb{1}_{a,q}(p)p^{-x}) = \sum_{p\in\mathbb{P}}\sum_{k=1}^{\infty} \frac{\mathbb{1}_{a,q}(p)^k p^{-kx}}{k}.$$

where the last equality follows from the Taylor series expansion of $\log(1-y) = -\sum_{k=1}^{\infty} \frac{y^k}{k}$ for $0 \leq y < 1$. And as before, since

$$\sum_{k=2}^{\infty} \frac{\mathbb{1}_{a,q}(p)^k p^{-kx}}{k} \leq \sum_{k=2}^{\infty} p^{-k} = \frac{p^{-2}}{1-p^{-1}} = \frac{1}{p(p-1)},$$

we have

$$\log L(x;\mathbb{1}_{a,q}) = \sum_{p\in\mathbb{P}} \left(\frac{\mathbb{1}_{a,q}(p)p^{-x}}{1} + \sum_{k=2}^{\infty} \frac{\mathbb{1}_{a,q}(p)^k p^{-kx}}{k} \right) = \sum_{\substack{p\in\mathbb{P}\\ p\equiv a \mod q}} \frac{1}{p^x} + E(x) \tag{1.2}$$

where $E(x) = \sum_{p\in\mathbb{P}} \frac{1}{p(p-1)} \leq 1$ is a convergent series as before. Therefore,

$$\log L(x;\mathbb{1}_{a,q}) \leq \sum_{\substack{p\in\mathbb{P}\\ p\equiv a \mod q}} \frac{1}{p^x} + 1.$$

As $x \to 1^+$, $L(x;\mathbb{1}_{a,q}) \to \infty$, so $\log L(x;\mathbb{1}_{a,q}) \to \infty$ by continuity of logarithm. Hence,

$$\sum_{\substack{p\in\mathbb{P}\\ p\equiv a \mod q}} \frac{1}{p^x} \to \infty \quad \text{as } x \to 1^+.$$

and thus,

$$\sum_{\substack{p\in\mathbb{P}\\ p\equiv a \mod q}} \frac{1}{p} = \infty.$$

Note *Note that if we want to show*

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x} = \prod_{p\in\mathbb{P}} \frac{1}{1-\mathbb{1}_{a,q}(p)p^{-x}} \quad \textit{for } x > 1.$$

in the above Lemma 4.1. Examining the proof of Euler Product Formula 1.4, we need to show that $\mathbb{1}_{a,q}(n)$ is completely multiplicative. (i.e. $\mathbb{1}_{a,q}(p^j q^k) = \mathbb{1}_{a,q}(p)^j \mathbb{1}_{a,q}(q)^k$). But unfortunately, $\mathbb{1}_{a,q}(n)$ is not completely multiplicative on arithmetic progression. So Lemma 4.1 just gives the idea of Dirichlet's proof of Theorem 4.5, but we need more advanced tools to complete the proof.

One way to fix this problem is to introduce Dirichlet characters χ modulo q, which will be discussed in later chapters. The core idea is to show the formula

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^x} = \prod_{p\in\mathbb{P}} \frac{1}{1-\chi(p)p^{-x}} \quad \textit{for } x > 1.$$

and then using the similar argument as in Lemma 4.1 but replacing $\mathbb{1}_{a,q}(n)$ by Dirichlet characters χ modulo q: $1_{a,q}(n) = \sum_{\chi} c_{\chi}\chi(n)$ (which will be discussed in later chapters) to get

$$\log L(x;\chi) = -\sum_{p\in\mathbb{P}} \log(1-\chi(p)p^{-x}) = \sum_{p\in\mathbb{P}}\sum_{k=1}^{\infty} \frac{\chi(p)^k p^{-kx}}{k}.$$

as before, where $L(x;\chi) := \sum_{n=1}^{\infty} \frac{\chi(n)}{n^x}$ *and*

$$\sum_{\chi} c_\chi \log L(x;\chi) = \sum_{\substack{p\in\mathbb{P} \\ p\equiv a \mod q}} \frac{1}{p^x} + E(x)$$

as in equation (1.2). At last, we just need $\sum_\chi c_\chi \log L(x;\chi) \to \infty$ *as* $x \to 1^+$*, the following Lemma 1.3 give us* $\log L(x;\chi_0) \to \infty$*, and later chapters will show that:*

- $L(x;\chi)$ *is bounded for all* $\chi \neq \chi_0$ *as* $x \to 1^+$.
- $L(x;\chi) \neq 0$ *for all* $\chi \neq \chi_0$ *as* $x \to 1^+$.

by using these two properties, we can show that $\sum_\chi c_\chi \log L(x;\chi) \to \infty$ *as* $x \to 1^+$*. Finally, we arrive at the conclusion of Dirichlet's Theorem 4.5.*

Lemma 1.3

For principal Dirichlet character χ_0 *modulo* q

$$\chi_0(n) = \begin{cases} 1 & \textit{if } \gcd(n,q) = 1, \\ 0 & \textit{otherwise}, \end{cases}$$

we have

$$L(x;\chi_0) = \sum_{n=1}^{\infty} \frac{\chi_0(n)}{n^x} \to \infty \quad \textit{as } x \to 1^+.$$

♡

Proof By the definition of principal Dirichlet character χ_0 modulo q, we have

$$L(x;\chi_0) = \sum_{n=1}^{\infty} \frac{\chi_0(n)}{n^x} = \sum_{m=0}^{\infty} \sum_{\substack{b=1 \\ \gcd(b,q)=1}}^{q} \frac{1}{(mq+b)^x} \geq \sum_{m=0}^{\infty} \sum_{\substack{b=1 \\ \gcd(b,q)=1}}^{q} \frac{1}{(mq+q)^x} = \phi(q) \sum_{m=1}^{\infty} \frac{1}{(mq)^x} = \frac{\phi(q)}{q^x} \sum_{m=1}^{\infty} \frac{1}{m^x}.$$

where the second equality follows from the definition of $\chi_0(n)$, and $\gcd(b,q) = 1$ if and only if $\gcd(mq+b,q) = 1$ for any $m \in \mathbb{N}_0$.

Thus, as $x \to 1^+$, we have

$$L(x;\chi_0) \geq \frac{\phi(q)}{q^x} \sum_{m=1}^{\infty} \frac{1}{m^x} \to \infty.$$

✍ **Exercise 1.6** The Principal Dirichlet Character χ_0 modulo q is defined as

$$\chi_0(n) = \begin{cases} 1 & \text{if } \gcd(n,q) = 1, \\ 0 & \text{otherwise}. \end{cases}$$

Show that

$$\chi_0(mn) = \chi_0(m)\chi_0(n) \quad \text{for all } m,n \in \mathbb{N}.$$

and

$$\chi_0(b+nq) = \chi_0(b) \quad \text{for all } b,n \in \mathbb{N}.$$

Solution *First, we show that* $\chi_0(mn) = \chi_0(m)\chi_0(n)$ *for all* $m,n \in \mathbb{N}$*. We consider two cases:*

(C1) If $\gcd(m,q) = 1$ *and* $\gcd(n,q) = 1$*, then* $\gcd(mn,q) = 1$*. Therefore,*

$$\chi_0(mn) = 1 = 1 \cdot 1 = \chi_0(m)\chi_0(n).$$

(C2) If either $\gcd(m,q) > 1$ *or* $\gcd(n,q) > 1$*, then* $\gcd(mn,q) > 1$*. Therefore,*

$$\chi_0(mn) = 0 = 0 \cdot \chi_0(n) = \chi_0(m)\chi_0(n) \quad \textit{or} \quad \chi_0(mn) = 0 = \chi_0(m) \cdot 0 = \chi_0(m)\chi_0(n).$$

Thus, in both cases, we have $\chi_0(mn) = \chi_0(m)\chi_0(n)$.

Next, we show that $\chi_0(b+nq) = \chi_0(b)$ *for all* $b,n \in \mathbb{N}$*. We consider two cases:*

(C1) If $\gcd(b,q) = 1$*, then* $\gcd(b+nq,q) = 1$*. Therefore,*

$$\chi_0(b+nq) = 1 = \chi_0(b).$$

(C2) If $\gcd(b, q) > 1$*, then* $\gcd(b + nq, q) > 1$*. Therefore,*

$$\chi_0(b + nq) = 0 = \chi_0(b).$$

Thus, in both cases, we have $\chi_0(b + nq) = \chi_0(b)$*.*

Definition 1.8 (Dirichlet L-Function (Outline))

Given a Dirichlet character χ *modulo* q*, the Dirichlet* L*-function* $L(s, \chi)$ *is defined for complex numbers* s *with* $\Re(s) > 1$ *by the series*

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

Similar to the Riemann zeta function, the Dirichlet L*-function can be analytically continued to the entire complex plane except for a simple pole at* $s = 1$ *when* χ *is the principal character.*

♣

1.4 Chebyshev's bounds

Lemma 1.4 (Legendre's Formula)

For any integer n *and prime* p*, consider the prime factorization of* $m!$*:*

$$m! = p^\alpha \cdot (\textit{other prime factors not equal to } p).$$

Then we have

$$\alpha = \sum_{k=1}^{\infty} \left\lfloor \frac{m}{p^k} \right\rfloor.$$

and denote it as $\alpha(p) = \alpha$ *of the prime* p *in the prime factorization of* $m!$*.*

♡

Note *The series* $\sum_{k=1}^{\infty} \left\lfloor \frac{m}{p^k} \right\rfloor$ *is actually finite since* $\left\lfloor \frac{m}{p^k} \right\rfloor = 0$ *for all* $k > \log_p m$*.*

Proof Notice that the prime factors of $m!$ are exactly the all prime factors from 1 to m. Since

$$\left\lfloor \frac{m}{p^k} \right\rfloor = \#\{np^k : n \in \mathbb{N}, np^k \leq m\}$$

counts the number of multiples of p^k that are less than or equal to m, each multiple contributes at least k to the exponent α of p in the prime factorization of $m!$. Therefore, summing over all k gives the total exponent of p in $m!$:

$$\alpha = \sum_{k=1}^{\infty} \left\lfloor \frac{m}{p^k} \right\rfloor.$$

Exercise 1.7 Show that $2^n \leq \binom{2n}{n} < 4^n$ for all $n \in \mathbb{N}$.

Solution *We can prove this inequality using the definition of the binomial coefficient:*

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2}.$$

We want to show that

$$2^n \leq \frac{(2n)!}{(n!)^2}.$$

Multiplying both sides by $(n!)^2$*, we need to prove that*

$$2^n (n!)^2 \leq (2n)!.$$

We can rewrite $(2n)!$ *as the product of the first* $2n$ *integers:*

$$(2n)! = 1 \cdot 2 \cdot 3 \cdots (2n).$$

Now, we can group the terms in pairs:

$$(2n)! = (1 \cdot 2) \cdot (3 \cdot 4) \cdots ((2n-1) \cdot (2n)).$$

Each pair $(2k-1)\cdot(2k)$ *for* $k=1,2,\ldots,n$ *is greater than or equal to* $2k$*, since*

$$(2k-1)\cdot(2k)=4k^2-2k=2k(2k-1)\geq 2k \quad \text{for } k\geq 1.$$

Therefore,

$$(2n)!\geq(2\cdot 1)\cdot(2\cdot 2)\cdots(2\cdot n)=2^n(1\cdot 2\cdots n)=2^n n!.$$

This shows that

$$2^n(n!)^2\leq(2n)!.$$

For the upper bound, by using the binomial theorem, we have

$$4^n=(1+1)^{2n}=\sum_{k=0}^{2n}\binom{2n}{k}\geq\binom{2n}{n}.$$

Thus, we have shown that

$$2^n\leq\binom{2n}{n}<4^n.$$

Lemma 1.5 (Chebyshev's Product Bounds lemma)

For n *sufficiently large, we have*

$$\left(\frac{4}{3}\right)^n\leq\prod_{\substack{p\in\mathbb{P}\\ p\leq 2n}} \quad \text{and} \quad \prod_{\substack{p\in\mathbb{P}\\ n<p\leq 2n}} p\leq 4^n.$$

♡

Proof By Lemma 1.4, and notice that the prime factors of $m!$ are exactly the all prime factors from 1 to m, we have

$$n!=\prod_{\substack{p\in\mathbb{P}\\ p\leq n}} p^{\alpha(p)} \quad \text{where } \alpha(p)=\sum_{k=1}^{\infty}\left\lfloor\frac{n}{p^k}\right\rfloor.$$

Thus, by the definition of binomial coefficient, we have

$$\binom{2n}{n}=\frac{(2n)!}{(n!)^2}=\frac{\prod_{\substack{p\in\mathbb{P}\\ p\leq 2n}} p^{\sum_{k=1}^{\infty}\left\lfloor\frac{2n}{p^k}\right\rfloor}}{\left(\prod_{\substack{p\in\mathbb{P}\\ p\leq n}} p^{\sum_{k=1}^{\infty}\left\lfloor\frac{n}{p^k}\right\rfloor}\right)^2}=\prod_{\substack{p\in\mathbb{P}\\ p\leq 2n}} p^{\beta(p)}$$

where

$$\beta(p)=\sum_{k=1}^{\infty}\left(\left\lfloor\frac{2n}{p^k}\right\rfloor-2\left\lfloor\frac{n}{p^k}\right\rfloor\right).$$

By the Note, $\lfloor\frac{2n}{p^k}\rfloor=0$ if $p^k>2n$ or equivalently $k>\log_p(2n)$, and since $0\leq\lfloor 2x\rfloor-2\lfloor x\rfloor\leq 1$ for all $x\geq 0$, we have

$$\beta(p)\leq\sum_{k=1}^{\lfloor\log_p(2n)\rfloor}1=\lfloor\log_p(2n)\rfloor\leq\log_p(2n) \quad \text{and thus} \quad p^{\beta(p)}=p^{\log_p(2n)}\leq 2n.$$

Furthermore, when $p>\sqrt{2n}$, $\lfloor\frac{2n}{p^k}\rfloor=0$ and $\lfloor\frac{n}{p^k}\rfloor=0$ for all $k\geq 2$, so

$$\beta(p)=\left\lfloor\frac{2n}{p}\right\rfloor-2\left\lfloor\frac{n}{p}\right\rfloor\leq 1 \quad \text{for } p>\sqrt{2n}.$$

Hence,

$$\begin{aligned}\binom{2n}{n} &= \prod_{\substack{p\in\mathbb{P}\\ p\le 2n}} p^{\beta(p)} = \prod_{\substack{p\in\mathbb{P}\\ p\le\sqrt{2n}}} p^{\beta(p)}\cdot \prod_{\substack{p\in\mathbb{P}\\ \sqrt{2n}<p\le 2n}} p^{\beta(p)} \le \prod_{\substack{p\in\mathbb{P}\\ p\le\sqrt{2n}}} 2n\cdot \prod_{\substack{p\in\mathbb{P}\\ \sqrt{2n}<p\le 2n}} p\\ &\le (2n)^{\sqrt{2n}}\cdot \prod_{\substack{p\in\mathbb{P}\\ \sqrt{2n}<p\le 2n}} p\\ &\le \left(\frac{3}{2}\right)^n \prod_{\substack{p\in\mathbb{P}\\ p\le 2n}} p\end{aligned}$$

where the last inequality holds for n sufficiently large since

$$\lim_{n\to\infty}\frac{(2n)^{\sqrt{2n}}}{(3/2)^n} = \lim_{n\to\infty}\frac{e^{\sqrt{2n}\log(2n)}}{e^{n\log(3/2)}} = \lim_{n\to\infty} e^{n\left(\frac{\log(2n)}{\sqrt{2n}}-\log(3/2)\right)} = 0 \quad \text{since } \lim_{n\to\infty}\frac{\log(2n)}{\sqrt{2n}} = \lim_{n\to\infty}\frac{1/n}{1/\sqrt{2n}} = 0.$$

where the last equality follows from L'Hôpital's rule if we treat n as a real variable and then replace it back to integer variable. Therefore, by Exercise 1.7, we have

$$2^n \le \binom{2n}{n} \le \left(\frac{3}{2}\right)^n \prod_{\substack{p\in\mathbb{P}\\ p\le 2n}} p \implies \left(\frac{4}{3}\right)^n \le \prod_{\substack{p\in\mathbb{P}\\ p\le 2n}} p.$$

For the second inequality, notice that

$$\beta(p) = \left\lfloor\frac{2n}{p}\right\rfloor - 2\left\lfloor\frac{n}{p}\right\rfloor = 1-0=1 \quad \text{for } n<p\le 2n,$$

Combining with Exercise 1.7 again, we have

$$4^n \ge \binom{2n}{n} = \prod_{\substack{p\in\mathbb{P}\\ p\le 2n}} p^{\beta(p)} \ge \prod_{\substack{p\in\mathbb{P}\\ n<p\le 2n}} p.$$

Theorem 1.6 (Chebyshev's Bounds on $\pi(N)$)

There exist positive constants $c_1, c_2 > 0$ such that

$$c_1\frac{N}{\log N} \le \pi(N) \le c_2\frac{N}{\log N} \quad \textit{for } N \textit{ sufficiently large}.$$

Proof Chebyshev used the bounds in Lemma 1.5 to prove the theorem. By using the first inequality in Lemma 1.5, we have

$$\left(\frac{4}{3}\right)^n \le \prod_{\substack{p\in\mathbb{P}\\ p\le 2n}} p \le \prod_{\substack{p\in\mathbb{P}\\ p\le 2n}} 2n = (2n)^{\pi(2n)}.$$

Taking logarithm on both sides, we have

$$n\log\left(\frac{4}{3}\right) \le \pi(2n)\log(2n) \implies \frac{\log(4/3)}{2}\cdot\frac{2n}{\log(2n)} \le \pi(2n).$$

Thus, for $N=2n$ sufficiently large, it gives the lower bound. And for $N=2n+1$ sufficiently large, we have

$$\pi(N) = \pi(2n+1) \ge \pi(2n) \ge \frac{\log(4/3)}{2}\cdot\frac{2n}{\log(2n)} \ge \frac{\log(4/3)}{4}\cdot\frac{2n+1}{\log(2n+1)}$$

where the last inequality holds as $\frac{x}{\log x}$ is increasing for $x\ge e$ ($[\frac{x}{\log x}]' = \frac{\log x-1}{(\log x)^2} > 0$ for $x>e$). Hence, we have the lower bound for all sufficiently large N with $c_1 = \frac{\log(4/3)}{4}$.

For the upper bound, by extend the second inequality in Lemma 1.5 (Let $n = \lfloor\frac{1}{2}x\rfloor$,$x>0$), we have

$$\prod_{\substack{p\in\mathbb{P}\\ \lfloor x/2\rfloor<p\le 2\lfloor x/2\rfloor}} p \le 4^{\lfloor x/2\rfloor} \le 4^{x/2} = 2^x$$

and since there are at most one prime between $2\lfloor x/2\rfloor$ and x. For x sufficiently large, we have

$$\left(\frac{x}{2}\right)^{\pi(x)-\pi(x/2)} \leq \left(\frac{x}{2}\right)^{\pi(x)-\pi(\lfloor x/2\rfloor)} \leq \prod_{\substack{p\in\mathbb{P}\\ \lfloor x/2\rfloor<p\leq x}} \frac{x}{2} \leq \prod_{\substack{p\in\mathbb{P}\\ \lfloor x/2\rfloor<p\leq x}} p \leq x\cdot 2^x \leq 3^x.$$

Taking logarithm on both sides, we have

$$(\pi(x)-\pi(x/2))\log\left(\frac{x}{2}\right) \leq x\log 3 \implies \pi(x)-\pi(x/2) \leq \frac{x\log 3}{\log(x/2)}.$$

Replacing x by $\frac{x}{2^{j-1}}$ for $j=1,2,\ldots,m$

$$\pi\left(\frac{x}{2^{j-1}}\right)-\pi\left(\frac{x}{2^j}\right) \leq \frac{\frac{x}{2^{j-1}}\log 3}{\log\left(\frac{x}{2^{j-1}}\right)} \leq \log(9)\cdot\frac{x2^{-j}}{\log(x2^{-j})}.$$

Summing over $j=1,2,\ldots,m$ gives

$$\pi(x) \leq \pi\left(\frac{x}{2^m}\right)+\log(9)\sum_{j=1}^{m}\frac{x2^{-j}}{\log(x2^{-j})}.$$

As for $j\leq m$, we have $\log(x2^{-j}) \geq \log(x2^{-m}) \geq \log(x)/2$, $\pi(\frac{x}{2^m}) \leq \frac{x}{2^m} \leq 2\sqrt{x}$ and $\sqrt{x} \leq \frac{x}{\log(x)}$ for x sufficiently large, it gives

$$\pi(x) \leq 2\sqrt{x}+\log(9)\sum_{j=1}^{m}\frac{x2^{-j}}{\log(x)/2} = 2\sqrt{x}+\frac{2x\log(9)}{\log(x)}\sum_{j=1}^{m}2^{-j} \leq 2\sqrt{x}+\frac{2x\log(9)}{\log(x)} \leq (1+2\log(9))\frac{x}{\log(x)}$$

By setting $c_2 = 1+2\log(9)$, and $x=N$, we have the upper bound for all sufficiently large N.

Exercises

1. For each $n\in\mathbb{N}$, show that the prime factors of $n(n+1)$ have more than the prime factors of n itself.
   And deduce that there are infinitely many primes by considering the sequence $\{x_n\}_{n=1}^{\infty}$ where $x_n = x_{n-1}(x_{n-1}+1)$.
   (Hint: show that $\gcd(n,n+1)=1$.)
   Solution *By Euclid's algorithm, we have* $\gcd(n,n+1)=\gcd(n+1-n,n)=\gcd(1,n)=1$.
   Thus, n and $n+1$ are coprime, which means they share no common prime factors. Therefore, the prime factors of $n(n+1)$ are the union of the prime factors of n and the prime factors of $n+1$, which means

   $$\#\textit{prime factors of } n(n+1) = \#\textit{prime factors of } n + \#\textit{prime factors of } (n+1).$$

   and clearly, #prime factors of $(n+1)\geq 1$ as it is either prime itself or has at least one prime factor. Hence, the prime factors of $n(n+1)$ have more than the prime factors of n itself.
   For the sequence $\{x_n\}_{n=1}^{\infty}$ defined by $x_n = x_{n-1}(x_{n-1}+1)$ with $x_1=1$. By previous result, x_n has more (strictly) prime factors than x_{n-1}. Since $x_1=1$ has no prime factors, it follows that x_n has at least $n-1$ distinct prime factors. Therefore, as n increases, the number of distinct prime factors of x_n also increases without bound, which implies that there are infinitely many primes.
2. Fix any positive integer N, let $P_k = 2,3,5,\ldots,p_k$ denote the set of first k primes. Define

   $$E_N^k = \{n\in\mathbb{N} : n\leq N \text{ and } n \text{ only divisible by any prime in } P_k\}.$$

   Show that
   (a) Every $n\in E_N^k$ can be expressed as

   $$n = p_1^{e_1}p_2^{e_2}\cdots p_k^{e_k}s^2$$

   for some non-negative integers $e_1,e_2,\ldots,e_k\in\{0,1\}$ and some $s\in\mathbb{N}$.
   (b) Then deduce that there are at most $2^k\sqrt{N}$ elements in E_N^k.
   (c) Finally, for any fixed N, let $n\in\mathbb{N}$ such that $P_n=\{p_1,p_2,\ldots,p_n\}$ will exact all the primes less than or equal to N. i.e.

   $$P_n = \{p\in\mathbb{P} : p\leq N\} = \{2,3,5,\ldots,p_n\}.$$

   So that $\pi(N)=n$ and note that $E_N^n=\{1,2,\ldots,N\}$ (prove it!). From above results, we have $N=\#E_N^n\leq$

$2^n\sqrt{N}$ or equivalently

$$\pi(N) = n \geq \frac{\log N}{2\log 2}.$$

By using this, we can show that $\pi(N) \to \infty$ as $N \to \infty$. And compare this bound with our previous bound for $\pi(N)$ in Proposition 1.1, which is

$$\pi(N) \geq \log\log N$$

Solution

(a) *By the Fundamental Theorem of Arithmetic 1.1, every $n \in E_N^k$ can be expressed as*

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

for some non-negative integers $a_1, a_2, \ldots, a_k$. For each a_i, by using Euclid's division algorithm, we can write

$$a_i = 2d_i + r_i \quad \text{where } d_i \in \mathbb{N}_0 \text{ and } r_i \in \{0,1\}.$$

Thus,

$$n = p_1^{2d_1+r_1} p_2^{2d_2+r_2} \cdots p_k^{2d_k+r_k} = (p_1^{d_1} p_2^{d_2} \cdots p_k^{d_k})^2 \cdot p_1^{r_1} p_2^{r_2} \cdots p_k^{r_k}.$$

and we get the desired form by letting $s = p_1^{d_1} p_2^{d_2} \cdots p_k^{d_k}$.
For the uniqueness of this representation, suppose there are two representations of n:

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} s^2 = p_1^{f_1} p_2^{f_2} \cdots p_k^{f_k} t^2$$

where $e_i, f_i \in \{0,1\}$ and $s, t \in \mathbb{N}$. Rearranging gives

$$p_1^{e_1-f_1} p_2^{e_2-f_2} \cdots p_k^{e_k-f_k} = \left(\frac{t}{s}\right)^2.$$

The left side is an integer, so $\frac{t}{s}$ must also be an integer, say m. Thus,

$$p_1^{e_1-f_1} p_2^{e_2-f_2} \cdots p_k^{e_k-f_k} = m^2.$$

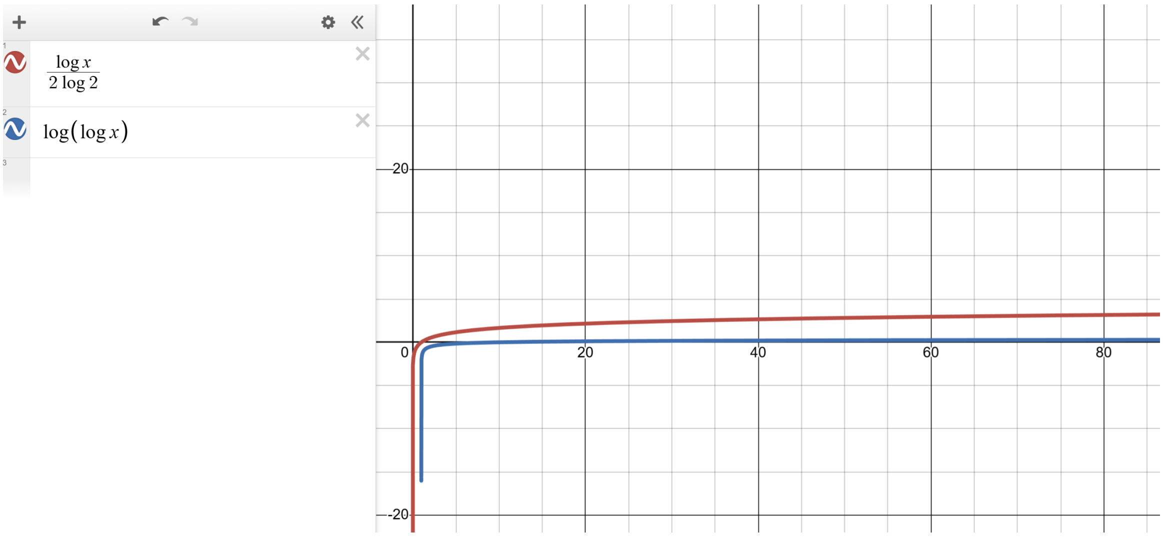
Since the right side is a perfect square, all exponents on the left side must be even. Given that $e_i, f_i \in \{0,1\}$, the only way for $e_i - f_i$ to be even is if $e_i = f_i$ for all i. This implies that the exponents are unique.

(b) *By the above result, for each $n \in E_N^k$, there are 2^k choices for the exponents $e_1, e_2, \ldots, e_k$ and since $n \leq N$, we have*

$$s^2 \leq n \leq N \implies s \leq \sqrt{N}.$$

Thus, there are at most $\sqrt{N}$ choices for s. Therefore, the total number of elements in E_N^k is at most $2^k\sqrt{N}$.

(c) *To show that $E_N^n = \{1, 2, \ldots, N\}$, notice that any integer m such that $1 \leq m \leq N$ can only have prime factors less than or equal to N. Since P_n contains all primes less than or equal to N, it follows that every integer m in this range is included in E_N^n. Conversely, any number in E_N^n is by definition less than or equal to N. Thus, we have $E_N^n = \{1, 2, \ldots, N\}$.*
For the lower bound on $\pi(N)$, this bounds shows a quicker growth rate compared to the previous bound. Can be seen as follows:



3. This question show how previous exercises implies that $\sum_{p\in\mathbb{P}} \frac{1}{p} = \infty$.
Suppose by contradiction that $\sum_{p\in\mathbb{P}} \frac{1}{p}$ converges. Then the tails of the series must tend to zero, and so we can find a positive integer k such that

$$\sum_{i=k+1}^{\infty} \frac{1}{p_i} < \frac{1}{2}.$$

where we denote the i-th prime by p_i.
Now, fix a positive integer N and fix a prime p then we have

$$\#\{0 < n \leq N : p \mid n, n \in \mathbb{N}\} \leq \frac{N}{p}. \quad \text{(show it!)}$$

Then consider E_N^k defined in the previous exercise, we have

$$N - \#E_N^k \leq \frac{N}{p_{k+1}} + \frac{N}{p_{k+2}} + \frac{N}{p_{k+3}} + \cdots < \frac{N}{2}. \quad \text{(show it!)}$$

As $\#E_N^k \leq 2^k\sqrt{N}$ from previous exercise, we have

$$\frac{N}{2} < 2^k\sqrt{N} \implies N < 2^{2k+2}$$

which is impossible as N can be arbitrarily large. Hence, we conclude that $\sum_{p\in\mathbb{P}} \frac{1}{p} = \infty$.

Solution

(a) *For a fixed prime p and positive integer N, the integers n such that $1 \leq n \leq N$ and $p \mid n$ are precisely the multiples of p: $p, 2p, 3p, \ldots, mp$ where $mp \leq N$. Thus, the largest integer m satisfying this condition is $\lfloor \frac{N}{p} \rfloor$. Therefore we have the inequality.*

(b) *For a integer $0 \leq n \leq N$, if $n \notin E_N^k$, then n must be divisible by at least one prime p_i where $i > k$. Thus, by the previous result, the number of integers n such that $1 \leq n \leq N$ and $n \notin E_N^k$ is at most $\frac{N}{p_{k+1}} + \frac{N}{p_{k+2}} + \frac{N}{p_{k+3}} + \cdots$. By our assumption, this sum is less than $\frac{N}{2}$. Therefore, we have the inequality.*

4. This is a topological way to show there are infinitely many primes. There are some basic concepts in topology we need to introduce first.
Suppose that a collection $\mathcal{B}$ of subsets of a set X (i.e. $\mathcal{B} \subseteq \mathcal{P}(X)$) satisfies the following two properties:
   (i) Every $x \in X$, there is at least one $B \in \mathcal{B}$ such that $x \in B$;
   (ii) For any $B_1, B_2 \in \mathcal{B}$ and any $x \in B_1 \cap B_2$, there is a $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

then we call $\mathcal{B}$ a basis for a topology τ on X it generates. Here a open set of τ is either the empty set or it is a union of sets in $\mathcal{B}$ and a set $F \subset X$ is called closed if its complement $X \setminus F$ is open. We now consider the a topology on $X = \mathbb{Z}$.
   (a) Let $\mathcal{B} = \{\mathrm{S}_{a,q} : a, q \in \mathbb{Z}, q > 0\}$ where $\mathrm{S}_{a,q} = \{a + nq : n \in \mathbb{Z}\}$ denotes the arithmetic progression. Show that $\mathcal{B}$ is a basis for a topology τ on $\mathbb{Z}$ and any finite set $S \subset \mathbb{Z}$ cannot be open in this topology.
   (b) Show that $S_{a,q}$ is both open and closed for any $a, q \in \mathbb{Z}, q > 0$.

(c) Finally, show that

$$\mathbb{Z} \setminus \{-1, 1\} = \bigcup_{p \in \mathbb{P}} S_{0,p}.$$

and thus if $\#\mathbb{P} < \infty$, then the right hand side is a closed set but the left hand side cannot be closed. Hence, we conclude that there are infinitely many primes.

Solution

(a) For any $x \in \mathbb{Z}$, by letting $a = x$ and $q = 1$, then $x \in S_{a,q}$. and hence property (i) holds. For any $B_1 = S_{a_1,q_1}, B_2 = S_{a_2,q_2} \in \mathcal{B}$ and any $x \in B_1 \cap B_2$, then $x = a_1 + n_1 q_1 = a_2 + n_2 q_2$ and by letting $q_3 = \operatorname{lcm}(q_1, q_2)$ and $a_3 = x$. Then $x \in S_{a_3,q_3}$ and for any $y \in S_{a_3,q_3}$, we have

$$y = a_3 + n_3 q_3 = x + n_3 \operatorname{lcm}(q_1, q_2) = x + n_3 m_1 q_1 = x + n_3' m_2 q_2$$

for some integers m_1, m_2, n_3, n_3'. Thus, $y \in B_1 \cap B_2$ and hence property (ii) holds. Therefore, $\mathcal{B}$ is a basis for a topology τ on $\mathbb{Z}$.

For any finite set $S \subset \mathbb{Z}$. Suppose by contradiction that S is open, then S is either the empty set (which is not the case here) or it is a union of sets in $\mathcal{B}$. Since each set in $\mathcal{B}$ is infinite, any union of sets in $\mathcal{B}$ must also be infinite. This contradicts the assumption that S is finite. Hence, any finite set $S \subset \mathbb{Z}$ cannot be open in this topology.

(b) For any $S_{a,q} \in \mathcal{B}$, it is open as it is a union of itself. To show that it is closed, consider its complement

$$\mathbb{Z} \setminus S_{a,q} = \bigcup_{r=0}^{q-1} S_{r,q} \setminus S_{a,q} = \bigcup_{\substack{r=0 \\ r \neq a \mod q}}^{q-1} S_{r,q}$$

which is a union of sets in $\mathcal{B}$ and hence is open. Therefore, $S_{a,q}$ is closed.

(c) for each $n \in \mathbb{Z} \setminus \{-1, 1\}$, if $n = 0$, then $n \in S_{0,2}$; if $|n| > 1$, then by the Fundamental Theorem of Arithmetic 1.1, n has at least one prime factor p, so that $n \in S_{0,p}$. Thus $\mathbb{Z} \setminus \{-1, 1\} \subseteq \bigcup_{p \in \mathbb{P}} S_{0,p}$.

Conversely, for any $n \in \bigcup_{p \in \mathbb{P}} S_{0,p}$, then there exists a prime p such that $n \in S_{0,p}$, which means $p \mid n$. Therefore, n cannot be ± 1. Thus, $\bigcup_{p \in \mathbb{P}} S_{0,p} \subseteq \mathbb{Z} \setminus \{-1, 1\}$. Therefore, we get the desired equality.

Chapter 2 Infinite Products and Series

2.1 Infinite Products, Uniform Convergence Criteria and M-test

Definition 2.1 (Infinite Product)

A infinite product $\prod_{n=1}^{\infty} a_n$ *converges if the sequence of partial products*

$$P_N = \prod_{n=1}^{N} a_n$$

converges to a a limit P *as* $N \to \infty$. *In this case, we write*

$$\prod_{n=1}^{\infty} a_n = P.$$

♣

Theorem 2.1 (n-term Test, Divergence Test for Infinite Products)

If the infinite product $\prod_{n=1}^{\infty} a_n$ *converges to a nonzero limit, then* $\lim_{n\to\infty} a_n = 1$.
And thus we usually write infinite products in the form

$$\prod_{n=1}^{\infty} (1 + b_n)$$

and then the n-term test becomes: if $\prod_{n=1}^{\infty}(1 + b_n)$ *converges, then* $\lim_{n\to\infty} b_n = 0$.

♡

Proof Suppose that the infinite product $\prod_{n=1}^{\infty} a_n$ converges to a nonzero limit P. Then the sequence of partial products

$$P_N = \prod_{n=1}^{N} a_n \to P \neq 0 \quad \text{as } N \to \infty.$$

Then, $a_n \neq 0$ for any $n \in \mathbb{N}$, and we have

$$a_n = \frac{P_n}{P_{n-1}} \to \frac{P}{P} = 1 \quad \text{as } n \to \infty.$$

Definition 2.2 (Bounded Complex-Value Function)

A complex-value function $G : X \to \mathbb{C}$ *defined on a set* X *is bounded If*

$$\|G\|_\infty := \sup_{x\in X} |G(x)| < \infty.$$

and the number $\|G\|_\infty$ *is the least upper bound of* $|G(x)|$ *for all* $x \in X$.

♣

Definition 2.3

A sequence of complex-value functions $S_n : X \to \mathbb{C}$ *converges uniformly to a complex-value function* $S : X \to \mathbb{C}$ *on a set* X *if*

$$\lim_{n\to\infty} \|S_n - S\|_\infty = \lim_{n\to\infty} \sup_{x\in X} |S_n(x) - S(x)| = 0.$$

♣

Exercise 2.1 If a sequence of complex-value functions $S_n : X \to \mathbb{C}$ converges uniformly to a complex-value function $S : X \to \mathbb{C}$ on a set X, then S is a bounded function on X.

Solution *Since* S_n *converges uniformly to* S *on* X, *for any* $\varepsilon > 0$, *there exists an integer* N *such that for all* $n \geq N$,

$$\|S_n - S\|_\infty = \sup_{x\in X} |S_n(x) - S(x)| < \varepsilon.$$

This implies that for all $x \in X$ *and all* $n \geq N$,

$$|S(x)| \leq |S_n(x)| + |S_n(x) - S(x)| < |S_n(x)| + \varepsilon.$$

Since S_N is a bounded function on X, there exists a constant M_N such that

$$|S_N(x)| \leq M_N \quad \text{for all } x \in X.$$

Therefore, for all $x \in X$,

$$|S(x)| < M_N + \varepsilon.$$

Thus,

$$\sup_{x \in X} |S(x)| \leq M_N + \varepsilon < \infty.$$

Hence, S is a bounded function on X.

Exercise 2.2 Show that if $F, G : X \to \mathbb{C}$ are bounded complex-value functions on a set X, then $\|F + G\|_\infty \leq \|F\|_\infty + \|G\|_\infty$ and $\|FG\|_\infty \leq \|F\|_\infty \|G\|_\infty$.

Solution *For any $x \in X$, by the triangle inequality, we have*

$$|F(x) + G(x)| \leq |F(x)| + |G(x)| \leq \|F\|_\infty + \|G\|_\infty.$$

Taking the supremum over all $x \in X$, we get

$$\|F + G\|_\infty = \sup_{x \in X} |F(x) + G(x)| \leq \|F\|_\infty + \|G\|_\infty.$$

Similarly, for the product, we have

$$|F(x)G(x)| = |F(x)||G(x)| \leq \|F\|_\infty \|G\|_\infty.$$

Taking the supremum over all $x \in X$, we get

$$\|FG\|_\infty = \sup_{x \in X} |F(x)G(x)| \leq \|F\|_\infty \|G\|_\infty.$$

Definition 2.4 (Uniform Convergence of Series of Functions)

A series of complex-value bounded functions $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on a set X if the sequence of partial sums

$$S_N(x) = \sum_{n=1}^{N} f_n(x)$$

converges uniformly on X.

Definition 2.5 (Uniform Convergence of Products of Functions)

A product of complex-value bounded functions $\prod_{n=1}^{\infty} f_n(x)$ converges uniformly on a set X if the sequence of partial products

$$P_N(x) = \prod_{n=1}^{N} f_n(x)$$

converges uniformly on X.

Theorem 2.2 (Cauchy Criterion for Uniform Convergence of Series of Functions)

A series of complex-value bounded functions $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on a set X if the series of norms $\sum_{n=1}^{\infty} \|f_n\|_\infty$ converges.

$$S_N(x) = \sum_{n=1}^{N} f_n(x)$$

satisfies

$$\|S_N - S\|_\infty \to 0 \quad \text{as } N \to \infty.$$

Proof For any $\varepsilon > 0$, since the series of norms $\sum_{n=1}^{\infty} \|f_n\|_\infty$ converges, there exists an integer N such that for all

$m > n \geq N$,

$$\sum_{k=n+1}^{m} \|f_k\|_\infty < \varepsilon.$$

Then, for any $x \in X$ and all $m > n \geq N$, we have

$$|S_m(x) - S_n(x)| = \left| \sum_{k=n+1}^{m} f_k(x) \right| \leq \sum_{k=n+1}^{m} |f_k(x)| \leq \sum_{k=n+1}^{m} \|f_k\|_\infty < \varepsilon.$$

Taking the supremum over all $x \in X$, we get

$$\|S_m - S_n\|_\infty = \sup_{x \in X} |S_m(x) - S_n(x)| < \varepsilon.$$

This shows that the sequence of partial sums $\{S_N\}$ is uniformly Cauchy on X. Since the space of bounded functions on X is complete with respect to the supremum norm, it follows that $\{S_N\}$ converges uniformly to a bounded function $S : X \to \mathbb{C}$ on X. Hence, the series $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on X.

Theorem 2.3

A product of complex-value bounded functions $\prod_{n=1}^{\infty}(1 + f_n(x))$ converges uniformly on a set X if the series of norms $\sum_{n=1}^{\infty} \|f_n\|_\infty$ converges.

$$P_N(x) = \prod_{n=1}^{N} (1 + f_n(x))$$

satisfies

$$\|P_N - P\|_\infty \to 0 \quad \text{as } N \to \infty.$$

♡

Proof Let

$$P_N(x) = \prod_{k=1}^{N} \big(1 + f_k(x)\big), \qquad x \in X.$$

Fix $\varepsilon > 0$. Since $\sum_{n=1}^{\infty} \|f_n\|_\infty$ converges, define

$$A := \sum_{n=1}^{\infty} \|f_n\|_\infty < \infty, \qquad M := e^A.$$

For any $n \in \mathbb{N}$ and any $x \in X$, we have

$$|P_n(x)| = \prod_{k=1}^{n} |1 + f_k(x)| \leq \prod_{k=1}^{n} \big(1 + |f_k(x)|\big) \leq \prod_{k=1}^{n} \big(1 + \|f_k\|_\infty\big) \leq \exp\Big(\sum_{k=1}^{n} \|f_k\|_\infty\Big) \leq M.$$

Now choose $\delta > 0$ such that

$$e^\delta - 1 < \frac{\varepsilon}{M}.$$

Since $\sum_{n=1}^{\infty} \|f_n\|_\infty$ converges, there exists $N \in \mathbb{N}$ such that for all $m > n \geq N$,

$$\sum_{k=n+1}^{m} \|f_k\|_\infty < \delta.$$

For any $x \in X$ and $m > n \geq N$, we write

$$|P_m(x) - P_n(x)| = \left| \prod_{k=1}^{n} (1 + f_k(x)) \right| \left| \prod_{k=n+1}^{m} (1 + f_k(x)) - 1 \right|.$$

Using the inequality $|1 + z| \leq e^{|z|}$ for $z \in \mathbb{C}$, we obtain

$$\left| \prod_{k=n+1}^{m} (1 + f_k(x)) \right| \leq \exp\Big(\sum_{k=n+1}^{m} |f_k(x)| \Big),$$

and hence

$$\left| \prod_{k=n+1}^{m} (1 + f_k(x)) - 1 \right| \leq e^{\sum_{k=n+1}^{m} |f_k(x)|} - 1 < e^\delta - 1.$$

Combining the estimates, we obtain
$$|P_m(x) - P_n(x)| \le M\left(e^{\delta} - 1\right) < \varepsilon.$$
Taking the supremum over all $x \in X$, we get
$$\|P_m - P_n\|_\infty = \sup_{x\in X} |P_m(x) - P_n(x)| < \varepsilon.$$
Thus $\{P_N\}$ is a uniformly Cauchy sequence on X. Since the space of bounded functions on X is complete with respect to the supremum norm, there exists a bounded function $P : X \to \mathbb{C}$ such that
$$\|P_N - P\|_\infty \to 0 \quad (N \to \infty).$$
Therefore, the infinite product $\prod_{n=1}^{\infty}(1 + f_n(x))$ converges uniformly on X.

Theorem 2.4 (Uniform Convergence and Zeros of Infinite Products)

Let $u_n : X \to \mathbb{C}$ be a sequence of bounded complex-value functions on a set X such that the series
$$\sum_{n=1}^{\infty} |u_n| \text{ converges uniformly on } X.$$
Then the infinite product
$$f(x) = \prod_{n=1}^{\infty}(1 + u_n(x))$$
converges uniformly on X.
Furthermore,
$$f(x_0) = 0 \text{ for some } x_0 \in X \iff u_n(x_0) = -1 \text{ for some } n \in \mathbb{N}.$$
i.e. f has a zero at x_0 if and only if one of the factors in the product is zero at x_0.

Note *We can use this theorem even in the case where u_n is constant function. And in this case, it states that*
$$\sum_{n=1}^{\infty} |u_n| \text{ converges and } u_n \neq -1 \text{ for all } n \in \mathbb{N} \implies \prod_{n=1}^{\infty}(1 + u_n) \text{ converges to a nonzero limit.}$$

Exercise 2.3 Show that $e^x \ge x + 1$ for all $x \ge 0$

Solution *By applying Mean Value Theorem to the function $f(x) = e^x$ on the interval $[0, x]$ for any $x \ge 0$, there exists some $c \in (0, x)$ such that*
$$f'(c) = \frac{f(x) - f(0)}{x - 0} \implies e^c = \frac{e^x - 1}{x}.$$
Since $e^c \ge 1$ for all $c \ge 0$, we have
$$\frac{e^x - 1}{x} \ge 1 \implies e^x - 1 \ge x \implies e^x \ge x + 1.$$

Exercise 2.4 Show that for any sequence of complex numbers $\{a_n\}_{n=1}^{\infty}$, we have
$$\left|\prod_{n=1}^{N}(1 + a_n) - 1\right| \le \exp\left(\sum_{n=1}^{N} |a_n|\right) - 1.$$

Solution *We prove this by induction on N. For $N = 1$, we have*
$$|(1 + a_1) - 1| = |a_1| \le e^{|a_1|} - 1, \text{ by Exercise 2.3.}$$

Now, assume the inequality holds for some $N \geq 1$. Then for $N+1$, we have

$$\begin{aligned}\left|\prod_{n=1}^{N+1}(1+a_n)-1\right| = \left|(1+a_{N+1})\prod_{n=1}^{N}(1+a_n)-1\right| &= \left|(1+a_{N+1})\left(\prod_{n=1}^{N}(1+a_n)-1\right)+a_{N+1}\right| \\ &\leq |1+a_{N+1}|\left|\prod_{n=1}^{N}(1+a_n)-1\right|+|a_{N+1}| \\ &\leq (1+|a_{N+1}|)\left(e^{\sum_{n=1}^{N}|a_n|}-1\right)+|a_{N+1}|, \text{ by the induction hypothesis} \\ &= e^{\sum_{n=1}^{N}|a_n|}-1+|a_{N+1}|e^{\sum_{n=1}^{N}|a_n|}-|a_{N+1}|+|a_{N+1}| \\ &= (1+|a_{N+1}|)e^{\sum_{n=1}^{N}|a_n|}-1 \\ &\leq e^{|a_{N+1}|}e^{\sum_{n=1}^{N}|a_n|}-1, \text{ by Exercise 2.3} \\ &= e^{\sum_{n=1}^{N+1}|a_n|}-1.\end{aligned}$$

Thus, by induction, the inequality holds for all $N \in \mathbb{N}$.

Proof By our assumption, as $\sum_{n=1}^{\infty}|u_n|$ converges uniformly on X, the partial sums $S_N(x)=\sum_{n=1}^{N}|u_n(x)|$ converge uniformly to $S=\sum_{n=1}^{\infty}|u_n(x)|$ on X. Thus, for a given $\varepsilon>0$, there is a N_0 (not depending on x) such that

$$\|S_N-S\|_\infty = \sup_{x\in X}\sum_{n=N+1}^{\infty}|u_n(x)|<\varepsilon, \qquad \text{for all } N \geq N_0.$$

and exercise 2.1 implies that S is bounded on X. Hence, $S(x)=\sum_{n=1}^{\infty}|u_n(x)|\leq M$ for some $M>0$ and all $x\in X$.

Let $P_N(x)=\prod_{n=1}^{N}(1+u_n(x))$ denote the N-th partial product of $f(x)$. Then by exercise 2.4, we have

$$|P_N(x)-1| \leq \exp\left(\sum_{n=1}^{N}|u_n(x)|\right)-1 \leq e^M-1 \quad \text{since } \sum_{n=1}^{N}|u_n(x)| \leq S(x) \leq M.$$

and thus $|P_N(x)|\leq e^M-1+1=e^M$ for all $x\in X$ and all $N\in\mathbb{N}$. Therefore, for any $M>N\geq N_0$ and any $x\in X$, using exercise 2.4 again, we have

$$|P_M(x)-P_N(x)| = |P_N(x)|\left|\prod_{n=N+1}^{M}(1+u_n(x))-1\right| \leq e^M\left(\exp\left(\sum_{n=N+1}^{M}|u_n(x)|\right)-1\right) < e^M(e^\varepsilon-1) \tag{2.1}$$

for all $x\in X$. Taking the supremum over all $x\in X$, we get

$$\|P_M-P_N\|_\infty = \sup_{x\in X}|P_M(x)-P_N(x)| < e^M(e^\varepsilon-1) < \varepsilon'$$

when we choose $\varepsilon=\log\left(\frac{\varepsilon'}{e^M}+1\right)>0$. Thus, $\{P_N\}$ is a uniformly Cauchy sequence on X, and hence $f(x)$ converges uniformly on X.

Now, do the similar argument as in equation (2.1) and choose $M\to\infty$, $N=N_0$, we have

$$|f(x)-P_{N_0}(x)| \leq |P_{N_0}|(e^\varepsilon-1) \qquad \text{for all } x\in X.$$

Setting $\varepsilon<\frac{1}{2}$, we have

$$|f(x)-P_{N_0}(x)| < |P_{N_0}(x)||\sqrt{e}-1|$$

thus,

$$|f(x)| = |P_{N_0}(x)-(f(x)-P_{N_0}(x))| \geq |P_{N_0}(x)|-|P_{N_0}(x)||\sqrt{e}-1| = |P_{N_0}(x)|(2-\sqrt{e}) \geq \frac{1}{3}|P_{N_0}(x)|$$

Therefore, if $f(x_0)=0$ for some $x_0\in X$, then $P_{N_0}(x_0)=0$ and hence $u_n(x_0)=-1$ for some $1\leq n\leq N_0$. The converse is trivial.

Theorem 2.5 (Weierstrass M-test for Infinite Series and Products)

If a sequence of complex function $u_n : X\to\mathbb{C}$ satisfies $|u_n(x)|\leq M_n$ and $\sum_{n=1}^{\infty}M_n$ converges, then the infinite series

$$\sum_{n=1}^{\infty}|u_n(x)| \text{ converges uniformly on } X.$$

hence the series

$$\sum_{n=1}^{\infty} u_n(x) \text{ converges uniformly on } X.$$

And by the previous theorem 2.4, the infinite product

$$f(x) = \prod_{n=1}^{\infty}(1 + u_n(x))$$

converges uniformly on X.

Proof Direct from the M-test in Complex Analysis and theorem 2.4.

Proposition 2.1 (Properties of Uniform Convergence)

1. *uniform limit of continuous functions is continuous.*
2. *uniform limit of bounded functions is bounded. (See exercise 2.1.)*
3. *uniform limit of analytic functions is analytic.*

Proof omitted.

Corollary 2.1 (Analyticity of Riemann Zeta Function and Euler Product)

Recall the Riemann zeta function and the Euler product formula

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} \quad \text{and} \quad \mathcal{P}(z) = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-z}} = \prod_{p \in \mathbb{P}} (1 + u_p(z)) \quad \text{for } \Re(z) > 1,$$

where $u_p(z) = \frac{1}{1-p^z} - 1 = \frac{p^{-z}}{1-p^{-z}}$.
Then both $\zeta(z)$ and $\mathcal{P}(z)$ are analytic on the half plane $\{z \in \mathbb{C} : \Re(z) > 1\}$. Further more, if

$$\zeta(z) = \mathcal{P}(z) \quad \text{for } \Re(z) > 1,$$

we can show that

$$|\zeta(z)| \neq 0 \quad \text{on } \{z \in \mathbb{C} : \Re(z) > 1\}$$

(i.e., $\zeta(z)$ has no zero on this half plane)

Note *For a^z, where $a > 0$ is a positive real number, if not specified, we always consider it as $e^{z \log a}$, where $\log a$ is using the real logarithm to avoid multi-valued issue. And thus a^z is analytic on the whole complex plane $\mathbb{C}$.*

Proof Notice that $\frac{1}{n^z}$ is an analytic function on $\mathbb{C}$ and $u_p(z)$ is analytic on $\{z \in \mathbb{C} : \Re(z) > 1\}$ for each prime p. Also,

$$|1 - p^{-z}| \geq 1 - |p^{-z}| = 1 - p^{-\Re(z)} \geq 1 - p^{-1} \geq \frac{1}{2} \quad \text{for } \Re(z) > 1.$$

Thus, for $\Re(z) \geq 1$, on the open set $X_\varepsilon = \{z \in \mathbb{C} : \Re(z) \geq 1 + \varepsilon\}$ for any $\varepsilon > 0$, we have

$$|u_p(z)| = \left| \frac{p^{-z}}{1 - p^{-z}} \right| \leq 2|p^{-z}| = 2p^{-\Re(z)} \leq 2p^{-(1+\varepsilon)} \quad \text{and} \quad \left| \frac{1}{n^z} \right| = n^{-\Re(z)} \leq n^{-(1+\varepsilon)}.$$

Since

$$\sum_{p \in \mathbb{P}} \frac{1}{p^{1+\varepsilon}} \leq \sum_{n=1}^{\infty} \frac{1}{n^{1+\varepsilon}} < \infty$$

the Weierstrass M-test 2.5 implies that both $\zeta(z)$ and $\mathcal{P}(z)$ converge uniformly on X_ε for any $\varepsilon > 0$. Therefore, by proposition 2.1, both $\zeta(z)$ and $\mathcal{P}(z)$ are analytic on $\{z \in \mathbb{C} : \Re(z) > 1\}$.

Further more, by theorem 2.4, since $u_p(z) \neq -1$ for all primes p and all z with $\Re(z) > 1$, we have $|\mathcal{P}(z)| \neq 0$ for all z with $\Re(z) > 1$. As $\zeta(z) = \mathcal{P}(z)$ for $\Re(z) > 1$, we have $|\zeta(z)| = |\mathcal{P}(z)| \neq 0$ for all z with $\Re(z) > 1$.

Note *The same analysis can be applied to Dirichlet L-functions and their Euler product representations*

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^z} \quad \text{and} \quad \prod_{p \in \mathbb{P}} \frac{1}{1 - \chi(p)p^{-z}}$$

but will discuss later once we have introduced Dirichlet characters.

2.2 Recap of Complex Analysis

Theorem 2.6 (Morera's Theorem)

Suppose $f : X \to \mathbb{C}$ is continuous function on an open set $X \subseteq \mathbb{C}$. If

$$\int_\Gamma f(z)\,dz = 0 \quad \text{for all loop } \Gamma \text{ in } X, \quad \text{then } f \text{ is analytic on } X.$$

♡

Proof Omitted.

Theorem 2.7 (Cauchy Integral Theorem)

Suppose $f : X \to \mathbb{C}$ is an analytic function inside and on a simple closed contour Γ in an open set $X \subseteq \mathbb{C}$. Then

$$\int_\Gamma f(z)\,dz = 0$$

♡

Proof Omitted.

✍ **Exercise 2.5** Use Morera's Theorem and Cauchy Integral Theorem to show the 3rd property in the above proposition 2.1.

Solution *Suppose $\{f_n\}$ is a sequence of analytic functions on an open set $X \subseteq \mathbb{C}$ that converges uniformly to a function f on X. We want to show that f is analytic on X.*

For any simple closed contour Γ in X, as f_n is analytic on X, by Cauchy Integral Theorem, we have

$$\int_\Gamma f_n(z)\,dz = 0 \quad \text{for all } n \in \mathbb{N}.$$

Since f_n converges uniformly to f on X, by the properties of uniform convergence, we can interchange the limit and the integral:

$$\int_\Gamma f(z)\,dz = \lim_{n\to\infty} \int_\Gamma f_n(z)\,dz = \lim_{n\to\infty} 0 = 0.$$

Thus, by Morera's Theorem, since f is continuous (as uniform limit of continuous functions is continuous) and the integral of f over any simple closed contour in X is zero, we conclude that f is analytic on X.

2.3 Euler Products

Definition 2.6

For a function $f : \mathbb{N} \to \mathbb{C}$ which is not identically zero, we say:

(a) f is multiplicative, if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$.

(b) f is completely multiplicative, if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$.

♣

✍ **Exercise 2.6** Show that the Möbius function $\mu(n)$ which is defined as

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1 \cdots p_k \text{ is a product of } k \text{ distinct primes}, \\ 0 & \text{otherwise}, \end{cases}$$

is a multiplicative function.

Solution *Let $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$. We want to show that $\mu(mn) = \mu(m)\mu(n)$.*

Case 1: If either m or n has a squared prime factor, then mn also has a squared prime factor. Thus, $\mu(m) = 0$ or $\mu(n) = 0$, and hence $\mu(mn) = 0 = \mu(m)\mu(n)$.

Case 2: If both m and n are products of distinct primes, say $m = p_1 p_2 \cdots p_k$ and $n = q_1 q_2 \cdots q_l$, where p_i and q_j are distinct primes. Since $\gcd(m, n) = 1$, the sets of primes $\{p_1, p_2, \ldots, p_k\}$ and $\{q_1, q_2, \ldots, q_l\}$ are disjoint. Therefore,

$mn = p_1 p_2 \cdots p_k q_1 q_2 \cdots q_l$ is a product of $k+l$ distinct primes. Thus,

$$\mu(mn) = (-1)^{k+l} = (-1)^k(-1)^l = \mu(m)\mu(n).$$

Case 3: If either m or n is equal to 1, then without loss of generality, let $m = 1$. Then $\mu(1) = 1$, and we have

$$\mu(mn) = \mu(n) = \mu(1)\mu(n).$$

In all cases, we have shown that $\mu(mn) = \mu(m)\mu(n)$ for all $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$. Therefore, the Möbius function $\mu(n)$ is multiplicative.

Proposition 2.2

If $f : \mathbb{N} \to \mathbb{C}$ is a (completely) multiplicative function, and $f(n_0) \neq 0$ for some $n_0 \in \mathbb{N}$, then $f(1) = 1$.

Note *If f is identically zero, then $f(mn) = f(m)f(n)$ holds trivially for all $m, n \in \mathbb{N}$. Thus, if not specified, we always assume that a (completely) multiplicative function is not identically zero. And then by the above proposition holds always for any (completely) multiplicative function.*

Exercise 2.7 Prove the above proposition.

Solution *Since f is (completely) multiplicative, we have*

$$f(n_0) = f(1 \cdot n_0) = f(1)f(n_0) \quad \textit{since } \gcd(1, n_0) = 1.$$

Since $f(n_0) \neq 0$, we can divide both sides by $f(n_0)$ to get

$$f(1) = 1.$$

Lemma 2.1 ("Euler Product")

Suppose $f : \mathbb{N} \to \mathbb{C}$ is a multiplicative function (not identically zero) and the series

$$\sum_{n=1}^{\infty} |f(n)| < \infty.$$

Then the "Euler Product" holds:

$$\sum_{n=1}^{\infty} f(n) = \prod_{p\in\mathbb{P}} \left(1 + f(p) + f(p^2) + \cdots\right) = \prod_{p\in\mathbb{P}} \sum_{k=0}^{\infty} f(p^k).$$

Note *The last equality holds since f is multiplicative function (not identically zero) implies that $f(1) = 1$ by proposition 2.2.*

Proof Let $u_p = \sum_{k\geq 1} f(p^k)$ for each prime p and set

$$P_N = \prod_{\substack{p\in\mathbb{P} \\ p\leq N}} (1 + u_p) = (1 + u_{p_1})(1 + u_{p_2}) \cdots (1 + u_{p_r}),$$

where $p_1, p_2, \ldots, p_r$ are all the primes less than or equal to N and suppose $p_1 < p_2 < \cdots < p_r$. We need to show u_p is convergent for each prime p to ensure P_N is well-defined (i.e. The limit of P_N exists) for each N. As $|u_p| \leq \sum_{k=1}^{\infty} |f(p^k)| \leq \sum_{n=1}^{\infty} |f(n)| < \infty$, each u_p is convergent.

Notice that $f(1) = 1$ by proposition 2.2. Thus,

$$P_N = \lim_{K\to\infty} \Big[\sum_{k_1=0}^{K} f(p_1^{k_1}) \sum_{k_2=0}^{K} f(p_2^{k_2}) \cdots \sum_{k_r=0}^{K} f(p_r^{k_r})\Big].$$

because if $\lim_{K\to\infty} a_K = A$ and $\lim_{K\to\infty} b_K = B$, then $\lim_{K\to\infty} a_K b_K = AB$. Rearranging the sums on the RHS, and using the multiplicative property of f, we have

$$\sum_{k_1=0}^{K} \sum_{k_2=0}^{K} \cdots \sum_{k_r=0}^{K} f(p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}) = \sum_{n\in F_{N,K}} f(n) \quad \text{since } \gcd(p_i^{k_i}, p_j^{k_j}) = 1 \text{ for } i \neq j.$$

where

$$F_{N,K} = \{n \in \mathbb{N} : n = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}, 0 \leq k_i \leq K \text{ for } 1 \leq i \leq r\}.$$

Recall u_p is convergent, P_N exists for each N. Thus

$$\lim_{K\to\infty} \sum_{n\in F_{N,K}} f(n) = P_N.$$

Since $F_{N,K} \subseteq \mathbb{N}$, we can choose a large $M_{N,K}$ such that $F_{N,K} \subseteq \{1, 2, \ldots, M_{N,K}\}$ for each N, K. Then

$$\left| \sum_{n=1}^{M_{N,K}} f(n) - \sum_{n\in F_{N,K}} f(n) \right| = \left| \sum_{\substack{n\le M_{N,K} \\ n\notin F_{N,K}}} f(n) \right| \le \sum_{n\ge N} |f(n)| + \sum_{n\ge 2^K} |f(n)| \tag{2.2}$$

where the last inequality holds since any $n \notin F_{N,K}$, it is either has a prime factor greater than N (So that $n > N$) or has a prime factor less than or equal to N with exponent greater than K (So that $n \ge p_i^K \ge 2^K$ for some $1 \le i \le r$).

Fix a value of N, and let $K \to \infty$, notice that

$$\lim_{K\to\infty} \sum_{n=1}^{M_{N,K}} f(n) = \sum_{n=1}^{\infty} f(n) \quad \text{and} \quad \lim_{K\to\infty} \sum_{n\in F_{N,K}} f(n) = P_N.$$

Thus, taking $K \to \infty$ in the above inequality (2.2), we obtain

$$\left| \sum_{n=1}^{\infty} f(n) - P_N \right| \le \sum_{n\ge N} |f(n)|.$$

Finally, letting $N \to \infty$. As $\sum_{n=1}^{\infty} |f(n)|$ converges, the tail $\sum_{n\ge N} |f(n)| \to 0$ as $N \to \infty$. Thus,

$$\sum_{n=1}^{\infty} f(n) = \lim_{N\to\infty} P_N = \prod_{p\in\mathbb{P}} \left(1 + f(p) + f(p^2) + \cdots\right).$$

Theorem 2.8 (Euler Product Formula)

Let $f : \mathbb{N} \to \mathbb{C}$ be a completely multiplicative function (not identically zero) and suppose the series

$$\sum_{n=1}^{\infty} |f(n)| < \infty.$$

Then the Euler Product holds:

$$\sum_{n=1}^{\infty} f(n) = \prod_{p\in\mathbb{P}} \left(1 + f(p) + f(p^2) + \cdots\right) = \prod_{p\in\mathbb{P}} \sum_{k=0}^{\infty} f(p)^k = \prod_{p\in\mathbb{P}} \frac{1}{1 - f(p)}.$$

♡

Proof We only need to prove the last equality holds as the first two equalities are direct from lemma 2.1.

Recall the geometric series formula,

$$1 + z + z^2 + \cdots = \frac{1}{1-z} \quad \text{for } |z| < 1.$$

one may want to use this to justify the last equality. However, we need to ensure that $|f(p)| < 1$ for all primes p to apply the geometric series formula. But this is indeed true since if there exists a prime p_0 such that $|f(p_0)| \ge 1$, then as f is completely multiplicative, we have

$$|f(p_0^k)| = |f(p_0)|^k \ge 1 \quad \text{for all } k \in \mathbb{N},$$

thus the series $\sum_{n=1}^{\infty} |f(n)|$ diverges, as the subseries $\sum_{k=1}^{\infty} |f(p_0^k)|$ diverges. This contradicts our assumption that $\sum_{n=1}^{\infty} |f(n)| < \infty$. Therefore, we must have $|f(p)| < 1$ for all primes p, and hence we can apply the geometric series formula to get the last equality.

Corollary 2.2 (Classical Euler Product Formula)

The Riemann zeta function and the Euler product formula

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} \quad \textit{and} \quad \mathcal{P}(z) = \prod_{p\in\mathbb{P}} \frac{1}{1 - p^{-z}} \quad \textit{for } \Re(z) > 1,$$

are equal, i.e.,

$$\zeta(z) = \mathcal{P}(z) \quad \text{for } \Re(z) > 1.$$

♡

Proof By applying theorem 2.8 to the completely multiplicative function $f(n) = \frac{1}{n^z}$ for $\Re(z) > 1$, and noticing that $\sum_{n=1}^{\infty} |f(n)| = \sum_{n=1}^{\infty} \frac{1}{n^{\Re(z)}}$ converges for $\Re(z) > 1$, we have

$$\sum_{n=1}^{\infty} \frac{1}{n^z} = \prod_{p\in\mathbb{P}} \frac{1}{1-\frac{1}{p^z}} = \prod_{p\in\mathbb{P}} \frac{1}{1-p^{-z}}.$$

Remark We will also want to apply $f(n) = \chi(n)n^{-z}$ to get the Euler product formula for Dirichlet L-functions later, where χ is a Dirichlet character. And extend the Euler product formula to

$$L(z,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^z} = \prod_{p\in\mathbb{P}} \frac{1}{1-\chi(p)p^{-z}} \quad \text{for } \Re(z) > 1.$$

2.4 Taking Logarithm

Definition 2.7 (Simple Connected)

An open set $X \subseteq \mathbb{C}$ is said to be simply connected if there is no hole in X, i.e., any loop in X can be continuously contracted to a point within X.

♣

Theorem 2.9 (Existence of Logarithm Branch)

Let $X \subseteq \mathbb{C}$ be a simply connected open set, and $g(z)$ is analytic on X with no zeros in X. Then there exists an analytic function $h(z)$ on X such that

$$e^{h(z)} = g(z) \quad \text{for all } z \in X.$$

and we say that $h(z)$ is a branch of logarithm of $g(z)$ on X.

♡

Note *There is a mini definition in the above theorem: A branch of logarithm of $g(z)$ on X is an analytic function $h(z)$ on X such that $e^{h(z)} = g(z)$ for all $z \in X$.*

Note *We may also write $h(z) = \log g(z)$ to denote a branch of the logarithm of $g(z)$ on X. However, one must be careful when performing algebraic operations with the logarithm, since $\log g(z)$ is defined only up to a choice of branch. Nevertheless, many familiar properties of the logarithm still hold for a fixed branch. For example, the derivative of $h(z)$ is given by*

$$h'(z) = \frac{g'(z)}{g(z)},$$

which is formally the same as the derivative of the real logarithm.

Indeed, differentiating both sides of the identity $e^{h(z)} = g(z)$ and applying the chain rule, we obtain

$$e^{h(z)}h'(z) = g'(z).$$

Since $e^{h(z)} = g(z)$, it follows that

$$h'(z) = \frac{g'(z)}{g(z)}.$$

Proof Since $g(z)$ has no zeros in X, we can define a function

$$f(z) := \int_{z_0}^{z} \frac{g'(w)}{g(w)}\, dw$$

for some fixed $z_0 \in X$. This integral is well-defined because X is simply connected, and $\frac{g'(w)}{g(w)}$ is analytic on X (as $g(w)$ has no zeros).

Now, consider $F(z) := g(z)e^{-f(z)}$. Notice that $F'(z) = g'(z)e^{-f(z)} - g(z)e^{-f(z)} \cdot \frac{g'(z)}{g(z)} = g'(z)e^{-f(z)} -$

$g'(z)e^{-f(z)} = 0$. Thus, $F(z)$ is constant on X. Evaluating at $z = z_0$, we have

$$F(z) = F(z_0) = g(z_0)e^{-f(z_0)} = g(z_0)e^0 = g(z_0) \implies g(z)e^{-f(z)} = F(z) = g(z_0). \implies g(z) = g(z_0)e^{f(z)}.$$

Therefore, we can set $h(z) := f(z) + c$ where $e^c = g(z_0)$, and we have $e^{h(z)} = g(z)$ for all $z \in X$.

Exercise 2.8 Consider the analytic function $f(z) = \mathrm{Log}(1-z)$ on the simply connected open set $X = \{z \in \mathbb{C} : |z| < 1\}$ where Log is the principal branch of logarithm defined on $\mathbb{C} \setminus (-\infty, 0]$. Show the $f(z)$ has the Taylor series expansion

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n = -\sum_{n=1}^{\infty} \frac{z^n}{n} \quad \text{valid for } z \in X,$$

and hence deduce that when $|z| \le \frac{1}{2}$, $|\mathrm{Log}(1-z)| \le 2|z|$.

Solution *Notice that for each $n \in \mathbb{N}$,*

$$f^{(n)}(z) = \frac{d^n}{dz^n}\mathrm{Log}(1-z) = \frac{d^{n-1}}{dz^{n-1}}\left(-\frac{1}{1-z}\right) = \frac{d^{n-2}}{dz^{n-2}}\left(-\frac{1}{(1-z)^2}\right) = \cdots = -\frac{(n-1)!}{(1-z)^n}, \quad \textit{for } n \ge 1.$$

Thus,

$$f^{(n)}(0) = -(n-1)! \quad \textit{for all } n \ge 1, \quad \textit{and} \quad f(0) = \mathrm{Log}(1-0) = \mathrm{Log}(1) = 0.$$

Therefore, the Taylor series expansion of $f(z)$ at $z = 0$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n = f(0) + \sum_{n=1}^{\infty} \frac{-(n-1)!}{n!} z^n = -\sum_{n=1}^{\infty} \frac{z^n}{n}.$$

where the last equality holds since $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n$ converges for $|z| < 1$.

Next, we show that when $|z| \le \frac{1}{2}$, we have $|\mathrm{Log}(1-z)| \le 2|z|$. Since $|z| < \frac{1}{2}$, we have

$$|\mathrm{Log}(1-z)| = \left| -\sum_{n=1}^{\infty} \frac{z^n}{n} \right| \le \sum_{n=1}^{\infty} \frac{|z|^n}{n} \le \sum_{n=1}^{\infty} |z|^n = \frac{|z|}{1-|z|} \le \frac{|z|}{1-\frac{1}{2}} = 2|z|.$$

Theorem 2.10 (Uniform Convergence of Logarithm of Infinite Product)

On a simply connected open set $X \subseteq \mathbb{C}$, the infinite product

$$f(z) = \prod_{n=1}^{\infty} (1 + u_n(z))$$

where $|u_n(z)| \le M_n < 1$ for all $z \in X$ and $\sum_{n=1}^{\infty} M_n < \infty$, is uniformly convergent on X. Further more, the function

$$L(z) := \sum_{n=1}^{\infty} \mathrm{Log}(1 + u_n(z)) \quad \textit{where } \mathrm{Log} \textit{ is principal branch of logarithm,}$$

is convergent uniformly on X and satisfies

$$e^{L(z)} = f(z) \quad \textit{for all } z \in X.$$

$L(z)$ is thus a branch of logarithm of $f(z)$ on X.
(i.e., "$L(z) = \log f(z)$" and $L(z)$ is analytic on X, which is an application of the previous theorem 2.9.)

Proof $f(z)$ is clearly uniformly convergent on X by theorem 2.5 since $\sum_{n=1}^{\infty} M_n < \infty$ and $|u_n(z)| \le M_n$ for all $z \in X$.

Next, we need to show that $L(z)$ is well-defined and uniformly convergent on X. For each $N \in \mathbb{N}$, we define

$$\widetilde{\log}\, P_N(z) := \sum_{n=1}^{N} \mathrm{Log}(1 + u_n(z)), \quad \text{where } P_N(z) = \prod_{n=1}^{N} (1 + u_n(z)).$$

Since $|u_n(z)| \le M_n < 1$ for all $z \in X$, we have $|1 + u_n(z)| \ge 1 - |u_n(z)| \ge 1 - M_n > 0$ for all $z \in X$. Thus, $\widetilde{\log}\, P_N(z)$ is well-defined and analytic on X. Further more, notice that

$$e^{\widetilde{\log}\, P_N(z)} = e^{\sum_{n=1}^{N} \mathrm{Log}(1+u_n(z))} = \prod_{n=1}^{N} e^{\mathrm{Log}(1+u_n(z))} = \prod_{n=1}^{N} (1 + u_n(z)) = P_N(z).$$

so that $\widetilde{\log}\, P_N(z)$ is a branch of logarithm of $P_N(z)$ on X. (i.e., $e^{\widetilde{\log}\, P_N(z)} = P_N(z)$ for all $z \in X$.) Since $|u_n(z)| \le M_n$ and $\sum_{n=1}^{\infty} M_n < \infty$, $u_n(z) \le \frac{1}{2}$ for all sufficiently large N_0 and all $z \in X$. Thus, by exercise 2.8, for all $N \ge N_0$ and all $z \in X$, we have $|\mathrm{Log}(1+u_n(z))| \leq 2|u_n(z)|$. Therefore, by the Weierstrass M-test 2.5, considering applying it to the series $\sum_{n=N_0}^{\infty} \mathrm{Log}(1+u_n(z))$, we have

$$\widetilde{\log}\, P_N(z) \to L(z) := \sum_{n=1}^{\infty} \mathrm{Log}(1+u_n(z)) \quad \text{uniformly on } X \text{ as } N \to \infty$$

and hence

$$P_N(z) = e^{\widetilde{\log}\, P_N(z)} \to e^{L(z)} \quad \text{uniformly on } X \text{ as } N \to \infty.$$

Recall that $P_N(z) \to f(z)$ uniformly on X as $N \to \infty$, we have

$$f(z) = e^{L(z)} \quad \text{for all } z \in X.$$

Exercise 2.9 Recall the Generalized Cauchy Integral Formula

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_\Gamma \frac{f(z)}{(z-z_0)^{n+1}}\, dz,$$

where f is analytic inside and on a simple closed contour Γ in an open set $X \subseteq \mathbb{C}$ and z_0 is a point inside Γ.

Use this formula to show that if a sequence of analytic functions $\{f_n\}$ converges uniformly to a function f on an open set $X \subseteq \mathbb{C}$, then the sequence of derivatives $\{f_n'\}$ converges uniformly to f' on X. i.e., show that

$$f_n' \to f' \quad \text{uniformly on } X.$$

Solution *Let $\{f_n\}$ be a sequence of analytic functions on an open set $X \subseteq \mathbb{C}$ that converges uniformly to a function f on X. We show that the sequence of derivatives $\{f_n'\}$ converges uniformly to f' on every compact subset of X.*

Let $K \subset X$ be an arbitrary compact set. Since X is open and K is compact, there exists $\delta > 0$ such that

$$\mathrm{dist}(K, \partial X) \geq \delta.$$

Fix $z_0 \in K$ and let Γ be the positively oriented circle

$$\Gamma := \{z \in \mathbb{C} : |z - z_0| = r\}, \quad r := \delta/2.$$

Then $\Gamma \subset X$ for all $z_0 \in K$. By the Generalized Cauchy Integral Formula, we have

$$f_n'(z_0) = \frac{1}{2\pi i} \int_\Gamma \frac{f_n(z)}{(z-z_0)^2}\, dz, \quad f'(z_0) = \frac{1}{2\pi i} \int_\Gamma \frac{f(z)}{(z-z_0)^2}\, dz.$$

Since $f_n \to f$ uniformly on X, for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n \geq N$ and all $z \in X$,

$$|f_n(z) - f(z)| < \varepsilon.$$

Hence, for $n \geq N$, using the M–L inequality we obtain

$$\begin{aligned} |f_n'(z_0) - f'(z_0)| &= \left| \frac{1}{2\pi i} \int_\Gamma \frac{f_n(z) - f(z)}{(z-z_0)^2}\, dz \right| \\ &\leq \frac{1}{2\pi} \max_{z \in \Gamma} \left| \frac{f_n(z) - f(z)}{(z-z_0)^2} \right| \cdot \mathrm{length}(\Gamma). \end{aligned}$$

Since $|z - z_0| = r$ on Γ and $\mathrm{length}(\Gamma) = 2\pi r$, we have

$$|f_n'(z_0) - f'(z_0)| \leq \frac{1}{2\pi} \cdot \frac{\varepsilon}{r^2} \cdot 2\pi r = \frac{\varepsilon}{r}.$$

The constant $r = \delta/2$ is independent of $z_0 \in K$. Therefore,

$$\sup_{z_0 \in K} |f_n'(z_0) - f'(z_0)| \leq \frac{\varepsilon}{r}, \quad \textit{for all } n \geq N.$$

This shows that $f_n' \to f'$ uniformly on K. Since $K \subset X$ was arbitrary, we conclude that

$$f_n' \to f' \quad \textit{uniformly on compact subsets of } X.$$

Corollary 2.3

Let $\zeta(z)$ be zeta function defined as in corollary 2.2, we have

$$\frac{\zeta'(z)}{\zeta(z)} = -\sum_{p \in \mathbb{P}} \frac{\log p}{p^z - 1} \quad \textit{for } \Re(z) > 1.$$

Note *It's a very important formula in the proof of the prime number theorem (PNT) later.*

Proof For the zeta function $\zeta(z)$, by corollary 2.2, we have

$$\zeta(z)=\prod_{p\in\mathbb{P}}\frac{1}{1-p^{-z}}=\prod_{p\in\mathbb{P}}(1+u_p(z)) \quad \text{where } u_p(z)=\frac{1}{p^z-1}.$$

Consider a simply connected open set $X_\varepsilon=\{z\in\mathbb{C}:\Re(z)>1+\varepsilon\}$ for some $\varepsilon>0$, we have

$$|u_p(z)|=\left|\frac{1}{p^z-1}\right|\leq 2|p^{-z}|=2p^{-\Re(z)}\leq 2p^{-(1+\varepsilon)} \quad \text{for all } z\in X_\varepsilon.$$

Let $M_p=2p^{-(1+\varepsilon)}$, we have $\sum_{p\in\mathbb{P}}M_p<\infty$ (Comparison test with the p-series test.) and $M_p<1$ for all primes p. Thus, by theorem 2.10, we have

$$L(z):=\sum_{p\in\mathbb{P}}\mathrm{Log}(1+u_p(z))=\sum_{p\in\mathbb{P}}\mathrm{Log}\left(\frac{1}{1-p^{-z}}\right)=-\sum_{p\in\mathbb{P}}\mathrm{Log}(1-p^{-z})$$

is uniformly convergent on X_ε and satisfies

$$e^{L(z)}=\zeta(z) \quad \text{for all } z\in X_\varepsilon.$$

Then, by exercise 2.9, considering the sequence of partial sums of $L(z)$, denoted by $S_N(z)=-\sum_{p\leq N}\mathrm{Log}(1-p^{-z})$, we have

$$S_N'(z)=-\sum_{p\leq N}\frac{p^{-z}\log p}{1-p^{-z}}=-\sum_{p\leq N}\frac{\log p}{p^z-1}\to L'(z) \quad \text{uniformly on } X_\varepsilon \text{ as } N\to\infty.$$

As $e^{L(z)}=\zeta(z)$ for all $z\in X_\varepsilon$, by the chain rule, we have

$$L'(z)e^{L(z)}=L'(z)\zeta(z)=\zeta'(z)\implies L'(z)=\frac{\zeta'(z)}{\zeta(z)}.$$

Therefore, for all z such that $\Re(z)>1$, we can choose $\varepsilon>0$ such that $z\in X_\varepsilon$, and hence

$$\frac{\zeta'(z)}{\zeta(z)}=L'(z)=-\sum_{p\in\mathbb{P}}\frac{\log p}{p^z-1} \quad \text{for } z\in X_\varepsilon.$$

Corollary 2.4 (Logarithm of Dirichlet L-function)

The Dirichlet L-function $L(z,\chi)$ for a Dirichlet character χ satisfies

$$L(z;\chi)=\sum_{n=1}^{\infty}\frac{\chi(n)}{n^z}=\prod_{p\in\mathbb{P}}\frac{1}{1-\chi(p)p^{-z}} \quad \text{for } \Re(z)>1,$$

(although we have not defined Dirichlet characters yet, and not proved the Euler product formula for Dirichlet L-functions, but we will do so later, just take it as given for now).

Then we have

$$\log L(z,\chi):=\sum_{p\in\mathbb{P}}\sum_{k=1}^{\infty}\frac{1}{k}\chi(p^k)p^{-kz} \quad \text{for } \Re(z)>1,$$

is a branch of logarithm of $L(z,\chi)$ on the open set $\{z\in\mathbb{C}:\Re(z)>1\}$, i.e.,

$$e^{\log L(z,\chi)}=L(z,\chi) \quad \text{for } \Re(z)>1.$$

♡

Note *Be careful that here $\log L(z,\chi)$ is defined as a branch of logarithm of $L(z,\chi)$ on the open set $\{z\in\mathbb{C}:\Re(z)>1\}$, not the natural logarithm of real numbers.*

Proof For the Dirichlet L-function $L(z,\chi)$, we have

$$L(z,\chi)=\prod_{p\in\mathbb{P}}\frac{1}{1-\chi(p)p^{-z}}=\prod_{p\in\mathbb{P}}(1+u_p(z)) \quad \text{where } u_p(z)=\frac{\chi(p)}{p^z-\chi(p)}.$$

Consider a simply connected open set $X_\varepsilon=\{z\in\mathbb{C}:\Re(z)>1+\varepsilon\}$ for some $\varepsilon>0$, we have

$$|u_p(z)|=\left|\frac{\chi(p)}{p^z-\chi(p)}\right|\leq 2|\chi(p)||p^{-z}|=2p^{-\Re(z)}\leq 2p^{-(1+\varepsilon)} \quad \text{for all } z\in X_\varepsilon.$$

Let $M_p=2p^{-(1+\varepsilon)}$, we have $\sum_{p\in\mathbb{P}}M_p<\infty$ (Comparison test with the p-series test.) and $M_p<1$ for all primes p.

Thus, by theorem 2.10, we have

$$L(z) := \sum_{p\in\mathbb{P}} \operatorname{Log}(1+u_p(z)) = \sum_{p\in\mathbb{P}} \operatorname{Log}\left(\frac{1}{1-\chi(p)p^{-z}}\right) = -\sum_{p\in\mathbb{P}} \operatorname{Log}(1-\chi(p)p^{-z})$$

is uniformly convergent on X_ε and satisfies

$$e^{L(z)} = L(z,\chi) \quad \text{for all } z \in X_\varepsilon.$$

Next, we expand $\operatorname{Log}(1-\chi(p)p^{-z})$ using the Taylor series expansion from exercise 2.8, $\operatorname{Log}(1-z) = -\sum_{n=1}^{\infty} \frac{z^n}{n}$ for $|z| < 1$, we have

$$\operatorname{Log}(1-\chi(p)p^{-z}) = -\sum_{k=1}^{\infty} \frac{(\chi(p)p^{-z})^k}{k} = -\sum_{k=1}^{\infty} \frac{1}{k}\chi(p^k)p^{-kz}.$$

Therefore,

$$L(z) = \sum_{p\in\mathbb{P}}\sum_{k=1}^{\infty} \frac{1}{k}\chi(p^k)p^{-kz} \quad \text{for } z \in X_\varepsilon.$$

Finally, for all z such that $\Re(z) > 1$, we can choose $\varepsilon > 0$ such that $z \in X_\varepsilon$, and hence

$$e^{\log L(z,\chi)} = L(z,\chi) \quad \text{for } \Re(z) > 1.$$

Exercises

1. Let $\{a_n\}$ be a sequence of complex number such that the infinite product $\prod_{n=1}^{\infty} a_n$ converges to a nonzero limit P. Show that
   (a) The infinite product $\prod_{n=1}^{\infty} a_n^{-1}$ converges and

   $$\prod_{n=1}^{\infty} \frac{1}{a_n} = \frac{1}{P}$$

   (b) Hence conclude that for $\operatorname{Re}(z) > 1$,

   $$\frac{1}{\zeta(z)} = \prod_{p \text{ prime}} (1-p^{-z})$$

Solution

(a) Since the infinite product $\prod_{n=1}^{\infty} a_n$ converges to a nonzero limit P, by definition, the sequence of partial products

$$P_N := \prod_{n=1}^{N} a_n \longrightarrow P \quad \text{as } N\to\infty \quad \text{and } a_n \neq 0 \text{ for all } n \in \mathbb{N}.$$

Now, using the rule of quotient of limits, we have

$$\prod_{n=1}^{N} \frac{1}{a_n} = \frac{1}{P_N} \longrightarrow \frac{1}{P} \quad \text{as } N \to \infty.$$

Therefore,

$$\prod_{n=1}^{\infty} \frac{1}{a_n} = \frac{1}{P}.$$

(b) By corollary 2.2, we have

$$\zeta(z) = \prod_{p \text{ prime}} \frac{1}{1-p^{-z}} \quad \text{for } \operatorname{Re}(z) > 1.$$

By part (a), we have

$$\frac{1}{\zeta(z)} = \prod_{p \text{ prime}} (1-p^{-z}) \quad \text{for } \operatorname{Re}(z) > 1.$$

2. The Möbius function $\mu : \mathbb{N} \to \mathbb{C}$ is defined as follows:

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1 \cdots p_k \text{ is a product of } k \text{ distinct primes}, \\ 0 & \text{otherwise}, \end{cases} .$$

Show that

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1. \end{cases}$$

(Hint: The case $n = 1$ is easy. For $n > 1$, consider the prime factorization of $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$ so that the divisors d of n are of the form $d = p_1^{\beta_1} p_2^{\beta_2} \cdots p_r^{\beta_r}$ where $0 \le \beta_i \le \alpha_i$ for all $1 \le i \le r$. The only divisors d which give a non-zero contribution to the sum are those d such that $\beta_i \in \{0, 1\}$ for all $1 \le i \le r$. So that

$$\sum_{d|n} \mu(d) = \mu(1) + (\mu(p_1) + \mu(p_2) + \cdots + \mu(p_r)) + (\mu(p_1p_2) + \mu(p_1p_3) + \cdots) + \cdots + \mu(p_1p_2 \cdots p_r).$$

Now, use the binomial theorem, compare this to $0 = (1-1)^r = \sum_{k=0}^{r} \binom{r}{k} (-1)^k$.)

Solution *For $n = 1$, we have*

$$\sum_{d|1} \mu(d) = \mu(1) = 1.$$

which satisfies the required condition.
Next, consider $n > 1$. By the hint, we have

$$\begin{aligned} \sum_{d|n} \mu(d) &= \mu(1) + (\mu(p_1) + \mu(p_2) + \cdots + \mu(p_r)) + (\mu(p_1p_2) + \mu(p_1p_3) + \cdots) + \cdots + \mu(p_1p_2 \cdots p_r) \\ &= 1 + \binom{r}{1}(-1) + \binom{r}{2}1 + \binom{r}{3}(-1) + \cdots + \binom{r}{r}(-1)^r \\ &= \sum_{k=0}^{r} \binom{r}{k}(-1)^k \\ &= (1-1)^r = 0. \end{aligned}$$

Therefore, for $n > 1$, we get the required condition as well.

Let $f : \mathbb{N} \to \mathbb{C}$, then

$$F(z) := \sum_{n=1}^{\infty} \frac{f(n)}{n^z}$$

is called the *Dirichlet series* associated to f. You may find that the previous exercises 3., 4. and 5. give some examples of Dirichlet series and there is a useful properties of Dirichlet series involving the Dirichlet convolution defined below.

Theorem: Suppose $F(z) = \sum_{n=1}^{\infty} \frac{f(n)}{n^z}$ and $G(z) = \sum_{n=1}^{\infty} \frac{g(n)}{n^z}$ are two Dirichlet series that converge absolutely for some $z \in \mathbb{C}$. Then the product $F(z)G(z)$

$$F(z)G(z) = \sum_{n=1}^{\infty} \frac{(f * g)(n)}{n^z}$$

is also a Dirichlet series, where

$$(f * g)(n) := \sum_{d|n} f(d) g\left(\frac{n}{d}\right).$$

is called the *Dirichlet convolution* of f and g, which is a finite sum for each $n \in \mathbb{N}$.
This theorem will be proved later in exercise 6. and you can use it to solve the following exercises 3., 4. and 5..

3. Let $f(n) = \mu(n)$ denote the Möbius function as defined in the previous exercise and let $g(n) = 1$ for all $n \in \mathbb{N}$. Show that

   (a)
$$f * g := \sum_{d|n} f(d) g\left(\frac{n}{d}\right) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1. \end{cases}$$

   (b) Hence conclude that
$$\frac{1}{\zeta(z)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^z} \quad \text{for } \mathrm{Re}(z) > 1.$$

Solution

(a) From the previous exercise 2., we have
$$\sum_{d|n} f(d) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1. \end{cases}$$

Since $g\left(\frac{n}{d}\right) = 1$ for all $d|n$, we directly have the result.

(b) By the theorem stated above, we have
$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n^z} \cdot \sum_{n=1}^{\infty} \frac{1}{n^z} = \sum_{n=1}^{\infty} \frac{(f * g)(n)}{n^z} = 1 \quad \text{for } \mathrm{Re}(z) > 1.$$

where the last equality follows from part (a), $0 < |\frac{\mu}{n^z}| \leq |\frac{1}{n^z}| \leq \frac{1}{n^{\mathrm{Re}(z)}}$ for all $n \in \mathbb{N}$ and by the comparison test and the p-series test, these two Dirichlet series converge for $\mathrm{Re}(z) > 1$.

Therefore,
$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n^z} = \frac{1}{\zeta(z)} \quad \text{for } \mathrm{Re}(z) > 1.$$

4. Let $d(n) := \#$ of divisors of n, and let $f(n) = g(n) = 1$ for all $n \in \mathbb{N}$. Show that

   (a)
$$f * g := \sum_{d|n} f(d) g\left(\frac{n}{d}\right) = d(n).$$

   (b) Hence conclude that
$$\zeta^2(z) = \sum_{n=1}^{\infty} \frac{d(n)}{n^z} \quad \text{for } \mathrm{Re}(z) > 1.$$

Solution

(a) Since $f(d) = 1$ and $g\left(\frac{n}{d}\right) = 1$ for all $d|n$, we have
$$f * g := \sum_{d|n} f(d) g\left(\frac{n}{d}\right) = \sum_{d|n} 1 = \#\{d : d|n\} = d(n).$$

(b) By the theorem stated above and part (a), we have
$$\sum_{n=1}^{\infty} \frac{1}{n^z} \cdot \sum_{n=1}^{\infty} \frac{1}{n^z} = \sum_{n=1}^{\infty} \frac{(f * g)(n)}{n^z} = \sum_{n=1}^{\infty} \frac{d(n)}{n^z} \quad \text{for } \mathrm{Re}(z) > 1.$$

where both Dirichlet series on the left-hand side converge for $\mathrm{Re}(z) > 1$ by the p-series test. Therefore,
$$\zeta^2(z) = \sum_{n=1}^{\infty} \frac{d(n)}{n^z} \quad \text{for } \mathrm{Re}(z) > 1.$$

5. Let $\phi(n) := \#$ of integers $1 \leq k \leq n$ such that $\gcd(k, n) = 1$ denote the Euler totient function. Show that

   (a) For $n \in \mathbb{N}$,
$$\sum_{d|n} \phi(d) = n.$$

(b) Let $f(n) = \phi(n)$ and $g(n) = 1$ for all $n \in \mathbb{N}$. Show that

$$f * g := \sum_{d|n} f(d)g\left(\frac{n}{d}\right) = n.$$

(c) Hence conclude that

$$\frac{\zeta(z-1)}{\zeta(z)} = \sum_{n=1}^{\infty} \frac{\phi(n)}{n^z} \quad \text{for } \mathrm{Re}(z) > 2.$$

Solution

(a) (Method I.) For each integer d such that $d|n$, consider the set

$$A_d := \{k \in \mathbb{N} : 1 \le k \le n, \gcd(k,n) = d\}.$$

Notice that for any two divisors d_1 and d_2 of n with $d_1 \neq d_2$, the sets A_{d_1} and A_{d_2} are disjoint. (If not, there exists some integer k such that $\gcd(k,n) = d_1$ and $\gcd(k,n) = d_2$, which is a contradiction.) Also, the union of all such sets A_d for all divisors d of n is equal to the set $\{k \in \mathbb{N} : 1 \le k \le n\}$. (If k is any integer such that $1 \le k \le n$, then $\gcd(k,n)$ is a divisor of n, say d. So that $k \in A_d$.) Therefore, we have a partition of the set $\{k \in \mathbb{N} : 1 \le k \le n\}$ into disjoint subsets A_d for each divisor d of n:

$$\{k \in \mathbb{N} : 1 \le k \le n\} = \bigcup_{d|n} A_d \quad \Longrightarrow \quad n = \sum_{d|n} |A_d|.$$

Now, we compute $|A_d|$ for each divisor d of n. Notice that if $\gcd(k,n) = d$, then we can write $k = dk'$ for some integer k'. Since $d|n$, we can write $n = dm$ for some integer m. Therefore,

$$\gcd(k,n) = d \iff \gcd(dk', dm) = d \iff d \cdot \gcd(k', m) = d \iff \gcd(k', m) = 1.$$

So that A_d can be rewritten as

$$A_d = \{dk' : 1 \le k' \le m, \gcd(k',m) = 1\} = d \cdot \{k' : 1 \le k' \le m, \gcd(k',m) = 1\} \quad \text{where } m = \frac{n}{d}.$$

Therefore, $|A_d| = \phi(m) = \phi\left(\frac{n}{d}\right)$. Hence,

$$n = \sum_{d|n} |A_d| = \sum_{d|n} \phi\left(\frac{n}{d}\right).$$

(Method II.) For any $n \in \mathbb{N}$, factorization of n into primes is given by $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$. So that the divisors d of n are of the form $d = p_1^{\beta_1} p_2^{\beta_2} \cdots p_r^{\beta_r}$ where $0 \le \beta_i \le \alpha_i$ for all $1 \le i \le r$. So once we understand $\phi(d)$ for each divisor d of n, we can compute the sum $\sum_{d|n} \phi(d)$. In order to compute $\phi(d)$ for each divisor d of n, we need some properties of the Euler totient function ϕ:

Claim.

- *If p is a prime number and k is a positive integer, then*

$$\phi(p^k) = p^k - p^{k-1} = p^k\left(1 - \frac{1}{p}\right).$$

- *The Euler totient function ϕ is multiplicative, i.e., if $\gcd(m,n) = 1$, then*

$$\phi(mn) = \phi(m)\phi(n).$$

Proof of Claim. For the first property, the integers from 1 to p^k that are not coprime to p^k are precisely the multiples of p, which are $p, 2p, 3p, \ldots, p^{k-1}p$. There are p^{k-1} such multiples. Therefore,

$$\phi(p^k) = p^k - p^{k-1} = p^k\left(1 - \frac{1}{p}\right).$$

For the second property, let m and n be coprime positive integers. Notice that if a number $k \in \mathbb{N}$ is coprime to mn (i.e., $\gcd(k, mn) = 1$), if and only if it is coprime to both m and n (i.e., $\gcd(k,m) = 1$ and $\gcd(k,n) = 1$). By the Chinese Remainder Theorem, the number of integers from 1 to mn that are coprime to both m and n is equal to the product of the number of integers from 1 to m that are coprime to m and the number of integers from 1 to n that are coprime to n. Therefore,

$$\phi(mn) = \phi(m)\phi(n).$$

Now, using the above properties of the Euler totient function ϕ, we can compute $\phi(d)$ for each divisor d of n.

Since $d = p_1^{\beta_1} p_2^{\beta_2} \cdots p_r^{\beta_r}$ *where* $0 \le \beta_i \le \alpha_i$ *for all* $1 \le i \le r$*, by the multiplicative property of* ϕ*, we have*

$$\phi(d) = \phi(p_1^{\beta_1})\phi(p_2^{\beta_2}) \cdots \phi(p_r^{\beta_r}).$$

Using the formula for $\phi(p^k)$*, we have*

$$\phi(p_i^{\beta_i}) = p_i^{\beta_i}\left(1 - \frac{1}{p_i}\right).$$

Therefore,

$$\phi(d) = p_1^{\beta_1}\left(1 - \frac{1}{p_1}\right) p_2^{\beta_2}\left(1 - \frac{1}{p_2}\right) \cdots p_r^{\beta_r}\left(1 - \frac{1}{p_r}\right).$$

Now, we can compute the sum $\sum_{d|n} \phi(d)$*. We have*

$$\begin{aligned}
\sum_{d|n} \phi(d) &= \sum_{\beta_1=0}^{\alpha_1} \sum_{\beta_2=0}^{\alpha_2} \cdots \sum_{\beta_r=0}^{\alpha_r} \phi(p_1^{\beta_1} p_2^{\beta_2} \cdots p_r^{\beta_r}) \\
&= \sum_{\beta_1=0}^{\alpha_1} p_1^{\beta_1}\left(1 - \frac{1}{p_1}\right) \sum_{\beta_2=0}^{\alpha_2} p_2^{\beta_2}\left(1 - \frac{1}{p_2}\right) \cdots \sum_{\beta_r=0}^{\alpha_r} p_r^{\beta_r}\left(1 - \frac{1}{p_r}\right) \\
&= \left(1 - \frac{1}{p_1}\right)\left(\frac{p_1^{\alpha_1+1} - 1}{p_1 - 1}\right)\left(1 - \frac{1}{p_2}\right)\left(\frac{p_2^{\alpha_2+1} - 1}{p_2 - 1}\right) \cdots \left(1 - \frac{1}{p_r}\right)\left(\frac{p_r^{\alpha_r+1} - 1}{p_r - 1}\right) \\
&= p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r} = n.
\end{aligned}$$

(b) *By part (a), and* $g(n) = 1$ *for all* $n \in \mathbb{N}$*, we have*

$$f * g = \sum_{d|n} f(d) g\left(\frac{n}{d}\right) = \sum_{d|n} \phi(d) = n.$$

(c) *By the theorem stated above and part (b), we have*

$$\sum_{n=1}^{\infty} \frac{\phi(n)}{n^z} \cdot \sum_{n=1}^{\infty} \frac{1}{n^z} = \sum_{n=1}^{\infty} \frac{(f*g)(n)}{n^z} = \sum_{n=1}^{\infty} \frac{n}{n^z} = \sum_{n=1}^{\infty} \frac{1}{n^{z-1}} \quad \textit{for } \operatorname{Re}(z) > 2.$$

6. Now, with the following steps, prove the above theorem on the product of two Dirichlet series involving the Dirichlet convolution.

   (a) Suppose the Dirichlet series $F(z) = \sum_{n=1}^{\infty} \frac{f(n)}{n^z}$ and $G(z) = \sum_{n=1}^{\infty} \frac{g(n)}{n^z}$ converge for some $z \in \mathbb{C}$. For $N > 1$, write

$$\begin{aligned}
\sum_{n=1}^{N} \frac{f(n)}{n^z} \cdot \sum_{m=1}^{N} \frac{g(m)}{m^z} &= \sum_{k=1}^{N^2} \frac{1}{k^z}\left[\sum_{\substack{m=1 \\ mn=k}}^{N} \sum_{n=1}^{N} f(n)g(m)\right] = \sum_{k=1}^{N^2} \frac{1}{k^z}\left[\sum_{\substack{n=1 \\ n|k}}^{N} f(n) g\left(\frac{k}{n}\right)\right]. \\
&= \sum_{k=1}^{N^2} \frac{1}{k^z}\left[\sum_{\substack{n=1 \\ n|k}}^{k} f(n) g\left(\frac{k}{n}\right) - \sum_{\substack{n>N \\ n|k}} f(n) g\left(\frac{k}{n}\right)\right] \\
&= \sum_{k=1}^{N^2} \frac{(f*g)(k)}{k^z} - \sum_{k=N+1}^{N^2} \frac{1}{k^z}\left[\sum_{\substack{n>N \\ n|k}} f(n) g\left(\frac{k}{n}\right)\right].
\end{aligned}$$

   Let

$$A_N := \sum_{k=N+1}^{N^2} \frac{1}{k^z}\left[\sum_{\substack{n>N \\ n|k}} f(n) g\left(\frac{k}{n}\right)\right].$$

   Show that

$$A_N = \sum_{\substack{n=N+1 \\ N+1 \le nm \le N^2}}^{\infty} \sum_{m=1}^{N} \frac{f(n)}{n^z} \cdot \frac{g(m)}{m^z} \to 0 \quad \text{as } N \to \infty.$$

(b) Hence conclude that

$$F(z)G(z) = \sum_{n=1}^{\infty} \frac{(f*g)(n)}{n^z}.$$

so that we have proved the theorem.

Solution

(a) Consider the finite sum of $F(z)$ and $G(z)$ up to N, by the question, we have

$$\sum_{n=1}^{N} \frac{f(n)}{n^z} \cdot \sum_{m=1}^{N} \frac{g(m)}{m^z} = \sum_{k=1}^{N^2} \frac{(f*g)(k)}{k^z} - A_N.$$

where

$$A_N = \sum_{k=N+1}^{N^2} \frac{1}{k^z} \left[\sum_{\substack{n>N \\ n|k}} f(n) g\left(\frac{k}{n}\right) \right] = \sum_{k=N+1}^{N^2} \frac{1}{k^z} \left[\sum_{\substack{n=N+1 \\ mn=k}}^{\infty} \sum_{m=1}^{N} f(n)g(m) \right]$$

$$= \sum_{\substack{n=N+1 \\ N+1 \le nm \le N^2}}^{\infty} \sum_{m=1}^{N} \frac{f(n)}{n^z} \cdot \frac{g(m)}{m^z}$$

Notice that when $N+1 \le nm \le N^2$ and $m \le N$, we have $n \le \frac{N^2}{m} \le \frac{N^2}{1} = N^2$. Therefore,

$$|A_N| \le \sum_{n=N+1}^{N^2} \left| \frac{f(n)}{n^z} \right| \cdot \sum_{m=1}^{N} \left| \frac{g(m)}{m^z} \right| \le \sum_{n=N+1}^{\infty} \left| \frac{f(n)}{n^z} \right| \cdot \sum_{m=1}^{\infty} \left| \frac{g(m)}{m^z} \right|.$$

Since both Dirichlet series $F(z)$ and $G(z)$ converge absolutely for some $z \in \mathbb{C}$, $\sum_{n=N+1}^{\infty} \left| \frac{f(n)}{n^z} \right| \to 0$ as $N \to \infty$ and $\sum_{m=1}^{\infty} \left| \frac{g(m)}{m^z} \right|$ is some finite constant. Therefore, $A_N \to 0$ as $N \to \infty$.

(b) Now we show that $\sum_{k=1}^{N^2} \frac{(f*g)(k)}{k^z}$ converges absolutely as $N \to \infty$ to ensure that it is well-defined as $N \to \infty$. Notice that

$$\sum_{k=1}^{N^2} \frac{|(f*g)(k)|}{|k^z|} = \sum_{k=1}^{N^2} \frac{1}{|k^z|} \left| \sum_{\substack{n=1 \\ n|k}}^{k} f(n) g\left(\frac{k}{n}\right) \right| \le \sum_{k=1}^{N^2} \frac{1}{|k^z|} \left[\sum_{\substack{n=1 \\ n|k}}^{k} |f(n)| \cdot \left| g\left(\frac{k}{n}\right) \right| \right]$$

$$\le \sum_{n=1}^{N^2} \sum_{m=1}^{N^2} \frac{|f(n)|}{|n^z|} \cdot \frac{|g(m)|}{|m^z|} \le \sum_{n=1}^{\infty} \frac{|f(n)|}{|n^z|} \cdot \sum_{m=1}^{\infty} \frac{|g(m)|}{|m^z|}.$$

Since both Dirichlet series $F(z)$ and $G(z)$ converge absolutely for some $z \in \mathbb{C}$, the right-hand side is some finite constant. Therefore, $\sum_{k=1}^{N^2} \frac{(f*g)(k)}{k^z}$ converges absolutely as $N \to \infty$.
Taking limit as $N \to \infty$ on both sides of the equation

$$\sum_{n=1}^{N} \frac{f(n)}{n^z} \cdot \sum_{m=1}^{N} \frac{g(m)}{m^z} = \sum_{k=1}^{N^2} \frac{(f*g)(k)}{k^z} - A_N,$$

we have

$$F(z)G(z) = \sum_{n=1}^{\infty} \frac{(f*g)(n)}{n^z}.$$

where the left-hand side converges absolutely by the assumption.

Chapter 3 Dirichlet Characters

3.1 Definition and Basic Properties

Definition 3.1 (Dirichlet Character)

A function $\chi : \mathbb{Z} \to \mathbb{C}$ is called a Dirichlet character modulo q if the following conditions hold:

(i) χ is completely multiplicative, i.e., for all $m, n \in \mathbb{Z}$, $\chi(mn) = \chi(m)\chi(n)$.

(ii) χ is periodic with period q, i.e., for all $n \in \mathbb{Z}$, $\chi(n+q) = \chi(n)$.

(iii) $\chi(n) \neq 0$ if and only if $\gcd(n, q) = 1$.

♣

Note *By the definition of Dirichlet character, we have $\chi(1) = 1$ since $\chi(1) = \chi(1 \cdot 1) = \chi(1)\chi(1)$ and $\chi(1) \neq 0$ as $\gcd(1, q) = 1$. So $\chi(1) = 1$ for any Dirichlet character χ.*

Exercise 3.1 For $m, n \in \mathbb{Z}$ and q a positive integer, show that:

$$\gcd(mn, q) = 1 \iff \gcd(m, q) = 1 \text{ and } \gcd(n, q) = 1.$$

Solution *($\Rightarrow$) Suppose that $\gcd(mn, q) = 1$. Then there are integers a, b such that*

$$a(mn) + bq = 1.$$

Thus, any common divisor of m and q must divide the right-hand side, which is 1. Therefore, $\gcd(m, q) = 1$. Similarly, any common divisor of n and q must also divide 1, so $\gcd(n, q) = 1$.

($\Leftarrow$) Conversely, suppose that $\gcd(m, q) = 1$ and $\gcd(n, q) = 1$. Then there exist integers a_1, b_1 such that

$$a_1 m + b_1 q = 1,$$

and integers a_2, b_2 such that

$$a_2 n + b_2 q = 1.$$

Multiplying these two equations together, we get

$$(a_1 m + b_1 q)(a_2 n + b_2 q) = 1.$$

Expanding the left-hand side, we have

$$a_1 a_2 mn + (a_1 b_2 m + a_2 b_1 n + b_1 b_2 q)q = 1.$$

This shows that any common divisor of mn and q must divide the right-hand side, which is 1. Therefore, $\gcd(mn, q) = 1$.

Proposition 3.1

Let χ be a Dirichlet character modulo q. Consider a function $\widetilde{\chi} : \mathbb{Z}/q\mathbb{Z} \to \mathbb{C}$ defined by

$$\widetilde{\chi}([n]) := \chi(n)$$

where $[n]$ is the equivalence class of n in $\mathbb{Z}/q\mathbb{Z}$.

Then, $\widetilde{\chi}$ is well-defined on $\mathbb{Z}/q\mathbb{Z}$.

♠

Remark Recall that $\mathbb{Z}/q\mathbb{Z}$ is a finite cyclic group under addition modulo q, and further more, if we add the multiplication operation modulo q, then it becomes a finite ring.

One may consider $\widetilde{\chi}$ as a function defined on the finite ring $\mathbb{Z}/q\mathbb{Z}$ but note that $\widetilde{\chi}$ is not a ring homomorphism since it is not additive in general, i.e., $\widetilde{\chi}([m] + [n]) \neq \widetilde{\chi}([m]) + \widetilde{\chi}([n])$ in general.

Note *After this proposition, we will often identify χ with $\widetilde{\chi}$ and simply write $\chi : \mathbb{Z}/q\mathbb{Z} \to \mathbb{C}$. Hopefully this abuse of notation will not cause any confusion.*

Exercise 3.2 Prove the above proposition.

Solution *To show that $\widetilde{\chi}$ is well-defined, we need to verify that if $[n] = [m]$ in $\mathbb{Z}/q\mathbb{Z}$, then $\widetilde{\chi}([n]) = \widetilde{\chi}([m])$.*

By the definition of equivalence classes in $\mathbb{Z}/q\mathbb{Z}$, we have $[n] = [m]$ if and only if $n \equiv m \mod q$. This means that *there exists an integer k such that*

$$n = m + kq.$$

Since χ is periodic with period q, we have

$$\chi(n) = \chi(m + kq) = \chi(m).$$

Therefore,

$$\widetilde{\chi}([n]) = \chi(n) = \chi(m) = \widetilde{\chi}([m]).$$

This shows that $\widetilde{\chi}$ is well-defined on $\mathbb{Z}/q\mathbb{Z}$.

Definition 3.2 (Group of Units of $\mathbb{Z}/q\mathbb{Z}$)

*Let $\mathbb{Z}/q\mathbb{Z}$ be the finite **ring** of integers modulo q. The group of units of $\mathbb{Z}/q\mathbb{Z}$, denoted by G_q, is the set of all elements in $\mathbb{Z}/q\mathbb{Z}$ that have a multiplicative inverse. We regard G_q as a finite **group** under multiplication modulo q.*

♣

Example 3.1 Here, we review some properties in elementary number theory to introduce the following lemma.

Let $I_{m,n} = \{mx + ny : x, y \in \mathbb{Z}\} = \langle m, n\rangle$ be the ideal generated by m and n in the ring of integers $\mathbb{Z}$. Then it is a well-known result that

$$I_{m,n} = d\mathbb{Z} \quad \text{where } d = \gcd(m, n)$$

(i.e. $\langle m, n\rangle = \langle \gcd(m, n)\rangle$). If you are familiar with the concept of a principal ideal domain (PID), then this result follows directly from the fact that $\mathbb{Z}$ is a PID. Here, however, we give a fundamental proof without using the concept of PID. Our goal is first to show that $I_{m,n} = d\mathbb{Z}$ for some $d \in \mathbb{Z}$, and then to show that this d is indeed $\gcd(m, n)$.

First, we show that $I_{m,n} \subseteq d\mathbb{Z}$ for some $d \in \mathbb{Z}$. Let d be the smallest positive integer in $I_{m,n}$, so there exist integers x_0, y_0 such that $d = mx_0 + ny_0$. For any element $a \in I_{m,n}$, we have $a = mx_1 + ny_1$ for some integers x_1, y_1. By the division algorithm, there exist unique integers q, r such that $a = qd + r \quad \text{with } 0 \le r < d$. Thus,

$$r = a - qd = (mx_1 + ny_1) - q(mx_0 + ny_0) = m(x_1 - qx_0) + n(y_1 - qy_0) \in I_{m,n}.$$

Since d is the smallest positive integer in $I_{m,n}$, we must have $r = 0$. Therefore, $a = qd \in d\mathbb{Z}$, and hence $I_{m,n} \subseteq d\mathbb{Z}$.

Now we show that $d\mathbb{Z} \subseteq I_{m,n}$. Since $d = mx_0 + ny_0$, for any integer k we have

$$kd = k(mx_0 + ny_0) = m(kx_0) + n(ky_0) \in I_{m,n}.$$

Thus, $d\mathbb{Z} \subseteq I_{m,n}$, and hence $I_{m,n} = d\mathbb{Z}$.

Furthermore, since both m and n belong to $I_{m,n}$, d divides both m and n. Thus, d is a common divisor of m and n. If there exists a common divisor d' of m and n such that $d' < d$, then d' divides every element of $I_{m,n}$, which contradicts the definition of d as the smallest positive integer in $I_{m,n}$. Therefore, $d = \gcd(m, n)$.

Lemma 3.1 (Characterization of Group of Units of $\mathbb{Z}/q\mathbb{Z}$)

The group of units of $\mathbb{Z}/q\mathbb{Z}$ is given by

$$G_q = \{[n] \in \mathbb{Z}/q\mathbb{Z} : \gcd(n, q) = 1\}.$$

Sometimes, we also write $G_q = \{0 \le r \le q - 1 : \gcd(r, q) = 1\}$ by using r to denote the representative of the equivalence class $[r]$ in $\mathbb{Z}/q\mathbb{Z}$.
Furthermore, by definition of Euler's totient function $\phi(q)$, we have

$$|G_q| = \phi(q).$$

♡

Proof For a given $[n] \in \mathbb{Z}/q\mathbb{Z}$, we have $[n] \in G_q$ if and only if there exists $[m] \in \mathbb{Z}/q\mathbb{Z}$ such that

$$[n][m] = [1] \implies [nm] = [1] \implies nm \equiv 1 \mod q.$$

This is equivalent to saying that for which $n \in \mathbb{Z}$ the linear congruence equation $nx \equiv 1 \mod q$ has a solution. And this is equivalent to solve the linear Diophantine equation

$$nx - qy = 1$$

for integers x, y. By example 3.1, we know that $nx - qy \in \gcd(n,q)\mathbb{Z}$ for all integers x, y. Thus, the above equation has integer solutions only if $\gcd(n,q) = 1$. (Otherwise, $1 \notin \gcd(n,q)\mathbb{Z}$.) Thus, we have shown that

$$G_q \subseteq \{[n] \in \mathbb{Z}/q\mathbb{Z} : \gcd(n,q) = 1\}.$$

Conversely, suppose that $\gcd(n,q) = 1$. Then $1 \in I_{n,q}$ by example 3.1, so there exist integers x_0, y_0 such that $nx_0 + qy_0 = 1$. The equation $nx \equiv 1 \mod q$ has a solution $x \equiv x_0 \mod q$. Thus, there exists $[m] \in \mathbb{Z}/q\mathbb{Z}$ such that $[n][m] = [1]$, and hence $[n] \in G_q$.

Corollary 3.1 (Euler's Theorem)

For any integer n such that $\gcd(n,q) = 1$, we have

$$n^{\phi(q)} \equiv 1 \mod q.$$

♡

Proof Since $\gcd(n,q) = 1$, we have $[n] \in G_q$. By Corollary of Lagrange's Theorem (If $g \in G$, G is a finite group, then $g^{|G|} = e$ is the identity element of G), we have

$$[n]^{|G_q|} = [1] \implies [n]^{\phi(q)} = [1] \implies n^{\phi(q)} \equiv 1 \mod q.$$

Corollary 3.2 (Property of Dirichlet Characters from Euler's Theorem)

Let χ be a Dirichlet character modulo q. Then for any $n \in \mathbb{Z}$ such that $\gcd(n,q) = 1$, we have

$$\chi(n)^{\phi(q)} = 1$$

♡

Proof Direct from corollary 3.1 and the fact that χ is completely multiplicative, we have

$$\chi(n)^{\phi(q)} = \chi(n^{\phi(q)}) = \chi(1) = 1.$$

Corollary 3.3 (Dirichlet Characters are Roots of Unity)

Let χ be a Dirichlet character modulo q. Then we have

$$\chi(n) = \begin{cases} 0 & \text{if } \gcd(n,q) \neq 1, \\ e^{\frac{2\pi i \cdot k_q(n)}{\phi(q)}} & \text{for some } k_q(n) \in \{0, 1, 2, \ldots, \phi(q) - 1\} \text{ if } \gcd(n,q) = 1. \end{cases}$$

♡

Proof Direct from corollary 3.2, we have $\chi(n)^{\phi(q)} = 1$ for all n such that $\gcd(n,q) = 1$. Thus, $\chi(n)$ is a $\phi(q)$-th root of unity. The $\phi(q)$-th roots of unity are given by

$$e^{2\pi i k/\phi(q)} \quad \text{for } k = 0, 1, 2, \ldots, \phi(q) - 1.$$

and for n such that $\gcd(n,q) \neq 1$, we have $\chi(n) = 0$ by definition of Dirichlet characters.

Remark Clearly, $k = k_q(n)$ may depend on both q and n. Thus, understanding a Dirichlet character χ mod q boils down to understanding the map

$$n \mapsto k_q(n) \quad \text{for } n \text{ such that } \gcd(n,q) = 1.$$

3.2 Characters in Group Theory

In this section and following sections, we mainly focus on characters from the **finite** abelian group theory point of view. The theory of characters of finite abelian groups is a fundamental tool in number theory, particularly in the study of Dirichlet characters and L-functions.

Definition 3.3 (Character of a Finite Abelian Group)

Let G be a finite abelian group. A character of G is a group homomorphism $\chi : G \to \mathbb{C} \setminus \{0\}$, where $\mathbb{C} \setminus \{0\}$ is regarded as a group under multiplication.

♣

Note *Recall that a group homomorphism $\varphi : G \to H$ between two groups G and H is a map such that for all $g_1, g_2 \in G$,*

$$\varphi(g_1 g_2) = \varphi(g_1)\varphi(g_2).$$

Proposition 3.2 (Characters have Modulus 1)

Let χ be a character of a finite abelian group G. Then, $|\chi(g)| = 1$, for all $g \in G$.

Exercise 3.3 Prove the above proposition.

Solution *(Recall that for a homomorphism $\varphi : G \to H$, we have $\varphi(e_G) = e_H$, where e_G and e_H are the identity elements of G and H respectively.) Thus, for any $g \in G$ with order d, we have*

$$1 = \chi(e_G) = \chi(g^{|G|}) = (\chi(g))^{|G|} \implies (\chi(g))^{|G|} = 1 \implies |\chi(g)| = 1.$$

Definition 3.4 (Principal Character)

Let G be a finite abelian group. The character $\chi_0 : G \to \mathbb{C} \setminus \{0\}$ defined by

$$\chi_0(g) = 1 \quad \text{for all } g \in G$$

is called the Principal Character of G.

Definition 3.5 (Group of Characters)

Let G be a finite abelian group. The set of all characters of G is denoted by $\widehat{G}$, i.e.,

$$\widehat{G} := \{\chi : \chi \text{ is a character of } G\}.$$

Exercise 3.4 Let G be a finite abelian group and $g^* \in G$ be any element. Show that $H := \{gg^* : g \in G\}$ has the same cardinality as G. (i.e. Elements of G and H are same.)

Solution *We only need to show that the map $f : G \to G$ defined by $f(g) = gg^*$ is bijective. Clearly it is surjective so we only need to show that it is injective.*

Suppose that $f(g_1) = f(g_2)$ for some $g_1, g_2 \in G$. Then, we have

$$g_1 g^* = g_2 g^* \implies g_1 g^* (g^*)^{-1} = g_2 g^* (g^*)^{-1} \implies g_1 = g_2$$

since G is a group and hence every element has a multiplicative inverse. Thus, f is injective.

Theorem 3.1 (Orthogonality Relations of Characters)

For a finite abelian group G, let $\chi_1, \chi_2 \in \widehat{G}$, define the inner product of χ_1 and χ_2 by

$$\langle \chi_1, \chi_2 \rangle := \frac{1}{|G|} \sum_{g \in G} \chi_1(g)\overline{\chi_2(g)}.$$

Then, we have

$$\langle \chi_1, \chi_2 \rangle = \begin{cases} 1 & \text{if } \chi_1 = \chi_2, \\ 0 & \text{if } \chi_1 \neq \chi_2. \end{cases}$$

Proof We omitted the inner product proof of $\langle \chi_1, \chi_2 \rangle$ here. Just show the last part that $\langle \chi_1, \chi_2 \rangle = \delta_{\chi_1,\chi_2}$ where δ is the Kronecker delta.

If $\chi_1 = \chi_2$, $\chi_1(g)\overline{\chi_2(g)} = |\chi_1(g)|^2 = 1$ for all $g \in G$ by proposition 3.2. Thus, sum of them is $|G|$ and hence $\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \cdot |G| = 1$.

If $\chi_1 \neq \chi_2$, Let $\psi(g) := \chi_1(g)\overline{\chi_2(g)}$ is also a character of G. If ψ is the principal character, then $\psi(g) = 1$ for all $g \in G$, which implies that for all $g \in G$,

$$\chi_1(g)\overline{\chi_2(g)} = 1 \implies \chi_1(g)\overline{\chi_2(g)}\chi_2(g) = \chi_2(g) \implies \chi_1(g) = \chi_2(g)$$

contradicting the assumption that $\chi_1 \neq \chi_2$. Thus, ψ is not the principal character, so there exists some $g_0 \in G$ such that

$\psi(g_0) \neq 1$. Then, by exercise 3.4, we have

$$\sum_{g\in G} \psi(g) = \sum_{g\in G} \psi(gg_0) = \psi(g_0) \sum_{g\in G} \psi(g).$$

Since $\psi(g_0) \neq 1$, we have $\sum_{g\in G} \psi(g) = 0$. Therefore, $\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \cdot 0 = 0$.

Exercise 3.5 Let G be a finite set and consider the set $\mathbb{C}^G := \{f : G \to \mathbb{C} | f \text{ is any function}\}$. Show that $\mathbb{C}^G$ is a vector space over the field $\mathbb{C}$ with pointwise addition and scalar multiplication, i.e., for $f_1, f_2 \in \mathbb{C}^G$ and $\alpha \in \mathbb{C}$,

$$(f_1 + f_2)(g) := f_1(g) + f_2(g), \quad (\alpha f_1)(g) := \alpha \cdot f_1(g) \quad \text{for all } g \in G.$$

Solution *We need to verify the vector space conditions for $\mathbb{C}^G$.*

(i) (Closure under addition) For any $f_1, f_2 \in \mathbb{C}^G$, $(f_1+f_2)(g) = f_1(g)+f_2(g) \in \mathbb{C}$ for all $g \in G$. Thus, $f_1+f_2 \in \mathbb{C}^G$.

(ii) (Closure under scalar multiplication) For any $f_1 \in \mathbb{C}^G$ and $\alpha \in \mathbb{C}$, $(\alpha f_1)(g) = \alpha \cdot f_1(g) \in \mathbb{C}$ for all $g \in G$. Thus, $\alpha f_1 \in \mathbb{C}^G$.

(iii) (Associativity of addition) For any $f_1, f_2, f_3 \in \mathbb{C}^G$, we have

$$((f_1+f_2)+f_3)(g) = (f_1+f_2)(g)+f_3(g) = f_1(g)+f_2(g)+f_3(g) = f_1(g)+(f_2+f_3)(g) = (f_1+(f_2+f_3))(g).$$

Thus, $(f_1 + f_2) + f_3 = f_1 + (f_2 + f_3)$.

(iv) (Commutativity of addition) For any $f_1, f_2 \in \mathbb{C}^G$, we have

$$(f_1 + f_2)(g) = f_1(g) + f_2(g) = f_2(g) + f_1(g) = (f_2 + f_1)(g).$$

Thus, $f_1 + f_2 = f_2 + f_1$.

(v) (Existence of additive identity) Define the zero function $0 : G \to \mathbb{C}$ by $0(g) = 0$ for all $g \in G$. Then, for any $f \in \mathbb{C}^G$, we have

$$(f + 0)(g) = f(g) + 0(g) = f(g).$$

Thus, $f + 0 = f$.

(vi) (Existence of additive inverses) For any $f \in \mathbb{C}^G$, define $-f : G \to \mathbb{C}$ by $(-f)(g) = -f(g)$ for all $g \in G$. Then, we have

$$(f + (-f))(g) = f(g) + (-f)(g) = f(g) - f(g) = 0.$$

Thus, $f + (-f) = 0$.

(vii) (Distributivity of scalar multiplication over vector addition) For any $\alpha \in \mathbb{C}$ and $f_1, f_2 \in \mathbb{C}^G$, we have

$$(\alpha(f_1 + f_2))(g) = \alpha(f_1 + f_2)(g) = \alpha(f_1(g) + f_2(g)) = \alpha f_1(g) + \alpha f_2(g) = (\alpha f_1 + \alpha f_2)(g).$$

Thus, $\alpha(f_1 + f_2) = \alpha f_1 + \alpha f_2$.

(viii) (Distributivity of scalar multiplication over field addition) For any $\alpha, \beta \in \mathbb{C}$ and $f \in \mathbb{C}^G$, we have

$$((\alpha + \beta)f)(g) = (\alpha + \beta)f(g) = \alpha f(g) + \beta f(g) = (\alpha f + \beta f)(g).$$

Thus, $(\alpha + \beta)f = \alpha f + \beta f$.

(ix) (Compatibility of scalar multiplication with field multiplication) For any $\alpha, \beta \in \mathbb{C}$ and $f \in \mathbb{C}^G$, we have

$$((\alpha\beta)f)(g) = (\alpha\beta)f(g) = \alpha(\beta f(g)) = (\alpha(\beta f))(g).$$

Thus, $(\alpha\beta)f = \alpha(\beta f)$.

(x) (Existence of multiplicative identity) For any $f \in \mathbb{C}^G$, we have

$$(1f)(g) = 1 \cdot f(g) = f(g).$$

Thus, $1f = f$.

Therefore, $\mathbb{C}^G$ is a vector space over the field $\mathbb{C}$.

Exercise 3.6 Show that the vector space $\mathbb{C}^G$ over the field $\mathbb{C}$ which is defined in exercise 3.5 has dimension $|G|$.

Solution *Consider the set of functions $\{e_g : g \in G\}$ defined by*

$$e_g(h) = \begin{cases} 1 & \text{if } h = g, \\ 0 & \text{if } h \neq g. \end{cases}$$

for all $h \in G$. We will show that this set forms a basis for the vector space $\mathbb{C}^G$.

First, we show that the set $\{e_g : g \in G\}$ spans $\mathbb{C}^G$. For any function $f \in \mathbb{C}^G$, consider the linear combination

$$g = \sum_{g \in G} f(g)e_g.$$

Evaluating this linear combination at any $h \in G$, we have

$$g(h) = \sum_{g \in G} f(g)e_g(h) = f(h)e_h(h) + \sum_{\substack{g \in G \\ g \neq h}} f(g)e_g(h) = f(h) \cdot 1 + \sum_{\substack{g \in G \\ g \neq h}} f(g) \cdot 0 = f(h).$$

Thus, $g(h) = f(h)$ for all $h \in G$, which implies that $g = f$. This shows that any function in $\mathbb{C}^G$ can be written as a linear combination of the functions e_g, so the set spans $\mathbb{C}^G$.

Next, we show that the set $\{e_g : g \in G\}$ is linearly independent. Suppose that there exist scalars $\alpha_g \in \mathbb{C}$ such that

$$\sum_{g \in G} \alpha_g e_g = 0,$$

where 0 is the zero function. Evaluating this equation at each $h \in G$, we have

$$\sum_{g \in G} \alpha_g e_g(h) = 0.$$

Since $e_g(h) = 1$ if $h = g$ and 0 otherwise, this simplifies to

$$\alpha_h = 0.$$

Since this holds for all $h \in G$, we conclude that all coefficients α_g must be zero. Therefore, the set $\{e_g : g \in G\}$ is linearly independent.

Since the set $\{e_g : g \in G\}$ spans $\mathbb{C}^G$ and is linearly independent, it forms a basis for $\mathbb{C}^G$. The number of elements in this basis is equal to the cardinality of G, which is $|G|$. Therefore, the dimension of the vector space $\mathbb{C}^G$ is $|G|$.

Corollary 3.4 (Number of Characters is at most the Order of the Group)

For a finite abelian group G, and its group of characters $\widehat{G}$, we have

$$|\widehat{G}| \leq |G|.$$

♡

Proof We can view $\widehat{G}$ as a subset of the vector space $\mathbb{C}^G$ over the field $\mathbb{C}$ defined in exercise 3.5. By theorem 3.1, the characters in $\widehat{G}$ are orthonormal with respect to the inner product defined there. Thus, the characters in $\widehat{G}$ are linearly independent in the vector space $\mathbb{C}^G$. This is one of the standard results in linear algebra: Any orthonormal set in an inner product space is linearly independent and here is the proof: Suppose that there exist scalars $\alpha_i \in \mathbb{C}$ such that $\sum_{\chi_i \in \widehat{G}} \alpha_i \chi_i = 0$, then

$$\langle \sum_{\chi_i \in \widehat{G}} \alpha_i \chi_i, \chi_j \rangle = 0 \implies \sum_{\chi_i \in \widehat{G}} \alpha_i \langle \chi_i, \chi_j \rangle = 0 \implies \sum_{\chi_i \in \widehat{G}} \alpha_i \delta_{\chi_i, \chi_j} = 0 \implies \alpha_j = 0.$$

Therefore, by exercise 3.6, we have

$$|\widehat{G}| = \dim(\mathrm{span}(\widehat{G})) \leq \dim(\mathbb{C}^G) = |G|.$$

Theorem 3.2 (Number of Characters of Cyclic Group and Dirichlet Characters)

If $G = \mathbb{Z}/q\mathbb{Z}$, then there are exactly $|G| = q$ characters of G, i.e., $|\widehat{G}| = q$ and they are given by

$$\chi_k([n]) = e^{\frac{2\pi i k n}{q}} \quad \text{for } k = 0, 1, 2, \ldots, q-1.$$

Furthermore, $\widehat{G} \cong G$.

♡

Proof First, we show that $\chi_k([n])$ is indeed a character of G. For any $[m], [n] \in G$, we have

$$\chi_k([m] + [n]) = \chi_k([m+n]) = e^{\frac{2\pi i k(m+n)}{q}} = e^{\frac{2\pi i k m}{q}} \cdot e^{\frac{2\pi i k n}{q}} = \chi_k([m]) \cdot \chi_k([n]).$$

Thus, χ_k is a group homomorphism from G to $\mathbb{C} \setminus \{0\}$, so it is a character of G.

Next, we show that the characters χ_k for $k = 0, 1, 2, \ldots, q-1$ are distinct. Since $\chi_k = \chi_{k+q}$ for all integers k, it suffices to show that $\chi_k \neq \chi_{k'}$ for $k, k' \in \{0, 1, 2, \ldots, q-1\}$ with $k \neq k'$. Suppose that $\chi_k = \chi_{k'}$ for some

$k, k' \in \{0, 1, 2, \ldots, q-1\}$. Then, for all $[n] \in G$, we have

$$e^{\frac{2\pi i k n}{q}} = e^{\frac{2\pi i k' n}{q}}.$$

This implies that

$$e^{\frac{2\pi i (k-k')n}{q}} = 1 \quad \text{for all } n \in \mathbb{Z}.$$

In particular, for $n = 1$, we have

$$e^{\frac{2\pi i (k-k')}{q}} = 1.$$

This implies that $k - k' \equiv 0 \mod q$. Since both k and k' are in the range 0 to $q-1$, we must have $k = k'$. Thus, the characters χ_k are distinct. Therefore, we have found q distinct characters of G. By the previous corollary 3.4, we have $|\widehat{G}| \leq |G| = q$. Thus, we conclude that $|\widehat{G}| = q$.

For the $\widehat{G} \cong G$ part, we can define the map $\varphi : G \to \widehat{G}$ by $\varphi([k]) = \chi_k$. It is easy to check that φ is a group isomorphism.

3.3 Back to Dirichlet Characters

In the proposition 3.1, we have shown that a Dirichlet character χ modulo q induces a character $\widetilde{\chi}$ of the finite abelian group $\mathbb{Z}/q\mathbb{Z}$. On the other hand, if we start with a character χ of the finite abelian group $\mathbb{Z}/q\mathbb{Z}$, we can extend χ to all integers $\mathbb{Z}$ and obtain a Dirichlet character modulo q as follows:

Proposition 3.3

Let χ be a character of the finite abelian group $\mathbb{Z}/q\mathbb{Z}$. Define the map $\widehat{\chi} : \mathbb{Z} \to \mathbb{C}$ by

$$\widehat{\chi}(n) = \begin{cases} \chi([n]) & \text{if } \gcd(n,q) = 1, \\ 0 & \text{if } \gcd(n,q) \neq 1. \end{cases}$$

Then, $\widehat{\chi}$ is a Dirichlet character modulo q.

Exercise 3.7 Prove the above proposition.

Solution *We need to verify that $\widehat{\chi}$ satisfies the three properties of a Dirichlet character modulo q.*

(i) *(Periodicity) For any integers n and m such that $n \equiv m \mod q$, we have $[n] = [m]$ in $\mathbb{Z}/q\mathbb{Z}$. Thus,*

$$\widehat{\chi}(n) = \begin{cases} \chi([n]) & \text{if } \gcd(n,q) = 1, \\ 0 & \text{if } \gcd(n,q) \neq 1, \end{cases} = \begin{cases} \chi([m]) & \text{if } \gcd(m,q) = 1, \\ 0 & \text{if } \gcd(m,q) \neq 1, \end{cases} = \widehat{\chi}(m).$$

Thus, $\widehat{\chi}$ is periodic modulo q.

(ii) *(Zero at non-units) By definition, for any integer n such that $\gcd(n,q) \neq 1$, we have*

$$\widehat{\chi}(n) = 0.$$

Thus, $\widehat{\chi}$ is zero at non-units modulo q.

(iii) *(Completely multiplicative) For any integers m and n, we consider the following cases:*
- *If $\gcd(m,q) = 1$ and $\gcd(n,q) = 1$, then $\gcd(mn,q) = 1$. Thus,*

$$\widehat{\chi}(mn) = \chi([mn]) = \chi([m])\chi([n]) = \widehat{\chi}(m)\widehat{\chi}(n).$$

- *If $\gcd(m,q) \neq 1$ or $\gcd(n,q) \neq 1$, then $\gcd(mn,q) \neq 1$. Thus,*

$$\widehat{\chi}(mn) = 0 = \widehat{\chi}(m)\widehat{\chi}(n).$$

Therefore, in all cases, we have

$$\widehat{\chi}(mn) = \widehat{\chi}(m)\widehat{\chi}(n).$$

Thus, $\widehat{\chi}$ is completely multiplicative.

Since $\widehat{\chi}$ satisfies all three properties of a Dirichlet character modulo q, we conclude that $\widehat{\chi}$ is indeed a Dirichlet character modulo q.

Theorem 3.3 (Dirichlet Characters and Characters of Group of Units)

The Dirichlet characters modulo q are just the characters of the particular finite abelian group G_q, where G_q is the group of units of $\mathbb{Z}/q\mathbb{Z}$ with multiplication modulo q as the group operation.
Furthermore, the number of Dirichlet characters modulo q is $\phi(q)$ ♡

Proof Let ψ be the function we defined in proposition 3.1 and ϕ be the function we defined in the above proposition 3.3. Then, for any integer n, we have

$$\psi(e)(n) = \begin{cases} e([n]) & \text{if } \gcd(n,q)=1, \\ 0 & \text{if } \gcd(n,q)\neq 1, \end{cases} \quad \text{and} \quad \phi(\chi)(n) = \begin{cases} \chi([n]) & \text{if } \gcd(n,q)=1, \\ 0 & \text{if } \gcd(n,q)\neq 1. \end{cases}$$

where e is a Dirichlet character modulo q and χ is a character of the finite abelian group G_q. It is clear that ϕ and ψ are well-defined maps between the set of Dirichlet characters modulo q and the set of characters of the finite abelian group G_q.

Moreover, it is easy to check that ϕ and ψ are inverses of each other. Indeed, for any integer n, we have

$$\phi(\psi(e))(n) = e(n) \quad \text{and} \quad \psi(\phi(\chi))([n]) = \chi([n]).$$

These equalities follow directly from propositions 3.1 and 3.3. Thus, the set of Dirichlet characters modulo q is in bijection with the set of characters of the finite abelian group G_q.

Finally, by theorem 3.2, the number of characters of the finite abelian group G_q is equal to $|G_q| = \varphi(q)$. Therefore, the number of Dirichlet characters modulo q is also $\varphi(q)$.

Definition 3.6 (Primitive Root of Unity)

Let G_q be the group of units of $\mathbb{Z}/q\mathbb{Z}$ with multiplication modulo q as the group operation. If G_q is a cyclic group, then a generator of G_q, denoted by g, satisfies

$$G_q = \{1, g, g^2, \dots, g^{\phi(q)-1}\}.$$

is called a primitive root of unity modulo q. ♣

Definition 3.7 (Index base on Primitive Root of Unity)

Let G_q be the group of units of $\mathbb{Z}/q\mathbb{Z}$ with multiplication modulo q as the group operation. If G_q is a cyclic group generated by a primitive root of unity g, then we define the index base on g as

$$\mathrm{ind}_g(m) := k \quad \textit{if } g^k \equiv m \mod q \quad \textit{for } m\in\mathbb{Z} \textit{ such that } \gcd(m,q)=1.$$

i.e. $[m]$ is the k-th power of the primitive root of unity g in G_q, is called the index of m base on g. ♣

Definition 3.8 (Explicit Form of Dirichlet Characters (when corresponding Group of Units is Cyclic))

Let G_q be the group of units of $\mathbb{Z}/q\mathbb{Z}$ with multiplication modulo q as the group operation. If G_q is a cyclic group generated by a primitive root of unity g, then the Dirichlet characters modulo q are given by

$$\chi_n(m) = \begin{cases} 0 & \textit{if } \gcd(m,q)>1, \\ e^{\frac{2\pi i n\cdot \mathrm{ind}_g(m)}{\phi(q)}} & \textit{if } \gcd(m,q)=1, \end{cases}$$

♣

Remark It's direct from theorem 3.3, theorem 3.2 and our definition of index base on primitive root of unity 3.7.

3.4 Dirichlet Test

Lemma 3.2 (Summmation by Parts)

Let $A_n = \sum_{k=1}^{n} a_k$, $D_n = b_n - b_{n-1}$ for sequences $\{a_n\}$ and $\{b_n\}$. Setting $A_0 = 0$, then for any positive integer N, we have

$$\sum_{n=1}^{N} a_n b_n = A_N b_N - \sum_{n=1}^{N-1} A_n D_{n+1}.$$

♡

Proof By using the integration by parts idea, we have

$$\int_1^N a(x)b(x)\,dx = A(N)b(N) - \int_1^N A(x)b'(x)\,dx.$$

The A_n play the role of integration and the D_n play the role of differentiation. One may consider this lemma as a discrete version of integration by parts.

Now we prove it directly:

$$\begin{aligned}\sum_{n=1}^{N} a_n b_n &= \sum_{n=1}^{N}(A_n - A_{n-1})b_n = \sum_{n=1}^{N} A_n b_n - \sum_{n=1}^{N} A_{n-1} b_n \\ &= \sum_{n=1}^{N} A_n b_n - \sum_{n=0}^{N-1} A_n b_{n+1} = A_N b_N - \sum_{n=1}^{N-1} A_n (b_{n+1} - b_n) \\ &= A_N b_N - \sum_{n=1}^{N-1} A_n D_{n+1}.\end{aligned}$$

Theorem 3.4 (Dirichlet Test for Series Convergence)

Let $\{a_n\}$, $\{b_n\}$ be sequences of complex numbers. If the following conditions hold:

(i) The sequence of partial sums $A_N = \sum_{k=1}^{N} a_k$ is bounded.

(ii) The sequence $b_n \to 0$ as $n \to \infty$.

(iii) The series $\sum_{n=1}^{\infty} |b_{n+1} - b_n|$ converges.

Then, the series $\sum_{n=1}^{\infty} a_n b_n$ converges.

♡

Proof The three conditions imply that the series $\sum_{n=1}^{\infty} A_n D_{n+1}$ converges absolutely, where $D_n = b_n - b_{n-1}$, since

$$\sum_{n=1}^{\infty} |A_n D_{n+1}| \le M \sum_{n=1}^{\infty} |b_{n+1} - b_n| < \infty,$$

where M is a bound for the sequence of partial sums A_N.
Consider the partial sums of the series $\sum_{n=1}^{\infty} a_n b_n$:

$$S_N = \sum_{n=1}^{N} a_n b_n = A_N b_N - \sum_{n=1}^{N-1} A_n D_{n+1}, \quad \text{by lemma 3.2.}$$

Notice that as $N \to \infty$, the term $A_N b_N \to 0$ since A_N is bounded and $b_N \to 0$ and the series $\sum_{n=1}^{\infty} A_n D_{n+1}$ converges absolutely as shown before. Thus, the partial sums S_N converge to a finite limit as $N \to \infty$. Therefore, the series $\sum_{n=1}^{\infty} a_n b_n$ converges.

Example 3.2 We can use the Dirichlet test 3.4 to show the alternating series test: Let $\{a_n\}$ be a sequence of positive real numbers that is monotonically decreasing and converges to 0. Then, the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

converges.

We can apply the Dirichlet test 3.4 with $b_n = (-1)^{n+1}$ and a_n as given. The sequence of partial sums $A_N = \sum_{k=1}^{N} a_k$ is bounded since $\{a_n\}$ is positive and monotonically decreasing. The sequence $b_n = (-1)^{n+1}$ converges to 0 as $n \to \infty$. Finally, the series $\sum_{n=1}^{\infty} |b_{n+1} - b_n|$ converges since $|b_{n+1} - b_n| = 2$ for all n. Therefore, by the Dirichlet test, the alternating series $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ converges.

Theorem 3.5 (Convergence of Dirichlet L-function for Non-Principal Characters)

The Dirichlet L-function

$$L(z,\chi)=\sum_{n=1}^{\infty}\frac{\chi(n)}{n^z}$$

is convergent for every $z\in\mathbb{C}$ with $\Re(z)>0$, where χ is a non-principal Dirichlet character modulo q. ♡

Proof We can apply the Dirichlet test 3.4 to the Dirichlet L-function

$$L(z,\chi)=\sum_{n=1}^{\infty}\frac{\chi(n)}{n^z},$$

by taking $a_n=\chi(n)$ and $b_n=n^{-z}$. We verify the three conditions of the Dirichlet test for $\Re(z)>0$.

(i) **Boundedness of the partial sums $A_N=\sum_{k=1}^{N}\chi(k)$.** Since χ is a Dirichlet character modulo q, it is periodic with period q, and $\chi(n)=0$ whenever $\gcd(n,q)\neq 1$. As χ is not the principal character, we have

$$\sum_{r=1}^{q}\chi(r)=\sum_{\substack{1\le r\le q\\ \gcd(r,q)=1}}\chi(r)=\sum_{r\in G_q}\chi(r)=\sum_{r\in G_q}\chi(r)\overline{\chi_0(r)}=|G_q|\langle\chi,\chi_0\rangle=0,$$

by the orthogonality relations of characters 3.1. Using periodicity $\chi(mq+r)=\chi(r)$, we decompose

$$\begin{aligned}A_N=\sum_{k=1}^{N}\chi(k)&=\sum_{k=1}^{\lfloor N/q\rfloor q}\chi(k)+\sum_{k=\lfloor N/q\rfloor q+1}^{N}\chi(k)\\&=\sum_{m=0}^{\lfloor N/q\rfloor-1}\sum_{r=1}^{q}\chi(mq+r)+\sum_{k=\lfloor N/q\rfloor q+1}^{N}\chi(k)\\&=(\lfloor N/q\rfloor)\sum_{r=1}^{q}\chi(r)+\sum_{k=\lfloor N/q\rfloor q+1}^{N}\chi(k)\\&=\sum_{k=\lfloor N/q\rfloor q+1}^{N}\chi(k).\end{aligned}$$

Hence

$$|A_N|\le\sum_{k=\lfloor N/q\rfloor q+1}^{N}|\chi(k)|\le q,$$

since the last sum has at most q terms and $|\chi(k)|\le 1$. Therefore $\{A_N\}$ is bounded.

(ii) **Decay of b_n.** For $b_n=n^{-z}$, we have

$$|b_n|=|n^{-z}|=n^{-\Re(z)}\longrightarrow 0\quad(n\to\infty),$$

for any $\Re(z)>0$.

(iii) **Absolute convergence of $\sum_{n\ge1}|b_{n+1}-b_n|$.** Let $\gamma(t)=t^{-z}$. Then $\gamma'(t)=-z\,t^{-z-1}$, and thus

$$b_{n+1}-b_n=(n+1)^{-z}-n^{-z}=\gamma(n+1)-\gamma(n)=\int_n^{n+1}\gamma'(t)dt.$$

Taking absolute values and using $|t^{-z-1}|=t^{-\Re(z)-1}$ for $t>0$, we obtain

$$|b_{n+1}-b_n|\le\int_n^{n+1}|\gamma'(t)|dt=|z|\int_n^{n+1}t^{-\Re(z)-1}dt\le|z|\int_n^{n+1}n^{-\Re(z)-1}dt=\frac{|z|}{n^{\Re(z)+1}}.$$

Therefore,

$$\sum_{n=1}^{\infty}|b_{n+1}-b_n|\le|z|\sum_{n=1}^{\infty}\frac{1}{n^{\Re(z)+1}}<\infty\quad\text{for all }\Re(z)>0.$$

All three conditions of the Dirichlet test are satisfied when $\Re(z) > 0$. Hence the series

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^z}$$

converges for every $z \in \mathbb{C}$ with $\Re(z) > 0$, i.e. $L(z, \chi)$ is well-defined in the half-plane $\Re(z) > 0$.

Theorem 3.6 (Enhanced for previous Theorem)

The Dirichlet L-function

$$L(z, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^z}$$

defines an analytic function in the half-plane $\Re(z) > 0$, where χ is a non-principal Dirichlet character modulo q. In particular, as $z \to 1^+$, the function $L(z, \chi)$ remains bounded. ♡

Proof We now prove that $L(z, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^z}$ converges for $\Re(z) > 0$ and defines an analytic function in this half-plane, assuming χ is a nonprincipal Dirichlet character modulo q.
Define the sequences

$$a_n = \chi(n), \qquad b_n(z) = n^{-z}, \qquad A_n = \sum_{k=1}^{n} a_k, \qquad D_{n+1}(z) = b_{n+1}(z) - b_n(z).$$

Same as in the proof of theorem 3.5, the sequence of partial sums A_n is bounded:

$$|A_n| \leq q.$$

Fix $\varepsilon > 0$ and consider the half-plane $\{z \in \mathbb{C} : \Re(z) \geq \varepsilon\}$. Since $|b_N(z)| = |N^{-z}| = N^{-\Re(z)}$, we obtain

$$|A_N b_N(z)| \leq q\, N^{-\Re(z)} \leq q\, N^{-\varepsilon} \xrightarrow[N\to\infty]{} 0,$$

uniformly for $\Re(z) \geq \varepsilon$. Next, same as the proof of theorem 3.5, we have

$$|D_{n+1}(z)| \leq \frac{|z|}{n^{\Re(z)+1}}$$

Combining this with the boundedness of A_n, we get

$$|A_n D_{n+1}(z)| \leq q\, \frac{|z|}{n^{1+\varepsilon}} \qquad (\Re(z) \geq \varepsilon).$$

Fix z with $\Re(z) \geq \varepsilon$. Since

$$\sum_{n=1}^{\infty} n^{-1-\varepsilon} < \infty,$$

by using the Weierstrass M-test we conclude that the series

$$\sum_{n=1}^{\infty} A_n D_{n+1}(z)$$

converges uniformly in the half-plane $\{\Re(z) \geq \varepsilon\}$.

Moreover, since $|b_N(z)| = N^{-x}$ and $|A_N| \leq q$, we have

$$|A_N b_N(z)| \leq q\, N^{-x} \leq q\, N^{-\varepsilon} \xrightarrow[N\to\infty]{} 0,$$

uniformly for $\Re(z) \geq \varepsilon$.

By summation by parts 3.2, we have

$$\sum_{n=1}^{N} a_n b_n(z) = A_N b_N(z) - \sum_{n=1}^{N-1} A_n D_{n+1}(z).$$

Letting $N \to \infty$, the first term tends to 0 uniformly in $\{\Re(z) \geq \varepsilon\}$, while the second term converges uniformly. Hence the partial sums of

$$\sum_{n=1}^{\infty} a_n b_n(z) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^z}$$

converge uniformly in the half-plane $\{\Re(z) \geq \varepsilon\}$.

Since each term $z \mapsto \chi(n)n^{-z}$ is an analytic function, uniform convergence implies that $L(z,\chi)$ defines an analytic function in $\{\Re(z) \geq \varepsilon\}$. As $\varepsilon > 0$ is arbitrary, we conclude that $L(z,\chi)$ is analytic in the half-plane $\{\Re(z) > 0\}$.

In particular, when χ is a nonprincipal Dirichlet character modulo q, the function $L(z,\chi)$ remains bounded as $z \to 1$.

3.5 Putting Things Together

Theorem 3.7 (Characters Form an Orthonormal Basis)

For the vector space $\mathbb{C}^G$ defined in exercise 3.5, the characters in $\widehat{G}$ form an orthonormal basis of $\mathbb{C}^G$. ♡

Proof By orthogonality relations of characters 3.1, the characters in $\widehat{G}$ are orthonormal with respect to the inner product defined there. Thus, the characters in $\widehat{G}$ are linearly independent in the vector space $\mathbb{C}^G$. Recall that in theorem 3.2, we have shown that $|\widehat{G}| = |G|$ (since $\widehat{G} \cong G$). Therefore, by exercise 3.6, we have exactly $|G|$ linearly independent characters in $\widehat{G}$, which is equal to the dimension of the vector space $\mathbb{C}^G$. Thus, the characters in $\widehat{G}$ form a basis of $\mathbb{C}^G$.

Example 3.3 Let G_q be the group of units of $\mathbb{Z}/q\mathbb{Z}$ with multiplication modulo q as the group operation. Let $\delta_a : G_q \to \mathbb{C}$ be the function defined by $\delta_a([n]) = 1$ if $[n] = [a]$ and 0 otherwise. By theorem 3.7, we can express δ_a as a linear combination of the characters in $\widehat{G_q}$:

$$\delta_a = \sum_{\chi \in \widehat{G_q}} c_\chi \chi,$$

and we can easily find the coefficients c_χ using the inner product defined in theorem 3.1:

$$c_\chi = \langle \delta_a, \chi \rangle \quad \text{for all } \chi \in \widehat{G_q}.$$

The last part follows from the property of orthonormal basis in inner product spaces which we have learned in linear algebra, if you forget, here is the proof:

$$\langle \delta_a, \chi_j \rangle = \langle \sum_{\chi_i \in \widehat{G_q}} c_{\chi_i}\chi_i, \chi_j \rangle = \sum_{\chi_i \in \widehat{G_q}} c_{\chi_i} \langle \chi_i, \chi_j \rangle = \sum_{\chi_i \in \widehat{G_q}} c_{\chi_i} \delta_{\chi_i,\chi_j} = c_{\chi_j}.$$

Lemma 3.3 (Just one step to arrive at Dirichlet's Theorem)

We can extend our previous lemma 4.1 like below:

Once we have the dirichlet L-function

$$L(z,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^z} \quad \textit{for } \Re(z) > 1,$$

not equal to zero at $z = 1$ (i.e., $L(1,\chi) \neq 0$) for any non-principal Dirichlet character χ modulo q.

We can show that the series

$$\sum_{\substack{p \text{ prime} \\ p \equiv a \mod q}} \frac{1}{p} = \infty$$

and hence the Dirichlet's Theorem 4.5 holds. ♡

Proof **For any Dirichlet character χ modulo q:**

Let χ be a Dirichlet character modulo q. By the definition of Dirichlet character, we have χ is completely multiplicative. Thus,

$$\frac{\chi(n)}{n^z} \cdot \frac{\chi(m)}{m^z} = \frac{\chi(nm)}{(nm)^z} \quad \text{for all } n, m \in \mathbb{Z}^+.$$

which means that the function $a(n) = \frac{\chi(n)}{n^z}$ is also completely multiplicative. Notice that for any $z \in \mathbb{C}$ with $\Re(z) > 1$, we can choose $\varepsilon > 0$ such that $\Re(z) > 1 + \varepsilon$. Then, we have

$$\sum_{n=1}^{\infty} \left| \frac{\chi(n)}{n^z} \right| = \sum_{n=1}^{\infty} \frac{|\chi(n)|}{n^{\Re(z)}} \leq \sum_{n=1}^{\infty} \frac{1}{n^{\Re(z)}} \leq \sum_{n=1}^{\infty} \frac{1}{n^{1+\varepsilon}} < \infty.$$

Thus, for any $z \in \mathbb{C}$ with $\Re(z) > 1$, the series $\sum_{n=1}^{\infty} \frac{\chi(n)}{n^z}$ converges absolutely, and therefore by theorem 2.8, we have

$$L(z,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^z} = \prod_{p \text{ prime}} \frac{1}{1-\chi(p)p^{-z}} \quad \text{for } \Re(z) > 1.$$

Then, by Corollary 2.4, we have

$$\log L(z,\chi) := \sum_{p \text{ prime}} \sum_{k=1}^{\infty} \frac{1}{k}\chi(p^k)p^{-kz} \quad \text{for } \Re(z) > 1.$$

is a branch of the logarithm of $L(z,\chi)$ satisfying

$$e^{\log L(z,\chi)} = L(z,\chi) \quad \text{for } \Re(z) > 1.$$

Returning to the main proof of the lemma:

Let $\mathbb{1}_{a,q}(n)$ be the indicator function for the arithmetic progression $\mathrm{AP}_{a,q}$, defined as

$$\mathbb{1}_{a,q}(n) = \begin{cases} 1 & \text{if } n \equiv a \mod q, \\ 0 & \text{otherwise.} \end{cases}$$

then we can find that

$$\mathbb{1}_{a,q}(n) = \delta_a([n]) \quad \text{for all } n \in \mathbb{Z}$$

where δ_a is defined in example 3.3. Thus, by example 3.3, we have

$$\mathbb{1}_{a,q}(n) = \sum_{\chi \in \widehat{G_q}} c_\chi \chi([n]) \quad \text{for all } n \in \mathbb{Z}$$

where

$$c_\chi = \langle \delta_a, \chi \rangle = \frac{1}{|G_q|} \sum_{g \in G_q} \delta_a(g)\overline{\chi(g)} = \frac{1}{|G_q|}\overline{\chi([a])}.$$

Especially, $c_{\chi_0} = \frac{1}{|G_q|}\chi_0([a]) = \frac{1}{\varphi(q)}$ where χ_0 is the principal Dirichlet character modulo q. Therefore, we have

$$\sum_{\chi \in \widehat{G_q}} c_\chi \log L(z,\chi) = \sum_{\chi \in \widehat{G_q}} \sum_{p \text{ prime}} \sum_{k=1}^{\infty} \frac{1}{k} c_\chi \chi(p^k)p^{-kz} = \sum_{p \text{ prime}} \sum_{k=1}^{\infty} \frac{1}{k}p^{-kz} \sum_{\chi \in \widehat{G_q}} c_\chi \chi(p^k) = \sum_{p \text{ prime}} \sum_{k=1}^{\infty} \frac{1}{k}p^{-kz}\mathbb{1}_{a,q}(p^k)$$

do the same trick as in lemma 4.1, we have

$$\sum_{\chi \in \widehat{G_q}} c_\chi \log L(z,\chi) = \sum_{p \text{ prime}} \left(p^{-z}\mathbb{1}_{a,q}(p) + \sum_{k=2}^{\infty} \frac{1}{k}p^{-kz}\mathbb{1}_{a,q}(p^k) \right) = \sum_{\substack{p \text{ prime} \\ p \equiv a \mod q}} p^{-z} + E(z) \tag{3.1}$$

where

$$|E(z)| = \left| \sum_{p \text{ prime}} \sum_{k=2}^{\infty} \frac{1}{k}p^{-kz}\mathbb{1}_{a,q}(p^k) \right| \le \sum_{p \text{ prime}} \sum_{k=2}^{\infty} p^{-k\Re(z)} \le \sum_{p \text{ prime}} \sum_{k=2}^{\infty} p^{-k} = \sum_{p \text{ prime}} \frac{1}{p(p-1)} < 1 \quad \text{for } \Re(z) > 1.$$

Rearranging the above equation 3.1, we have

$$\sum_{\substack{p \text{ prime} \\ p \equiv a \mod q}} p^{-z} = c_{\chi_0} \log L(z,\chi_0) + \sum_{\substack{\chi \in \widehat{G_q} \\ \chi \neq \chi_0}} c_\chi \log L(z,\chi) - E(z) \quad \text{for } \Re(z) > 1.$$

where χ_0 is the principal Dirichlet character modulo q (i.e., $\chi_0(n) = 1$ if $\gcd(n,q) = 1$ and 0 otherwise).

Recall the Theorem 3.6 that for any non-principal Dirichlet character χ modulo q, the function $L(z,\chi)$ remains bounded as $z = x \to 1^+$, and we assume that $L(z,\chi)$ is nonzero at $z = 1$ for any non-principal Dirichlet character χ modulo q. Thus, we have

$$\sum_{\substack{p \text{ prime} \\ p \equiv a \mod q}} p^{-z} = c_{\chi_0} \log L(z,\chi_0) + M \quad \text{for } \Re(z) > 1,$$

where M is a constant that does not depend on z since the sum $\sum_{\substack{\chi \in \widehat{G_q} \\ \chi \neq \chi_0}} c_\chi \log L(z,\chi)$ remains bounded as $z = x \to 1^+$ and $E(z)$ is also bounded for $\Re(z) > 1$.

Notice that the principal Dirichlet character χ_0 modulo q is defined by $\chi_0(n) = 1$ if $\gcd(n,q) = 1$ and 0 otherwise.

Thus, we have

$$c_{\chi_0} = \langle \delta_a, \chi_0 \rangle = \frac{1}{|G_q|} \sum_{g \in G_q} \delta_a(g)\overline{\chi_0(g)} = \frac{1}{|G_q|} = \frac{1}{\varphi(q)} \neq 0.$$

and we also have $\log L(z, \chi_0) = \infty$ as $z = x \to 1^+$ by using the lemma 1.3. Thus, we conclude that

$$\sum_{\substack{p \text{ prime} \\ p \equiv a \mod q}} \frac{1}{p} = \infty.$$

Chapter 4 Dirichlet Theorem on Primes in Arithmetic Progressions

4.1 Complex Case

Lemma 4.1 (Dirichlet's Euler Product Idea)

The product of all Dirichlet L-functions modulo q

$$\prod_{\chi} L(x,\chi) \geq 1 \quad \textit{for } x > 1,$$

where x is a real number and the product is taken over all Dirichlet characters χ modulo q.

Proof By Corollary 2.4, for any Dirichlet character χ modulo q, we have

$$L(x,\chi) = \exp\left(\sum_{p \text{ prime}} \sum_{k=1}^{\infty} \frac{1}{k}\chi(p^k)p^{-kx}\right) \quad \text{for } x > 1.$$

Hence, we have

$$\prod_{\chi} L(x,\chi) = \exp\left(\sum_{\chi}\sum_{p \text{ prime}} \sum_{k=1}^{\infty} \frac{1}{k}\chi(p^k)p^{-kx}\right) = \exp\left(\sum_{p \text{ prime}} \sum_{k=1}^{\infty} \frac{1}{k}p^{-kx}\sum_{\chi}\chi(p^k)\right) \quad \text{for } x > 1.$$

where the product is taken over all Dirichlet characters χ modulo q.

Consider the sum $S(p^k) := \sum_{\chi}\chi(p^k)$. We consider the following two cases:

(**Case 1**) If there there exists a dirichlet character $\chi_\star$ such that $\chi_\star(p^k) \neq 1$. Then, by the orthogonality relations of characters 3.1, we have

$$S(p^k) = \sum_{\chi}\chi(p^k) = \sum_{\chi}[\chi\chi_\star](p^k) = \sum_{\chi}\chi(p^k)\chi_\star(p^k) = \chi_\star(p^k)S(p^k).$$

where the second equality follows from the exercise 3.4. Since $\chi_\star(p^k) \neq 1$, we have $(1-\chi_\star(p^k))S(p^k) = 0$, which implies that $S(p^k) = 0$.

(**Case 2**) If all dirichlet characters χ satisfy $\chi(p^k) = 1$. Then $S(p^k) = \sum_{\chi} 1 = |\widehat{G_q}| = |G_q| = \varphi(q)$.

Therefore, in either case, we have $S(p^k) \geq 0$. Hence, we conclude that

$$\sum_{p \text{ prime}} \sum_{k=1}^{\infty} \frac{1}{k}p^{-kz}\sum_{\chi}\chi(p^k) \geq 0 \quad \text{for } \Re(z) > 1,$$

which implies that

$$\prod_{\chi} L(x,\chi) = \exp\left(\sum_{p \text{ prime}} \sum_{k=1}^{\infty} \frac{1}{k}p^{-kx}\sum_{\chi}\chi(p^k)\right) \geq 1 \quad \text{for } x > 1.$$

Lemma 4.2 (Non-vanishing of Dirichlet L-functions at 1 for Complex Characters)

For every non-principal Dirichlet character χ modulo q, the Dirichlet L-function $L(z,\chi)$ at $z = 1$ is nonzero, i.e.

$$L(1,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n} \neq 0.$$

when χ is a complex character.

Proof We consider two cases: the real case where χ takes only real values so that $\chi(n) = \overline{\chi(n)}$ for all n, and the complex case where χ takes some non-real values. But here we only prove the complex case, the real case will be proved in the next section.

We prove that by using contradiction. Assume that there exists a non-principal Dirichlet character χ modulo q such that $L(1, \chi) = 0$. Consider the conjugate character $\overline{\chi}$ defined by $\overline{\chi}(n) = \overline{\chi(n)}$ for all n. It is easy to verify that $\overline{\chi}$ is also a non-principal Dirichlet character modulo q. Then, we have

$$L(z; \overline{\chi}) = \overline{L(x; \chi)} \quad \text{for } x > 1, \quad \text{and hence } L(1; \overline{\chi}) = \overline{L(1; \chi)} = 0.$$

Recall that both χ and $\overline{\chi}$ are analytic functions in a neighborhood of $z = 1$ by theorem 3.6. Thus, we can write the Taylor expansions of $L(z; \chi)$ and $L(z; \overline{\chi})$ at $z = 1$ as

$$L(z; \chi) = \sum_{n=0}^{\infty} a_n (z-1)^n, \quad L(z; \overline{\chi}) = \sum_{n=0}^{\infty} b_n (z-1)^n,$$

where $a_n = \frac{L^{(n)}(1;\chi)}{n!}$ and $b_n = \frac{L^{(n)}(1;\overline{\chi})}{n!}$ for all n. Since $L(1; \chi) = 0$ and $L(1; \overline{\chi}) = 0$, we have $a_0 = b_0 = 0$. Hence the Taylor expansions may be written as

$$L(z; \chi) = (z-1) \sum_{n=1}^{\infty} a_n (z-1)^{n-1}, \qquad L(z; \overline{\chi}) = (z-1) \sum_{n=1}^{\infty} b_n (z-1)^{n-1}.$$

Since both $L(z; \chi)$ and $L(z; \overline{\chi})$ are analytic in a neighborhood of $z = 1$, the power series

$$\sum_{n=1}^{\infty} a_n (z-1)^{n-1} \quad \text{and} \quad \sum_{n=1}^{\infty} b_n (z-1)^{n-1}$$

converge uniformly on some closed disk $|z - 1| \leq r$, where r is a sufficiently small positive number. In particular, there exists a constant $C > 0$ such that

$$\left| \sum_{n=1}^{\infty} a_n (z-1)^{n-1} \right| \leq C_1, \qquad \left| \sum_{n=1}^{\infty} b_n (z-1)^{n-1} \right| \leq C_2$$

for all $|z - 1| \leq r$. So that when $z \to 1$, we have

$$L(z; \chi) \leq C|z-1| \quad \text{and} \quad L(z; \overline{\chi}) \leq C|z-1|$$

for some constant $C > 0$.

On the other hand, by the first formula in the proof of lemma 1.3, we have

$$L(x; \chi_0) \leq \phi(q) \sum_{n=1}^{\infty} \frac{1}{n^x},$$

and by the Integral Test for convergence, we have

$$\sum_{n=1}^{\infty} \frac{1}{n^x} \leq 1 + \int_1^{\infty} \frac{1}{t^x} dt = \frac{x}{x-1} \quad \text{for } x > 1.$$

Hence, we have

$$L(x; \chi_0) \leq \phi(q) \frac{x}{x-1} \quad \text{for } x > 1.$$

Now, consider the product of all Dirichlet L-functions modulo q:

$$\prod_{\chi} L(x, \chi) = L(x; \chi_0) \cdot L(x; \chi) \cdot L(x; \overline{\chi}) \cdot \prod_{\substack{\chi' \in \widehat{G_q} \\ \chi' \neq \chi, \overline{\chi}, \chi_0}} L(x; \chi').$$

By using the above bounds for $L(x; \chi_0)$, $L(x; \chi)$ and $L(x; \overline{\chi})$, recall that for any non-principal Dirichlet character χ' modulo q, the function $L(z; \chi')$ remains bounded as $z = x \to 1^+$ by theorem 3.6, we have

$$\prod_{\chi} L(x, \chi) \leq C' \frac{x}{x-1} |x-1|^2 = C' x |x-1| \to 0 \quad \text{as } x \to 1^+$$

But this contradicts with the lemma 4.1 that $\prod_{\chi} L(x, \chi) \geq 1$ for $x > 1$. Hence, we conclude that $L(1, \chi) \neq 0$ for any non-principal Dirichlet character χ modulo q.

4.2 Real Case - Some Extra Work is Needed

Definition 4.1 (Divisor function)

For any positive integer n, we define the divisor function $d(n)$ as the number of positive divisors of n. For example,

$$d(1)=1,\quad d(2)=2,\quad d(3)=2,\quad d(4)=3,\quad d(5)=2,\quad d(6)=4,\quad d(12)=6.$$

♣

Theorem 4.1 (Formula for the Divisor Function)

For any positive integer n, we have

$$d(n)=\prod_{i=1}^{k}(1+\nu_i),$$

where

$$n=p_1^{\nu_1}p_2^{\nu_2}\cdots p_k^{\nu_k}$$

is the prime factorization of n.

♡

Proof Notice that any positive divisor d of n must be of the form

$$d=p_1^{\mu_1}p_2^{\mu_2}\cdots p_k^{\mu_k}$$

where $0\le\mu_i\le\nu_i$ for all $i=1,2,\ldots,k$. Hence, for each prime factor p_i, we have ν_i+1 choices for the exponent μ_i (i.e., $\mu_i=0,1,2,\ldots,\nu_i$). Therefore, the total number of positive divisors of n is given by

$$d(n)=(\nu_1+1)(\nu_2+1)\cdots(\nu_k+1)=\prod_{i=1}^{k}(1+\nu_i).$$

Proposition 4.1 (Multiplicativity of the Divisor Function)

For any two positive integers m and n, we have

$$d(mn)=\prod_{p \text{ prime}}(1+\nu_p(m)+\nu_p(n)),$$

where

$$m=\prod_{p \text{ prime}}p^{\nu_p(m)},\qquad n=\prod_{p \text{ prime}}p^{\nu_p(n)}$$

are the prime factorizations of m and n respectively.

♠

✍ **Exercise 4.1** Prove the above proposition.

Solution *Notice that the prime factorization of mn is given by*

$$mn=\prod_{p \text{ prime}}p^{\nu_p(m)+\nu_p(n)}.$$

Hence, by the formula for the divisor function in theorem 4.1, we have

$$d(mn)=\prod_{p \text{ prime}}(1+\nu_p(m)+\nu_p(n)).$$

Remark For the average value of the divisor function, define it as

$$A(N)=\frac{1}{N}\sum_{n=1}^{N}d(n).$$

Dirichlet proved that

$$A(N)=\log N+2\gamma-1+E(N),\qquad \text{where } |E(n)|\le\frac{C}{\sqrt{N}}$$

for some constant $C > 0$ and γ is the Euler-Mascheroni constant defined by

$$\gamma = 1 - \int_1^\infty \frac{t - \lfloor t \rfloor}{t^2} dt.$$

Definition 4.2 (Big O Notation)

Let f and g be two functions defined on some subset of the real numbers. We say that $f(x) = \mathcal{O}(g(x))$ as $x \to a$ if there exist positive constants C and δ such that

$$|f(x)| \leq C|g(x)| \quad \text{for all } x \text{ with } 0 < |x - a| < \delta.$$

For example, we have

$$A(N) = \log N + 2\gamma - 1 + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \quad \text{as } N \to \infty.$$

♣

Example 4.1 Let $A(N) = \frac{1}{N}\sum_{n=1}^{N} d(n)$ be the average value of the divisor function. We can prove that

$$A(N) = \log N + \mathcal{O}(1)$$

(Hint: You can use this useful fact that $\sum_{k=1}^{N}\frac{1}{k} = \log N + \gamma + \mathcal{O}\left(\frac{1}{N}\right)$ as $N \to \infty$).

Here is the proof: Notice that the divisor function $d(n)$ can be expressed as

$$d(n) = \sum_{d|n} 1 = \sum_{\substack{j \\ jk=n}} \sum_{k} 1$$

because we can consider the sum of all divisors of n as the sum of lattice points (j, k) in the first quadrant on a hyperbola $y = \frac{n}{x}$, where j and k are positive integers. So that we have

$$A(N) = \frac{1}{N}\sum_{n=1}^{N} d(n) = \frac{1}{N}\sum_{n=1}^{N}\left[\sum_{\substack{j \\ jk=n}}\sum_{k} 1\right] = \frac{1}{N}\sum_{j=1}^{N}\sum_{k=1}^{\lfloor N/j \rfloor} 1 = \frac{1}{N}\sum_{j=1}^{N}\left\lfloor \frac{N}{j} \right\rfloor.$$

Therefore,

$$A(N) = \frac{1}{N}\sum_{k=1}^{N}\frac{N}{k} + \mathcal{O}(1)$$

By using the hint, we have

$$A(N) = \frac{1}{N} \cdot N\left(\log N + \gamma + \mathcal{O}\left(\frac{1}{N}\right)\right) + \mathcal{O}(1) = \log N + \mathcal{O}(1) \quad \text{as } N \to \infty.$$

Theorem 4.2 (Lattice Point Count for the Divisor Function)

Define a sum

$$\mathcal{S}_N := NA(N) = \sum_{n=1}^{N}\left[\sum_{\substack{j \\ jk=n}}\sum_{k} 1\right]$$

where $A(N)$ is the average value of the divisor function defined by $A(N) = \frac{1}{N}\sum_{n=1}^{N} d(n)$. Then, we have

$$\mathcal{S}_N = 2\sum_{(j,k)\in L} 1 + \sum_{(j,k)\in S_q} 1$$

where

$$L = \{(j,k) \in \mathbb{N}^2 : 1 \leq j \leq \sqrt{N}, \sqrt{N} < k \leq N/j\}, \quad S_q = \{(j,k) \in \mathbb{N}^2 : (j,k) \in [0, \sqrt{N}]^2\}$$

♡

Proof We first rewrite $\mathcal{S}_N$ as a lattice point count

$$\mathcal{S}_N := NA(N) = \sum_{n=1}^{N}\left[\sum_{\substack{j \\ jk=n}}\sum_{k} 1\right]$$

Hence, if we define the region

$$H_N := \{(j,k) \in \mathbb{N}^2 : \ jk \le N\},$$

then

$$\mathcal{S}_N = \sum_{(j,k)\in H_N} 1.$$

Geometric decomposition. Let $Q = \lfloor \sqrt{N} \rfloor$. Since $jk \le N$ implies $j \le \sqrt{N}$ or $k \le \sqrt{N}$, every point of H_N lies either in the square $S_q = [1,Q] \times [1,Q]$ or in one of the two "rectangular" parts below/above the diagonal (see Figure 4.1). More precisely, define

$$U := \{(j,k) \in \mathbb{N}^2 : \ 1 \le j \le Q, \ Q < k \le N/j\}.$$

Then one checks that the three sets S_q, L, U are pairwise disjoint and

$$H_N = S_q \cup L \cup U.$$

Indeed:

- If $(j,k) \in S_q$, then $j \le Q$ and $k \le Q$.
- If $(j,k) \in L$, then $k \le Q$ and $j > Q$.
- If $(j,k) \in U$, then $j \le Q$ and $k > Q$.

These conditions are mutually exclusive, so the union is disjoint. Also, if $(j,k) \in H_N$ and $j \le Q$ and $k \le Q$, then $(j,k) \in S_q$; if $(j,k) \in H_N$ and $k \le Q < j$, then $j \le N/k$ so $(j,k) \in L$; and if $(j,k) \in H_N$ and $j \le Q < k$, then $k \le N/j$ so $(j,k) \in U$. Hence the disjoint union holds.

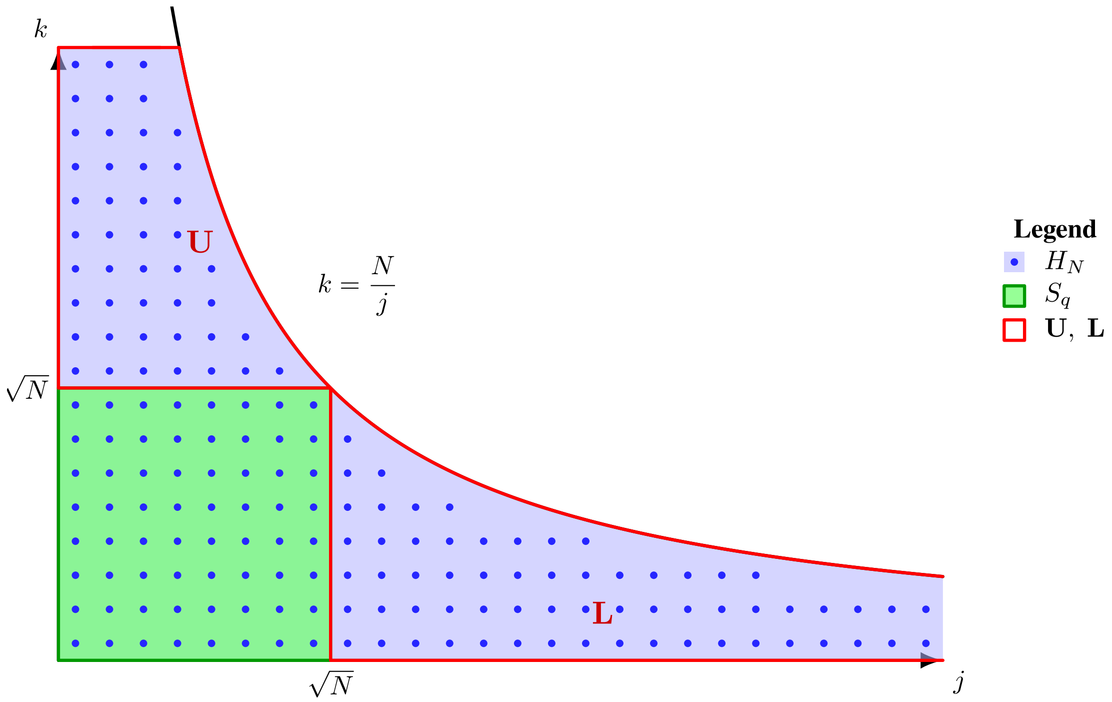


Figure 4.1: Decomposition of the region under $jk \le N$ into S_q and the symmetric parts U and L.

Using symmetry. Consider the involution $\sigma : \mathbb{N}^2 \to \mathbb{N}^2$ given by $\sigma(j,k) = (k,j)$. It preserves the condition $jk \le N$, hence $\sigma(H_N) = H_N$. Moreover, by definition, $\sigma(L) = U$ and $\sigma(U) = L$, so $|L| = |U|$, i.e.

$$\sum_{(j,k)\in L} 1 = \sum_{(j,k)\in U} 1.$$

Therefore, using the disjoint union $H_N = S_q \cup L \cup U$,

$$\mathcal{S}_N = \sum_{(j,k)\in H_N} 1 = \sum_{(j,k)\in S_q} 1 + \sum_{(j,k)\in L} 1 + \sum_{(j,k)\in U} 1 = \sum_{(j,k)\in S_q} 1 + 2\sum_{(j,k)\in L} 1,$$

which is exactly the claimed identity.

Theorem 4.3 (Asymptotic Formula for the Average Value of the Divisor Function)

The partial sums of the harmonic series satisfies

$$\sum_{n=1}^{N} \frac{1}{n} = \log N + \gamma + \mathcal{O}\left(\frac{1}{N}\right) \qquad \text{as } N \to \infty,$$

where γ is the Euler constant defined by

$$\gamma = 1 - \int_1^\infty \frac{t - \lfloor t \rfloor}{t^2} dt.$$

and more precisely, we have

$$A(N) = \frac{1}{N} \sum_{n=1}^{N} d(n) = \log N + 2\gamma - 1 + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \qquad \text{as } N \to \infty.$$

♡

Proof See Exercise set at the end of this chapter.

Theorem 4.4 (Asymptotic Formula for the Partial Sums of the Square Root Function)

Similarly as the above theorem, we can also prove that

$$\sum_{n=1}^{N} \frac{1}{\sqrt{n}} = 2\sqrt{N} + \sigma + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \qquad \text{as } N \to \infty,$$

where σ is a constant defined by

$$\sigma = 1 - \int_1^\infty \frac{t - \lfloor t \rfloor}{t^{3/2}} dt.$$

♡

Proof omitted.

4.3 Returning to the real case

Lemma 4.3 (Abel's Summation Formula)

Let $\{a_n\}$ and $\{b_n\}$ be two sequences of complex numbers. Define the partial sums $A_N = \sum_{n=1}^{N} a_n$ for $N \geq 1$. Then, for any integers $M \geq J \geq 1$,

$$\sum_{n=J}^{M} a_n b_n = A_M b_M - A_{J-1} b_J + \sum_{n=J}^{M-1} A_n (b_n - b_{n+1}),$$

where we define $A_0 = 0$.

♡

Proof We have

$$\begin{aligned}
\sum_{n=J}^{M} a_n b_n &= \sum_{n=J}^{M} (A_n - A_{n-1}) b_n \\
&= \sum_{n=J}^{M} A_n b_n - \sum_{n=J}^{M} A_{n-1} b_n \\
&= \sum_{n=J}^{M} A_n b_n - \sum_{n=J-1}^{M-1} A_n b_{n+1} \\
&= A_M b_M - A_{J-1} b_J + \sum_{n=J}^{M-1} A_n b_n - \sum_{n=J}^{M-1} A_n b_{n+1} \\
&= A_M b_M - A_{J-1} b_J + \sum_{n=J}^{M-1} A_n (b_n - b_{n+1}).
\end{aligned}$$

Lemma 4.4 (Bound for Partial Sums of Non-Principal Dirichlet Characters)

Let χ be a non-principal Dirichlet character modulo q. Then for $K \geq J \geq 1$,

$$\left|\sum_{n=J}^{K} \frac{\chi(n)}{\sqrt{n}}\right| \leq \frac{2q}{\sqrt{J}}.$$

♡

Proof Let G_q denote the multiplicative group of invertible elements of $\mathbb{Z}/q\mathbb{Z}$. By using the periodicity of Dirichlet character χ modulo q, we have

$$\sum_{r=1}^{q} \chi(r) = \sum_{\substack{1 \leq r \leq q \\ \gcd(r,q)=1}} \chi(r) = \sum_{r \in G_q} \chi(r) = \sum_{r \in G_q} \chi(r)\overline{\chi_0(r)} = |G_q|\langle \chi, \chi_0 \rangle = 0,$$

where χ_0 is the principal Dirichlet character modulo q and the last equality follows from the orthogonality of characters (see Theorem 3.1). Thus, if we use $A_N = \sum_{n=1}^{N} a_n$ where $a_n = \chi(n)$, to denote the partial sums of $\chi(n)$, then for any integer $N \geq 1$, we have

$$\begin{aligned}
A_N = \sum_{k=1}^{N} \chi(k) &= \sum_{k=1}^{\lfloor N/q \rfloor q} \chi(k) + \sum_{k=\lfloor N/q \rfloor q+1}^{N} \chi(k) \\
&= \sum_{m=0}^{\lfloor N/q \rfloor -1} \sum_{r=1}^{q} \chi(mq+r) + \sum_{k=\lfloor N/q \rfloor q+1}^{N} \chi(k) \\
&= (\lfloor N/q \rfloor) \sum_{r=1}^{q} \chi(r) + \sum_{k=\lfloor N/q \rfloor q+1}^{N} \chi(k) \\
&= \sum_{k=\lfloor N/q \rfloor q+1}^{N} \chi(k).
\end{aligned}$$

Hence

$$|A_N| \leq \sum_{k=\lfloor N/q \rfloor q+1}^{N} |\chi(k)| \leq q.$$

Let $b_n = n^{-1/2}$. By Lemma 3.2, for integers $K \geq J \geq 1$,

$$\sum_{n=J}^{K} \frac{\chi(n)}{\sqrt{n}} = A_K b_K - A_{J-1} b_J + \sum_{n=J}^{K-1} A_n (b_n - b_{n+1}).$$

Since $|A_N| \leq q$ for all $N \geq 1$ and $b_n = n^{-1/2}$ is decreasing, we have

$$\begin{aligned}
\left|\sum_{n=J}^{K} \frac{\chi(n)}{\sqrt{n}}\right| &\leq |A_K b_K| + |A_{J-1} b_J| + \sum_{n=J}^{K-1} |A_n|\,|b_n - b_{n+1}| \\
&\leq q\left(\frac{1}{\sqrt{K}} + \frac{1}{\sqrt{J}} + \sum_{n=J}^{K-1}\left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right)\right) \\
&= q\left(\frac{1}{\sqrt{K}} + \frac{1}{\sqrt{J}} + \left(\frac{1}{\sqrt{J}} - \frac{1}{\sqrt{K}}\right)\right) \\
&= \frac{2q}{\sqrt{J}}
\end{aligned}$$

Lemma 4.5

For any non-principal Dirichlet character χ modulo q. Consider the partial sums:

$$F_N(\chi) := \sum_{(m,n) \in H_N} \frac{\chi(n)}{\sqrt{mn}}$$

where $H_N = \{(m,n) \in \mathbb{N}^2 : mn \le N\}$. *Then, we have*

$$F_N(\chi) = 2\sqrt{N}L(1;\chi) + \mathcal{O}(1) \quad \textit{as } N \to \infty.$$

and for some $c > 0$, *we have*

$$F_N(\chi) \ge c\log N \quad \textit{for sufficiently large } N.$$

♡

Proof We begin with the proof of $F_N(\chi) \ge c\log N$ for some constant $c > 0$. Notice that for any non-principal real Dirichlet character χ modulo q, there exists an integer a such that $\gcd(a,q) = 1$ and $\chi(a) = -1$. Hence, we have

$$F_N(\chi) = \sum_{(m,n)\in H_N} \frac{\chi(n)}{\sqrt{mn}} = \sum_{k=1}^{N}\left[\sum_{\substack{m \quad n\\ mn=k}}\sum \frac{\chi(n)}{\sqrt{mn}}\right] = \sum_{k=1}^{N}\frac{1}{\sqrt{k}}\sum_{n|k}\chi(n)$$

Since χ is a real character, and we know that it's a root of unity from Theorem 3.2, we have $\chi(n) = \pm 1$ if $\gcd(n,q) = 1$. Hence, by factoring index k into prime factors $k = \prod_{p \text{ prime}} p^{\nu_p(k)}$, once n divides k, we can write n as $n = \prod_{p\text{ prime}} p^{\mu_p}$ where $0 \le \mu_p \le \nu_p(k)$ for all p. Hence,

$$\chi(n) = \chi\left(\prod_{p\text{ prime}} p^{\mu_p}\right) = \prod_{p\text{ prime}}\chi(p)^{\mu_p}.$$

and the sum over all divisors n of k can be written as

$$\sum_{n|k}\chi(n) = \prod_{p\text{ prime}}\left(1 + \chi(p) + \chi(p)^2 + \cdots + \chi(p)^{\nu_p(k)}\right).$$

Since $\chi(p) = 0$ for $\gcd(p,q) > 1$, and $\mu_p(k) = 0$ for all $p \not| k$, we can further simplify the above formula as

$$\sum_{n|k}\chi(n) = \prod_{\substack{p\text{ prime}\\ p|k\\ \gcd(p,q)=1}}\left(1 + \chi(p) + \chi(p)^2 + \cdots + \chi(p)^{\nu_p(k)}\right) = \prod_{\substack{p\text{ prime}\\ p|k\\ \gcd(p,q)=1}}\sum_{j=0}^{\nu_p(k)}\chi(p)^j.$$

Now, we discuss the value of $\sum_{j=0}^{\nu_p(k)}\chi(p)^j$ for each prime factor p of k with $\gcd(p,q) = 1$.

- If $\chi(p) = 1$, then $\sum_{j=0}^{\nu_p(k)}\chi(p)^j = \nu_p(k) + 1$.
- If $\chi(p) = -1$, and $\nu_p(k)$ is even, then $\sum_{j=0}^{\nu_p(k)}\chi(p)^j = 1$.
- If $\chi(p) = -1$, and $\nu_p(k)$ is odd, then $\sum_{j=0}^{\nu_p(k)}\chi(p)^j = 0$.

Thus, once we have $k = l^2$ is a perfect square, then $\nu_p(k)$ is even for all prime factors p of k, and we have

$$\sum_{n|l^2}\chi(n) = \prod_{\substack{p\text{ prime}\\ p|l^2\\ \gcd(p,q)=1}}\sum_{j=0}^{\nu_p(l^2)}\chi(p)^j \ge 1.$$

Return to the formula for $F_N(\chi)$, we have

$$\begin{aligned}
F_N(\chi) &= \sum_{k=1}^{N}\frac{1}{\sqrt{k}}\sum_{n|k}\chi(n) \ge \sum_{l=1}^{\lfloor\sqrt{N}\rfloor}\frac{1}{l}\sum_{n|l^2}\chi(n) \ge \sum_{l=1}^{\lfloor\sqrt{N}\rfloor}\frac{1}{l}\\
&\ge \log(\lfloor\sqrt{N}\rfloor) + \gamma + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \quad \text{as } N \to \infty \quad \text{(by the theorem 4.3)}\\
&= \frac{1}{2}\log N + \gamma + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right)\\
&\ge c\log N \quad \text{for sufficiently large } N. \quad \left(\text{since } \gamma > 0 \text{ and } \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \to 0 \text{ as } N \to \infty.\right)
\end{aligned}$$

Now, we turn to the proof of $F_N(\chi) = 2\sqrt{N}L(1;\chi) + \mathcal{O}(1)$ as $N \to \infty$. Doing the same lattice point count as in the proof of theorem 4.2, we have

$$H_N = S_q \cup L \cup U$$

where

$$S_q = \{(m,n) \in \mathbb{N}^2 : (m,n) \in [0, \sqrt{N}]^2\}$$

and

$$L = \{(m,n) \in \mathbb{N}^2 : 1 \le m \le \sqrt{N}, \sqrt{N} < n \le N/m\}, \quad U = \{(m,n) \in \mathbb{N}^2 : \sqrt{N} < m \le N/n, 1 \le n \le \sqrt{N}\}.$$

Hence, we have

$$F_N(\chi) = \sum_{(m,n)\in H_N} \frac{\chi(n)}{\sqrt{mn}} = \sum_{(m,n)\in L\cup S_q} \frac{\chi(n)}{\sqrt{mn}} + \sum_{(m,n)\in U} \frac{\chi(n)}{\sqrt{mn}} =: A + B.$$

Then, by using the lemma 4.4, we have

$$|B| = \left| \sum_{(m,n)\in U} \frac{\chi(n)}{\sqrt{mn}} \right| \le \sum_{n=1}^{\lfloor\sqrt{N}\rfloor} \frac{1}{\sqrt{n}} \sum_{m=\lfloor\sqrt{N}\rfloor+1}^{N/n} \frac{|\chi(n)|}{\sqrt{n}} \le \sum_{n=1}^{\lfloor\sqrt{N}\rfloor} \frac{1}{\sqrt{n}} \cdot \frac{2q}{\sqrt{\sqrt{N}+1}} \le \frac{2q}{N^{1/4}} \sum_{n=1}^{\lfloor\sqrt{N}\rfloor} \frac{1}{\sqrt{n}}$$

thus

$$\begin{aligned} |B| &\le \frac{2q}{N^{1/4}} \sum_{n=1}^{\lfloor\sqrt{N}\rfloor} \frac{1}{\sqrt{n}} \le \frac{2q}{N^{1/4}} \left(1 + \int_1^{\lfloor\sqrt{N}\rfloor} \frac{1}{\sqrt{t}} dt\right) = \frac{2q}{N^{1/4}} \left(1 + 2(\sqrt{\lfloor\sqrt{N}\rfloor} - 1)\right) \\ &\le \frac{2q}{N^{1/4}} \left(2 + 2(\sqrt{\sqrt{N}} - 1)\right) = 4q = \mathcal{O}(1) \quad \text{as } N \to \infty. \end{aligned}$$

For the term A, notice that $L \cup S_q = \{(m,n) \in \mathbb{N}^2 :\ 1 \le m \le \sqrt{N},\ 1 \le n \le N/m\}$, hence

$$A = \sum_{(m,n)\in L\cup S_q} \frac{\chi(n)}{\sqrt{mn}} = \sum_{m=1}^{\lfloor\sqrt{N}\rfloor} \frac{\chi(m)}{\sqrt{m}} \sum_{n=1}^{\lfloor N/m\rfloor} \frac{1}{\sqrt{n}} \le \sum_{m=1}^{\lfloor\sqrt{N}\rfloor} \frac{\chi(m)}{\sqrt{m}} P_{N/m} \quad \text{where } P_{N/m} = \sum_{n=1}^{N/m} \frac{1}{\sqrt{n}}.$$

By theorem 4.4, we have

$$P_{N/m} \le P_N = 2\sqrt{N} + \sigma + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \quad \text{as } N \to \infty.$$

Hence,

$$\begin{aligned} A &\le \sum_{m=1}^{\lfloor\sqrt{N}\rfloor} \frac{\chi(m)}{\sqrt{m}} P_N = P_N \sum_{m=1}^{\lfloor\sqrt{N}\rfloor} \frac{\chi(m)}{\sqrt{m}} \\ &= 2\sqrt{N} \sum_{m=1}^{\lfloor\sqrt{N}\rfloor} \frac{\chi(m)}{\sqrt{m}} + \sigma \sum_{m=1}^{\lfloor\sqrt{N}\rfloor} \frac{\chi(m)}{\sqrt{m}} + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \sum_{m=1}^{\lfloor\sqrt{N}\rfloor} \frac{\chi(m)}{\sqrt{m}}. \end{aligned}$$

As

$$\left| \sigma \sum_{m=1}^{\lfloor\sqrt{N}\rfloor} \frac{\chi(m)}{\sqrt{m}} \right| \le |\sigma| \cdot \frac{2q}{1} = \mathcal{O}(1) \quad \text{as } N \to \infty \quad \text{(by lemma 4.4)}$$

and

$$\mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \sum_{m=1}^{\lfloor\sqrt{N}\rfloor} \frac{\chi(m)}{\sqrt{m}} \le \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \cdot \frac{2q}{1} = \mathcal{O}(1) \quad \text{as } N \to \infty \quad \text{(by lemma 4.4)},$$

so we have

$$A = 2\sqrt{N} \sum_{m=1}^{\lfloor\sqrt{N}\rfloor} \frac{\chi(m)}{\sqrt{m}} + \mathcal{O}(1) \quad \text{as } N \to \infty.$$

Finally,

$$\begin{aligned} \left| 2\sqrt{N} \sum_{m=1}^{\lfloor\sqrt{N}\rfloor} \frac{\chi(m)}{\sqrt{m}} \right| &= 2\sqrt{N} \left| L(1;\chi) - \sum_{m=\lfloor\sqrt{N}\rfloor+1}^{\infty} \frac{\chi(m)}{\sqrt{m}} \right| \le 2\sqrt{N} \left(|L(1;\chi)| + \left| \sum_{m=\lfloor\sqrt{N}\rfloor+1}^{\infty} \frac{\chi(m)}{\sqrt{m}} \right| \right) \\ &\le 2\sqrt{N} \left(|L(1;\chi)| + \frac{2q}{\sqrt{\lfloor\sqrt{N}\rfloor + 1}} \right) = 2\sqrt{N} L(1;\chi) + \mathcal{O}(1) \quad \text{as } N \to \infty. \end{aligned}$$

Hence, we have

$$F_N(\chi) = A + B = 2\sqrt{N}L(1;\chi) + \mathcal{O}(1) \quad \text{as } N \to \infty.$$

Corollary 4.1 (Non-vanishing of Dirichlet L-functions at 1 for Real Characters)

For every non-principal Dirichlet character χ modulo q, the Dirichlet L-function $L(z,\chi)$ at $z = 1$ is nonzero, i.e.

$$L(1,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n} \neq 0.$$

when χ is a real character.

Proof By lemma 4.5, we have

$$F_N(\chi) = 2\sqrt{N}L(1;\chi) + \mathcal{O}(1) \quad \text{as } N \to \infty. \quad \text{and } F_N(\chi) \geq c\log N \quad \text{for sufficiently large } N.$$

Hence, we have

$$L(1;\chi) = \frac{F_N(\chi)}{2\sqrt{N}} + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right)$$

By definition of the Big O notation, there exists a constant $C > 0$ such that $\mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \leq \frac{C}{\sqrt{N}}$ for sufficiently large N. Hence, we have

$$L(1;\chi) \geq \frac{c\log N}{2\sqrt{N}} - \frac{C}{\sqrt{N}} = \frac{c\log N - 2C}{2\sqrt{N}} > 0 \quad \text{for sufficiently large } N.$$

Therefore, we conclude that $L(1;\chi) \neq 0$ for any non-principal real Dirichlet character χ modulo q.

Corollary 4.2 (Non-vanishing of Dirichlet L-functions at 1 for Non-principal Characters)

For every non-principal Dirichlet character χ modulo q, the Dirichlet L-function $L(z,\chi)$ at $z = 1$ is nonzero, i.e.

$$L(1,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n} \neq 0.$$

no matter χ is a real character or a complex character.

Proof Directly follows from lemma 4.2 and corollary 4.1.

Theorem 4.5 (Dirichlet's Theorem on Primes in Arithmetic Progressions)

Let a and q be positive integers with $\gcd(a,q) = 1$. *Then, there are infinitely many prime numbers in the arithmetic progression*

$$\mathrm{AP}_{a,q} = \{a + kq : k \in \mathbb{Z}^+\}.$$

Proof Directly follows from Lemma 3.3 and Corollary 4.2 by using the same argument as in the proof of Theorem **??**.

Exercises

1. Suppose that $f : \mathbb{R}^+ \to \mathbb{R}$ is continuously differentiable. By using the integral by parts, prove that for any $n \geq 2$,

$$\int_{n-1}^{n} tf'(t)dt = nf(n) - (n-1)f(n-1) - \int_{n-1}^{n} f(t)dt$$

and then use this to show that for any $n \geq 2$,

$$f(n) = \int_{n-1}^{n} f(t)dt + \int_{n-1}^{n} (t - \lfloor t \rfloor)f'(t)dt.$$

Finally, by using the above formula, prove that

$$\sum_{n=M}^{N} f(n) = \int_{M-1}^{N} f(t)dt + \int_{M-1}^{N} (t - \lfloor t \rfloor)f'(t)dt \quad \text{for any } 2 \leq M < N.$$

Solution *By using the integration by parts, for any $n \geq 2$, we have*

$$\int_{n-1}^{n} tf'(t)dt = tf(t)|_{n-1}^{n} - \int_{n-1}^{n} f(t)dt = nf(n) - (n-1)f(n-1) - \int_{n-1}^{n} f(t)dt.$$

Hence, we have

$$\begin{aligned}
nf(n)-(n-1)f(n-1) &= \int_{n-1}^{n} tf'(t)dt + \int_{n-1}^{n} f(t)dt \\
&= \int_{n-1}^{n} (t-\lfloor t\rfloor+\lfloor t\rfloor)f'(t)dt + \int_{n-1}^{n} f(t)dt \\
&= \int_{n-1}^{n} (t-\lfloor t\rfloor)f'(t)dt + \int_{n-1}^{n} f(t)dt + \int_{n-1}^{n} \lfloor t\rfloor f'(t)dt \\
&= \int_{n-1}^{n} (t-\lfloor t\rfloor)f'(t)dt + \int_{n-1}^{n} f(t)dt + \int_{n-1}^{n} (n-1)f'(t)dt \\
&= \int_{n-1}^{n} (t-\lfloor t\rfloor)f'(t)dt + \int_{n-1}^{n} f(t)dt + (n-1)f(n)-(n-1)f(n-1). \\
\implies nf(n) &= \int_{n-1}^{n} (t-\lfloor t\rfloor)f'(t)dt + \int_{n-1}^{n} f(t)dt + nf(n)-f(n) \\
\implies f(n) &= \int_{n-1}^{n} (t-\lfloor t\rfloor)f'(t)dt + \int_{n-1}^{n} f(t)dt.
\end{aligned}$$

Finally, by using the above formula, for any $2 \le M < N$, *we have*

$$\sum_{n=M}^{N} f(n) = \sum_{n=M}^{N}\left[\int_{n-1}^{n} f(t)dt + \int_{n-1}^{n}(t-\lfloor t\rfloor)f'(t)dt\right] = \int_{M-1}^{N} f(t)dt + \int_{M-1}^{N}(t-\lfloor t\rfloor)f'(t)dt.$$

2. By using the final formula in the above problem, show that

$$S(N) := \sum_{n=1}^{N}\frac{1}{n} = \log N + 1 - \int_{1}^{N}\frac{t-\lfloor t\rfloor}{t^2}dt.$$

and that for any $M \ge 1$, the limit

$$L(M) := \lim_{R\to\infty}\int_{M}^{R}\frac{t-\lfloor t\rfloor}{t^2}dt \quad \text{exists.}$$

and that

$$0 \le L(M) \le \frac{1}{M}.$$

Solution *By using the final formula in the above problem with* $f(n) = \frac{1}{n}$, *we have*

$$S(N) = \sum_{n=1}^{N}\frac{1}{n} = 1 + \sum_{n=2}^{N} f(n) = 1 + \int_{1}^{N}\frac{1}{t}dt - \int_{1}^{N}\frac{t-\lfloor t\rfloor}{t^2}dt = \log N + 1 - \int_{1}^{N}\frac{t-\lfloor t\rfloor}{t^2}dt.$$

For any $M \ge 1$, *we have*

$$\begin{aligned}
\int_{M}^{R}\frac{t-\lfloor t\rfloor}{t^2}dt &\le \int_{M}^{R}\frac{1}{t^2}dt \quad \textit{since } 0 \le t-\lfloor t\rfloor < 1 \\
&= \frac{1}{M} - \frac{1}{R} \to \frac{1}{M} \quad \textit{as} \quad R\to\infty.
\end{aligned}$$

Moreover, notice that $\frac{t-\lfloor t\rfloor}{t^2} \ge 0$ *for all* $t \ge 1$, *so that* $\int_M^R \frac{t-\lfloor t\rfloor}{t^2}dt$ *is an increasing and bounded above by* $\frac{1}{M}$ *as* $R\to\infty$. *Hence, by the monotone convergence theorem,* $L(M)$ *exists and clearly, we have* $0 \le L(M) \le \frac{1}{M}$.

3. Show that

$$S(N) = \log(N) + 1 - \int_{1}^{\infty}\frac{t-\lfloor t\rfloor}{t^2}dt + \int_{N}^{\infty}\frac{t-\lfloor t\rfloor}{t^2}dt.$$

(where the integral on the right-hand side converges by the previous problem) and hence conclude that

$$S(N) = \log N + \gamma + \mathcal{O}\left(\frac{1}{N}\right) \qquad \text{as } N\to\infty,$$

where

$$\gamma = 1 - \int_{1}^{\infty}\frac{t-\lfloor t\rfloor}{t^2}dt,$$

and S_N is defined by previously as $S(N) = \sum_{n=1}^{N}\frac{1}{n}$.

Solution *By previous problem, we have*

$$S(N) = \log N + 1 - \int_1^N \frac{t - \lfloor t \rfloor}{t^2} dt.$$

Consider the integral

$$\widetilde{S}_R := \log N + 1 - \int_1^R \frac{t - \lfloor t \rfloor}{t^2} dt + \int_N^R \frac{t - \lfloor t \rfloor}{t^2} dt.$$

By the previous problem, both $\int_1^R \frac{t-\lfloor t \rfloor}{t^2} dt$ *and* $\int_N^R \frac{t-\lfloor t \rfloor}{t^2} dt$ *converge as* $R \to \infty$. *Hence, we can take the limit as* $R \to \infty$ *on both sides to get*

$$\widetilde{S}_R \to \log N + 1 - \int_1^\infty \frac{t - \lfloor t \rfloor}{t^2} dt + \int_N^\infty \frac{t - \lfloor t \rfloor}{t^2} dt \quad \text{as } R \to \infty.$$

On the other hand, by the definition of $\widetilde{S}_R$, *we have*

$$\widetilde{S}_R = \log N + 1 - \int_1^N \frac{t - \lfloor t \rfloor}{t^2} dt + \int_N^R \frac{t - \lfloor t \rfloor}{t^2} dt.$$

Taking the limit as $R \to \infty$, *we get*

$$S(N) = \lim_{R \to \infty} \widetilde{S}_R = \log N + 1 - \int_1^\infty \frac{t - \lfloor t \rfloor}{t^2} dt + \int_N^\infty \frac{t - \lfloor t \rfloor}{t^2} dt.$$

Finally, since $0 \leq \int_N^\infty \frac{t-\lfloor t \rfloor}{t^2} dt \leq \frac{1}{N}$ *by the previous problem, it follows that* $\int_N^\infty \frac{t-\lfloor t \rfloor}{t^2} dt = \mathcal{O}\left(\frac{1}{N}\right)$ *as* $N \to \infty$. *Hence, we have*

$$S(N) = \log N + 1 - \int_1^\infty \frac{t - \lfloor t \rfloor}{t^2} dt + \int_N^\infty \frac{t - \lfloor t \rfloor}{t^2} dt = \log N + \gamma + \mathcal{O}\left(\frac{1}{N}\right) \quad \text{as } N \to \infty.$$

4. Finally, using the Theorem 4.2:

$$A(N) = \frac{1}{N} \sum_{n=1}^N d(n) \quad \text{and} \quad \mathcal{S}_N = \sum_{n=1}^N d(n) = NA(N) = 2 \sum_{(j,k) \in L} 1 + \sum_{(j,k) \in S_q} 1 =: A + B$$

where

$$L = \{(i,j) \in \mathbb{N}^2 : 1 \leq j \leq \sqrt{N}, \sqrt{N} < k \leq N/j\}, \quad S_q = \{(j,k) \in \mathbb{N}^2 : (j,k) \in [0, \sqrt{N}]^2\}$$

show that

$$A = 2 \sum_{k=1}^{\lfloor \sqrt{N} \rfloor} \left(\lfloor N/k \rfloor - \lfloor \sqrt{N} \rfloor \right)$$

and then use the previous problems: $S(N) = \sum_{n=1}^N \frac{1}{n} = \log N + \gamma + \mathcal{O}\left(\frac{1}{N}\right)$ as $N \to \infty$, to show that

$$A = N \log N + (2\gamma - 2)N + \mathcal{O}(\sqrt{N}) \quad \text{as } N \to \infty.$$

and also show that

$$B = N + \mathcal{O}(\sqrt{N}) \quad \text{as } N \to \infty.$$

Hence, conclude that

$$\frac{1}{N} \sum_{n=1}^N d(n) = A(N) = \log N + 2\gamma - 1 + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \quad \text{as } N \to \infty.$$

(Here $S(N), \gamma$ are defined in the previous problems.)

Solution *For the first part, by the definition of* L, *we have*

$$A = 2 \sum_{(j,k) \in L} 1 = 2 \sum_{k=1}^{\lfloor \sqrt{N} \rfloor} \sum_{j=\lfloor \sqrt{N} \rfloor + 1}^{\lfloor N/k \rfloor} 1 = 2 \sum_{k=1}^{\lfloor \sqrt{N} \rfloor} \left(\lfloor N/k \rfloor - \lfloor \sqrt{N} \rfloor \right).$$

For the second part, notice that $\lfloor N/k \rfloor - \lfloor \sqrt{N} \rfloor \geq 0$ *for all* $1 \leq k \leq \lfloor \sqrt{N} \rfloor$*, so that*

$$\begin{aligned}
|A| = A &\leq 2 \sum_{k=1}^{\lfloor \sqrt{N} \rfloor} N/k - 2 \sum_{k=1}^{\lfloor \sqrt{N} \rfloor} \lfloor \sqrt{N} \rfloor \\
&\leq 2N \left(\log \sqrt{N} + \gamma + \mathcal{O}\left(\frac{1}{N}\right)\right) - 2 \lfloor \sqrt{N} \rfloor^2 \\
&= N \log N + 2\gamma N + \mathcal{O}(1) - 2(\sqrt{N} - 1)^2 \\
&= N \log N + (2\gamma - 2)N + 2\sqrt{N} + 2 + \mathcal{O}(1) \\
&= N \log N + (2\gamma - 2)N + \mathcal{O}(\sqrt{N}) \quad \textit{as } N \to \infty.
\end{aligned}$$

To estimate B*, notice that*

$$|B| = B = \sum_{(j,k) \in S_q} 1 = \lfloor \sqrt{N} \rfloor^2 \leq (\sqrt{N} - 1)^2 = N - 2\sqrt{N} + 1 = N + \mathcal{O}(\sqrt{N}) \quad \textit{as } N \to \infty.$$

Finally, by the definition of $A(N)$*, we have*

$$\begin{aligned}
A(N) &= \frac{1}{N} \mathcal{S}_N = \frac{1}{N}(A + B) \\
&= \log N + (2\gamma - 2) + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) + 1 + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \\
&= \log N + 2\gamma - 1 + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \quad \textit{as } N \to \infty.
\end{aligned}$$

Chapter 5 The Riemann Zeta Function

Theorem 5.1 (Identity Theorem)

Let h be a holomorphic function on a domain $Y \subseteq \mathbb{C}$ such that $h = 0$ on some open subset $X \subseteq Y$. Then, $h = 0$ on the whole domain Y.

Proof omitted. (In Honour Complex Variables)

Theorem 5.2 (Principal of Analytic Continuation)

Given a complex function $f : X \to \mathbb{C}$ defined on a domain $X \subseteq \mathbb{C}$, if there exists a holomorphic function $F : Y \to \mathbb{C}$ defined on a domain $Y \subseteq \mathbb{C}$ such that $X \subseteq Y$ and $F = f$ on X, then we say that

$$F \text{ is an analytic continuation of } f \text{ from } X \text{ to } Y.$$

and the analytic continuation of f from X to Y is unique if it exists.

Proof Suppose that there are two analytic continuations F_1 and F_2 of f from X to Y. Then, we have

$$F_1 = F_2 \text{ on } X.$$

Hence, by the identity theorem 5.1, we have

$$F_1 = F_2 \text{ on } Y,$$

so the analytic continuation of f from X to Y is unique if it exists.

Theorem 5.3 (Sequence version of the Identity Theorem)

Let h be a holomorphic function on a domain $Y \subseteq \mathbb{C}$ such that there exists a sequence of distinct points $\{z_n\}_{n=1}^{\infty} \subseteq Y$ converging to some point $z_0 \in Y$ such that $h(z_n) = 0$ for all $n \in \mathbb{Z}^+$. Then, $h = 0$ on the whole domain Y.

Proof omitted. (In Honour Complex Variables)

Note *This Theorem is very useful since X may just be a piece of curve in Y, much smaller than an open subset of Y.*

Exercise 5.1 A small example is to show $\sin(z)^2 + \cos(z)^2 = 1$ for all $z \in \mathbb{C}$ by using the sequence version of the identity theorem.

As we know that $\sin(z)^2 + \cos(z)^2 = 1$ for all $z \in \mathbb{R}$, so we can take the sequence $\{z_n\}_{n=1}^{\infty} = \{n\pi\}_{n=1}^{\infty}$ which converges to $+\infty$ and satisfies $\sin(z_n)^2 + \cos(z_n)^2 - 1 = 0$ for all $n \in \mathbb{Z}^+$. Hence, by the sequence version of the identity theorem 5.3, we have $\sin(z)^2 + \cos(z)^2 - 1 = 0$ for all $z \in \mathbb{C}$, i.e. $\sin(z)^2 + \cos(z)^2 = 1$ for all $z \in \mathbb{C}$.

5.1 Analytic Continuation of the zeta function

Lemma 5.1 (Analytic Continuation of a Simple Integral Function)

Define a function

$$I(z) := \int_1^{\infty} \frac{1}{t^z} dt.$$

It will be shown that $I(z)$ converges for all $z \in \mathbb{C}$ with $\mathrm{Re}(z) > 1$ and defines an analytic function

$$\mathcal{I}(z) = \frac{1}{z-1} \quad \text{for all } z \in \mathbb{C} \setminus \{1\},$$

which is an analytic continuation of $I(z)$ from $\{z \in \mathbb{C} : \mathrm{Re}(z) > 1\}$ to $\mathbb{C} \setminus \{1\}$.

Proof For the function $I(z)$, notice that

$$I(z) = \int_1^{\infty} \frac{1}{t^z} dt = \lim_{N\to\infty} \int_1^{N} \frac{1}{t^z} dt = \lim_{N\to\infty} \left[\frac{t^{1-z}}{1-z}\right]_1^N = \lim_{N\to\infty} \frac{N^{1-z}-1}{1-z}.$$

Since

$$|N^{1-z} - 0| = |N^{1-z}| \leq N^{1-\mathrm{Re}(z)} \to 0 \quad \text{as } N \to \infty \quad \text{once} \quad \mathrm{Re}(z) > 1,$$

so the limit $\lim_{N\to\infty} \frac{N^{1-z}-1}{1-z} = \frac{1}{z-1}$ exists for all $z \in \mathbb{C}$ with $\mathrm{Re}(z) > 1$. Hence, we have

$$I(z) = \frac{1}{z-1} \qquad \text{for all } z \in \mathbb{C} \text{ with } \mathrm{Re}(z) > 1.$$

Moreover, the function $\mathcal{I}(z) = \frac{1}{z-1}$ is analytic on $\mathbb{C} \setminus \{1\}$ and clearly, $\mathcal{I}(z) = I(z)$ for all $z \in \mathbb{C}$ with $\mathrm{Re}(z) > 1$. Hence, $\mathcal{I}(z)$ is an analytic continuation of $I(z)$ from $\{z \in \mathbb{C} : \mathrm{Re}(z) > 1\}$ to $\{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}$

Exercise 5.2Absolute Value of an Integral Show that

$$\left|\int_a^b f(t)dt\right| \leq \int_a^b |f(t)|dt$$

for any continuous function $f : [a, b] \to \mathbb{C}$ and any $a, b \in \mathbb{R}$ with $a < b$.

Solution *By the definition of the absolute value of a complex number, we have*

$$\left|\int_a^b f(t)dt\right|^2 = \left(\int_a^b f(t)dt\right)\overline{\left(\int_a^b f(t)dt\right)} = \left(\int_a^b f(t)dt\right)\left(\int_a^b \overline{f(t)}dt\right).$$

By Fubini's theorem, we have

$$\left|\int_a^b f(t)dt\right|^2 = \int_a^b \int_a^b f(t)\overline{f(s)}dtds.$$

By the Cauchy-Schwarz inequality for integrals, we have

$$\left|\int_a^b f(t)dt\right|^2 = \int_a^b \int_a^b f(t)\overline{f(s)}dtds \leq \int_a^b |f(t)|^2 dt.$$

Hence, we have

$$\left|\int_a^b f(t)dt\right|^2 \leq \left(\int_a^b |f(t)|dt\right)^2,$$

which implies that

$$\left|\int_a^b f(t)dt\right| \leq \int_a^b |f(t)|dt.$$

Theorem 5.4 (Analytic Continuation of the Riemann Zeta Function)

The Riemann zeta function

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} = \frac{1}{z-1} - g(z) \quad \textit{for all } z \in \mathbb{C} \textit{ with } \mathrm{Re}(z) > 1,$$

where

$$g(z) = \sum_{n=1}^{\infty} g_n(z) \quad \textit{and} \quad g_n(z) := \int_n^{n+1} \left[\frac{1}{t^z} - \frac{1}{n^z}\right] dt.$$

is an analytic function on $\mathbb{C} \setminus \{1\}$

So that $\frac{1}{z-1} - g(z)$ *is an analytic continuation of* $\zeta(z)$ *from* $\{z \in \mathbb{C} : \mathrm{Re}(z) > 1\}$ *to* $\{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}$ ♡

Proof Consider the partial sums of zeta function and compare

$$\int_1^{N+1} \frac{1}{t^z}dt - \sum_{n=1}^{N} \frac{1}{n^z} = \sum_{n=1}^{N} \int_n^{n+1} \frac{1}{t^z}dt - \sum_{n=1}^{N} \int_n^{n+1} \frac{1}{n^z}dt = \sum_{n=1}^{N} \int_n^{n+1} \left[\frac{1}{t^z} - \frac{1}{n^z}\right] dt = \sum_{n=1}^{N} g_n(z),$$

where $g_n(z) = \int_n^{n+1} \left[\frac{1}{t^z} - \frac{1}{n^z}\right] dt$ is analytic on $\mathbb{C}$. We will use the complex Weierstrass M-test to see that $g(z) = \sum_{n=1}^{\infty} g_n(z)$ converges uniformly on $\mathbb{C}$ with $\mathrm{Re}(z) > 0$:

For any $z \in \mathbb{C}$ with $\mathrm{Re}(z) > 0$, we have

$$|g_n(z)| = \left|\int_n^{n+1} \left[\frac{1}{t^z} - \frac{1}{n^z}\right] dt\right| \leq \int_n^{n+1} \left|\frac{1}{t^z} - \frac{1}{n^z}\right| dt. \quad \text{by Exercise 5.2.}$$

Notice that for $n < t < n+1$,

$$\left|\frac{1}{t^z} - \frac{1}{n^z}\right| = \left|-z\int_n^t \frac{1}{s^{z+1}}ds\right| \le |z|\int_n^t \frac{1}{s^{\mathrm{Re}(z)+1}}ds \le |z|\int_n^{n+1} \frac{1}{s^{\mathrm{Re}(z)+1}}ds = |z|\frac{1}{n^{\mathrm{Re}(z)+1}}.$$

Hence, consider for any $\varepsilon > 0, R > 0$, on the open set $X_{\varepsilon,R} = \{z \in \mathbb{C} : \mathrm{Re}(z) > \varepsilon, |z| < R\}$, we have

$$|g_n(z)| \le |z|\frac{1}{n^{\mathrm{Re}(z)+1}} \le |z|\frac{1}{n^{\varepsilon+1}}.$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^{\varepsilon+1}}$ converges for $\varepsilon > 0$, we have that $g(z) = \sum_{n=1}^{\infty} g_n(z)$ converges uniformly on $\{z \in \mathbb{C} : \mathrm{Re}(z) > \varepsilon, |z| < R\}$ for any $\varepsilon > 0, R > 0$. Therefore, $g(z)$ is analytic on $X_{\varepsilon,R}$ for any $\varepsilon > 0, R > 0$, which implies that $g(z)$ is analytic on $\{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}$.

Finally, since $\frac{1}{z-1} - g(z)$ is an analytic function on $\{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}$ and $\frac{1}{z-1} - g(z) = \zeta(z)$ for all $z \in \mathbb{C}$ with $\mathrm{Re}(z) > 1$, we conclude that $\frac{1}{z-1} - g(z)$ is an analytic continuation of $\zeta(z)$ from $\{z \in \mathbb{C} : \mathrm{Re}(z) > 1\}$ to $\{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}$.

5.2 The Gamma Function

Definition 5.1 (Gamma Function)

For any $z \in \mathbb{C}$ with $\mathrm{Re}(z) > 0$, we define the Gamma function $\Gamma(z)$ by

$$\Gamma(z) = \int_0^{\infty} t^{z-1}e^{-t}dt.$$

♣

Proof The proof of the convergence of the integral defining $\Gamma(z)$ for $\mathrm{Re}(z) > 0$ is omitted.

Proposition 5.1 (Properties of the Gamma Function)

The Gamma function $\Gamma(z)$, $z \in \mathbb{C}$ with $\mathrm{Re}(z) > 0$, has the following properties:

1. $\Gamma(n+1) = n!$ *for all* $n \in \mathbb{Z}^+$.
2. $\Gamma(z+1) = z\Gamma(z)$ *for all* $z \in \mathbb{C}$ *with* $\mathrm{Re}(z) > 0$
3. $\Gamma(n+1) = n! \sim \sqrt{2\pi n}e^{-n}n^n$ *as* $n \to \infty$. *(Stirling's formula)*

♠

Solution *omitted. (In Advanced Methods in Applied Mathematics)*

Theorem 5.5 (Analytic Continuation of the Gamma Function)

There is a analytic continuation of the Gamma function $\Gamma(z)$ from $\{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}$ to $\mathbb{C} \setminus \{0, -1, -2, \ldots\}$ ♡

Proof By using the functional equation $\Gamma(z+1) = z\Gamma(z)$, we can redefine a new gamma function $\widetilde{\Gamma}(z)$ by

$$\widetilde{\Gamma}(z) = \frac{\Gamma(z+1)}{z},$$

which is an analytic continuation of $\Gamma(z)$ from $\{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}$ to $\{z \in \mathbb{C} : \mathrm{Re}(z) > -1\} \setminus \{0\}$. By repeating the above process, we can redefine a new gamma function $\widehat{\Gamma}_n(z)$ by

$$\widehat{\Gamma}_n(z) = \frac{\Gamma(z+n)}{z(z+1)\cdots(z+n-1)},$$

which is an analytic continuation of $\Gamma(z)$ from $\{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}$ to $\{z \in \mathbb{C} : \mathrm{Re}(z) > -n\} \setminus \{0, -1, \ldots, -n+1\}$ for any $n \in \mathbb{Z}^+$. Hence, by taking the limit as $n \to \infty$, we can get an analytic continuation of $\Gamma(z)$ from $\{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}$ to $\mathbb{C} \setminus \{0, -1, -2, \ldots\}$.

Theorem 5.6

After the analytic continuation we have constructed in Theorem 5.5, for any $z \in \mathbb{C} \setminus \mathbb{Z}$, we have

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

♡

Note *One can also rewrite the above formula as*

$$\frac{1}{\Gamma(z)} = \frac{\sin(\pi z)}{\pi}\Gamma(1-z),$$

and notice that $\sin(\pi z)$ *has zeros at* $z \in \mathbb{Z}$*, so that it can cancel the poles of* $\Gamma(z)$ *at* $z = 1, 2, \dots$ *and hence* $\frac{1}{\Gamma(z)}$ *is an entire function on* $\mathbb{C}$ *with zeros at* $z = 0, -1, -2, \dots$.

Proof Consider $z = a$ with $0 < a < 1$ and for $t > 0$. By using the change of variable $u = tv$ we have

$$\Gamma(1-a) = \int_0^\infty e^{-u}u^{-a}du = \int_0^\infty e^{-tv}(tv)^{-a}tdv = t^{1-a}\int_0^\infty e^{-tv}v^{-a}dv.$$

Hence, we have

$$\begin{aligned}
\Gamma(a)\Gamma(1-a) &= \int_0^\infty e^{-t}t^{a-1}\Gamma(1-a)dt = \int_0^\infty e^{-t}t^{a-1}\left[t^{1-a}\int_0^\infty e^{-tv}v^{-a}dv\right]dt \\
&= \int_0^\infty\int_0^\infty e^{-t(1+v)}v^{-a}dtdv \\
&= \int_0^\infty \frac{v^{-a}}{1+v}dv \\
&= \int_{-\infty}^\infty \frac{e^{(1-a)x}}{1+e^x}dx \quad \text{by the change of variable } v = e^x \\
&= \mathrm{Re}\left(\int_{-\infty}^\infty \frac{e^{(1-a)z}}{1+e^z}dz\right) \\
&= \frac{\pi}{\sin(\pi a)} \quad \text{by the Cauchy Residue Theorem}
\end{aligned}$$

and by the sequence version of the identity theorem 5.3, we have $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$ for all $z \in \mathbb{C} \setminus \mathbb{Z}$.

Theorem 5.7 (Connection between the Riemann zeta function and the Gamma function)

Fo any $z \in \mathbb{C} \setminus \{0, -2, -4, \dots\}$*, and any* $n \in \mathbb{Z}^+$*, we have*

$$\int_0^\infty e^{-\pi n^2 u}u^{z/2-1}du = \pi^{-z/2}\Gamma(z/2)n^{-z}$$

♡

Proof Simply by the change of variable $t = \pi n^2 u$, we have

$$\begin{aligned}
\int_0^\infty e^{-\pi n^2 u}u^{z/2-1}du &= \int_0^\infty e^{-t}\left(\frac{t}{\pi n^2}\right)^{z/2-1}\frac{dt}{\pi n^2} = \pi^{-z/2}n^{-z}\int_0^\infty e^{-t}t^{z/2-1}dt \\
&= \pi^{-z/2}\Gamma(z/2)n^{-z}.
\end{aligned}$$

Remark By summing over $n \in \mathbb{Z}^+$, we have

$$\pi^{-z/2}\Gamma(z/2)\zeta(z) = \sum_{n=1}^\infty \int_0^\infty e^{-\pi n^2 u}u^{z/2-1}du = \int_0^\infty \left(\sum_{n=1}^\infty e^{-\pi n^2 u}\right)u^{z/2-1}du,$$

where the interchange of the sum and the integral is justified by the complex Weierstrass M-test since for any $z \in \mathbb{C}$ with $\mathrm{Re}(z) > 0$ (doing in our homework)

5.3 The Theta Function